



STOCKTAKE REPORT FOR UGANDA'S UPDATED NDC

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List of Abbreviations

AFOLU	Agriculture, Forestry and Other Land Use
AR 5	Fifth Assessment Report of the IPCC
BAU	Business As Usual
BEV	Battery Electric Vehicle
BUR	Biennial Update Report
BRT	Bus Rapid Transport
CCD	Climate Change Department
CCS	Carbon Capture and Storage
CCUS	Carbon Capture, Utilization, and Storage
CSA	Climate Smart Agriculture
CTF	Common Tabular Format (UNFCCC)
EACOP	East African Crude Oil Pipeline
EF	Emission Factor
ERP	Electricity Regulatory Authority
ETF	Enhanced Transparency Framework
ETP	Energy Transition Plan
EV	Electric Vehicles
FY	Financial year
GDP	Gross Domestic Product
GEF	Grid Emission Factor
GHG	Greenhouse Gas
Gg	Giga Grammes
GIZ	Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ) GmbH
GKMA	Greater Kampala Metropolitan Area
GoU	Government of Uganda
GWP	Global Warming Potential
H-NAP	National Health Adaptation Plan

IEA	International Energy Agency
IPCC	Intergovernmental Panel on Climate Change
IPPU	Industrial Processes, and Product Use
LT-LEDS or LTS	Long-Term Climate Change Strategy
LULUCF	Land use, Land Use Change and Forestry
MDAs	Ministries, Departments, and Agencies
MEMD	Ministry of Energy and Mineral Development
MGR	Meter Gauge Railway
MoFPED	Ministry of Finance, Planning and Economic Development
MoJCA	Ministry of Justice and Constitutional Affairs
MWE	Ministry of Water and Environment
MoWT	Ministry of Works and Transport
MRV	Monitoring, Reporting and Verification
MtCO ₂ e	Million Tonnes of Carbon Dioxide Equivalent
MPGs	Modalities Procedures and Guidelines (for transparency under ETF)
MW	Megawatt
NA	Not Applicable
NAP	National Adaptation Plan
NAPA	National Adaptation Programme of Action
NC	National Communication
NCCAC	National Climate Change Advisory Committee
NDC	Nationally Determined Contribution
NDP	National Development Plan
NE	Not Evaluated
NGO	Non-governmental Organization
NMT	Non-Motorized Transport
NZE	Net Zero Emissions
NWSC	National Water and Sewerage Corporation
PPP	Public-Private Partnerships
PV	Photovoltaic

SDG	Sustainable Development Goals
SGR	Standard Gauge Railway
TORs	Terms of References
WWTP	Wastewater Treatment Plant
UNDP	United Nations Development Programme
URC	Uganda Railways Corporation
UNFCCC	United Nations Framework Convention on Climate Change
VA	Vulnerability Assessment

UGANDA NDC 2.0 STOCKTAKE REPORT

Executive Summary

ES.1 Purpose, scope, and assessment approach

This stocktake report provides a comprehensive assessment of Uganda's progress in implementing its Updated Nationally Determined Contribution (NDC 2.0) that covers the period 2020/21–2030/31. The review draws on Uganda's institutional framework under the Climate Change Act (2021), sectoral data from Ministries, Departments and Agencies (MDAs), especially under the lead of the Climate Change Department (CCD) under the Ministry of Water and Environment (MWE), stakeholder consultations, and complementary international reports. The analysis follows the Enhanced Transparency Framework (ETF) under the Paris Agreement, employing the Common Tabular Format (CTF) tables, and is enriched with a six-pillar analytical overlay (ambition, quality, process, implementability, investability, credibility) (UNFCCC, 2018; AfDB, 2023).

The report is a midpoint review of Uganda's NDC 2.0, designed to assess progress on mitigation, adaptation, and means of implementation (finance, technology, capacity building). It provides evidence to inform the preparation of Uganda's third NDC (NDC 3.0) while ensuring alignment with the global stocktake. A mixed-method approach was applied: review of policy and technical documents, customized data templates for sectoral reporting, and validation through multi-stakeholder consultations. Where data gaps persisted, triangulation with international publicly available sources e.g. the African Development Bank (AfDB), World Bank and United Nations Development Programme (UNDP), NDC Partnership, was used.

ES.2 Summary of findings on mitigation progress

Uganda's 2030 mitigation target is a 24.7% reduction in emissions relative to BAU (5.9% unconditional, 18.8% conditional). Progress has been achieved in renewable electricity generation (notably hydro and solar photovoltaic (PV) additions), forestry restoration under REDD+ initiatives, and climate-smart agriculture (MWE, 2022; UBOS, 2025). The state of progress for key mitigation components is given below.

Agriculture, Forestry and Other Land Use (AFOLU)

Several interventions are underway, including forest restoration, sustainable land management, climate-smart agriculture, and bioenergy woodlots. By 2024, measurable reductions include:

- Natural forest restoration: 1.68 MtCO₂e achieved emissions reductions by 2024 vs. 0.6 MtCO₂e target by 2030. However, due to difficulty in activity data disaggregation, these removals constitute removals from other strategic interventions including large scale commercial timber plantations sub-options such as commercial transmission pole and timber plantations, and commercial pole and sawlog plantations.
- Commercial small-holder and community bioenergy woodlots: 0.13 MtCO₂e achieved emissions reductions by 2024 vs. 2.9 MtCO₂e, ~4% of the 2030 target
- Most climate-smart agriculture interventions remain unimplemented and/or not reported, emissions reductions are far below the NDC 2.0 2030 target of 3.1 MtCO₂e due to lack of specific

activity data to allow for estimation. However, Uganda has launched the Climate Smart Agricultural Transformation project¹ with support from the World Bank whose implementation across 69 districts will enhance reporting and implementation of climate smart agriculture interventions.

- Non GHG targets: Wetlands coverage has increased from 8.9% in 2020 to 9.32% by the end of 2024. No verifiable data was provided to track changes in national deforestation rates while forest cover increased from 12.5% in 2020 to 12.6% in 2024 according to FSSD which is below the 15% target by 2025.

Energy (Stationary)

- Renewable energy generation: 2.43 million MWh/year by 2025, achieving ~0.319 MtCO₂e reductions, which is ~73% of the NDC 2.0 2030 target of 0.480 MtCO₂e and amounting to 15% of the overall NDC Stationary Energy target of 15%
- Improved charcoal production: 0.0166 MtCO₂e achieved vs. 0.48 MtCO₂e target by 2030 (~3.5% of target).
- Industrial efficiency and fuel switching: No measurable reductions reported; 2030 targets remain unaddressed.

Transport

- Key targets for 2030 in the NDC 2.0 comprise:
 - Fuel efficiency improvements: 1.86 MtCO₂e/year
 - Alternative fuel switching: 0.54 MtCO₂e/year
 - Non-motorized transport (NMT): 0.66 MtCO₂e/year
- Current achievements: 29 km GKMA rail operational, 400 km Meter Gauge Railway (MGR) rehabilitation (140 km operational), 1.95 km NMT infrastructure.
- No quantified emissions reductions are reported, meaning 0% progress toward mitigation targets.

Waste

- Methane recovery from wastewater: 0.003 MtCO₂e/year achieved vs. 0.017 MtCO₂e 2030 target (~18% of target).
- Solid waste composting: 0.194 MtCO₂e/year achieved vs. 1.1 MtCO₂e 2030 target (~18% of target).

Industrial Processes and Product Use (IPPU)

- Clinker substitution in cement: emission reductions projected at 0.1 MtCO₂e/year by 2030, but no reductions reported yet.
- Other industrial interventions remain in the planning or pilot stages.

Limited monitoring data for transport, waste, and IPPU sectors constrained the full assessment. Oil and gas developments pose a future challenge to emission trajectories (IEA, 2023). Accelerated implementation,

¹ <https://www.agriculture.go.ug/ucsatp/>

stronger Monitoring, Reporting and Verification (MRV) data and systems, and securing conditional financial support are needed to meet pledged reductions.

ES.3 Summary of findings on adaptation progress

Uganda's progress on climate adaptation actions under its NDC implementation (2021–2025) shows moderate overall advancement, with significant variation across sectors. However, data availability, quality, and baseline establishment in some sectors remain major challenges limiting full assessment. Across all sectors, average implementation progress is about 50–55%, signaling the need for strengthened monitoring, financing, and coordination mechanisms. Good progress is observed in the Transport (72%), Energy (61%) and Agriculture (58%) sectors while slow progress is observed in Cities, Health, DRR, Education, and Tourism and manufacturing sectors. Sectoral summaries are presented below:

Environment and Ecosystem

Overall progress in the sector stands at 50.1% with this progress mainly coming from the wetland subsector that has been supported by the GCF funded Wetlands restoration project. The major challenge is incomplete baseline data for several ecosystem indicators that were supposed to be determined during the NDC implementation period.

Water and Sanitation

Overall progress in the sector stands at 52.7% with strong improvements in water storage and sanitation access where some indicators such as population with access to basic sanitation and compliance with national water standards exceed the 2025 targets. The challenge was in data gaps for renewable (solar/wind)-powered water systems and rainwater harvesting. The stocktake shows strong sanitation progress but lagging urban water access and recommends expanding renewable water systems and data reporting in the sector.

Transport

Overall progress in the sector stands at a good value of 71.9% with urban road paving and green infrastructure showing strong gains mainly driven by the Uganda Cities and Municipalities Infrastructure Development program supported by the World Bank. The sector has also performed well by providing present information on all the indicators that have baselines in NDC2.0

Energy

Overall progress in the sector stands at 61.4% with significant gains in electricity access and transmission lines. However, clean cooking adoption remains low with still high values on the share of biomass energy used for cooking. Renewable off-grid energy tracking data missing to allow complete assessment of the sector. Clean cooking that lags in terms of progress requires targeted investment and behavior change initiatives for its improvement.

Agriculture

This was assessed based on 75% of the indicators with present data and overall progress at 57.9%. There are reduced post-harvest losses and increased irrigation as well as development of climate-resilient crop varieties. There are strong outcomes in post-harvest loss reduction and CSA promotion, but there is a need for quantifiable tracking for sustainable land management.

Forestry

There is significant progress in this sector with the only tracked indicator of tree planting exceeding the 2025 target and a significant portion of forestry land restored. Overall progress shows excellent reforestation results but baseline data for other indicators is needed to sustain long-term forest gains

Fisheries

The sector has not progressed well with the only quantitative indicator of Volume of fish stock having increased by only 25.5% from the baseline value. However, there are also initiatives on aquaculture training and regulations that are ongoing, but no quantitative tracking data is available.

Disaster Risk reduction

Overall progress in the sector stands at 40% with meteorological data accuracy and weather network automation improved. There is weak progress in local government action planning with only 2 districts out of 146 having developed their plans. There is a need to prioritize local-level DRR mainstreaming and accuracy verification methods for meteorological information.

Cities and built environment

Overall progress in the sector stands at 35.6% where drainage infrastructure has improved and slum population has reduced modestly. However, there is still weak data and overlaps with the transport sector that need harmonization.

Health sector

Progress in this sector is only based on 33.3% of the indicators with the National Health Adaptation Plan (H-NAP) and risk assessments completed. The implementation of H-NAP as well as climate-smart hospital transformation should form a strong basis in the development and implementation of NDC3.0.

Manufacturing, Industry and Mining

This was not assessed due to lack of current data and no established baselines. There is a strong need for establishing adaptation monitoring frameworks for industry and promotion of circular economy metrics.

Education

There was limited data to assess progress with the only indicator assessed being the integration of climate change education into the national curriculum. Climate change has been integrated in lower secondary revised curriculum and in some A level subject curricula such as Geography and Biology. Some tertiary institutions have some programs that address climate change from climate science to adaptation and mitigation.

Tourism

No measurable progress was assessed due to lack of current data with the only assessed indicator being climate change vulnerability assessment in the tourism sector that is planned for the financial year 2025/2026. There is a need to strengthen data systems and link adaptation with nature-based tourism resilience.

ES.4 Summary of findings on means of implementation and support

Implementation of NDC 2.0 requires an estimated USD 28.1 billion by 2030, of which only 15% (USD 4.1 billion) is expected from domestic resources. The remaining USD 24 billion is contingent on international support. While climate finance inflows have increased, Uganda has not yet developed a systematic pipeline of bankable projects nor fully leveraged opportunities under carbon markets (Article 6) (EIB, 2022; 2023). At the local level, adaptation financing remains scarce. Even where funding is secured, absorption capacity challenges delay implementation. Without enhanced access to international funds and development of domestic finance tools, NDC targets will remain at risk. Across sectors, persistent technical capacity shortages within MDAs continue to undermine effective NDC implementation.

To date, progress on mobilizing finance has been mixed: national budget support and external projects have covered multiple adaptation sectors (agriculture, health, forestry, water, transport, energy, and cities), while three of the five mitigation sectors (AFOLU, energy, transport) have attracted resources, leaving waste and IPPU unfunded. Technology transfer has advanced through renewable energy generation, resilient infrastructure, climate-smart agriculture, aquaculture innovations, and digital systems, while capacity-building has strengthened farmer groups, local governments, utilities, and sectoral institutions. Still several sectors (Environment & Ecosystems, Tourism, Disaster Risk Reduction) remain largely unfunded, and critical technology action plans have yet to be implemented. While there is evidence of functional capacity-building programs and technology deployment, the gap between NDC 2.0 financial needs and actual mobilized resources, as well as incomplete sectoral coverage, indicates that Uganda is only partially on track to meet the means-of-implementation targets for 2030.

Uganda has made significant institutional strides through the Climate Change Act (2021), which designates the CCD as the MRV control center. However, clear mandates for MDAs and local governments in implementing and reporting NDC actions are lacking, and off-budget actions by non-state actors are difficult to capture in a centralized system (MWE, 2024).

ES.5 Key Barriers and Gaps

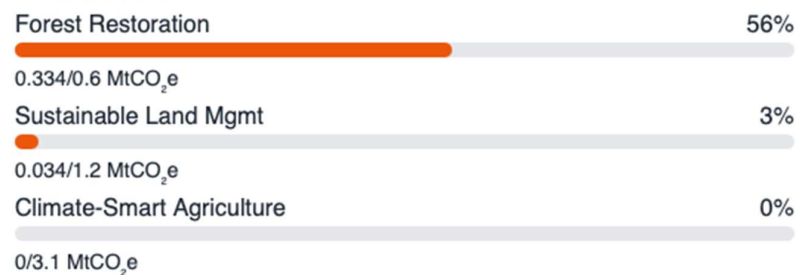
Persistent data limitations, weak MRV systems, limited capacity of technical skills, overreliance on external finance, and limited private sector participation hinder full implementation. Key barriers such as weak regulatory enforcement, inadequate funding (with 85% of conditional costs unmet), procurement delays, and socio-economic constraints have slowed project execution and restricted adoption of climate-resilient practices, especially in energy, AFOLU, and transport sectors.

Ambition is constrained by the absence of an explicit peak year, limited unconditional contributions, and insufficient integration of just transition, gender, and inclusivity principles.

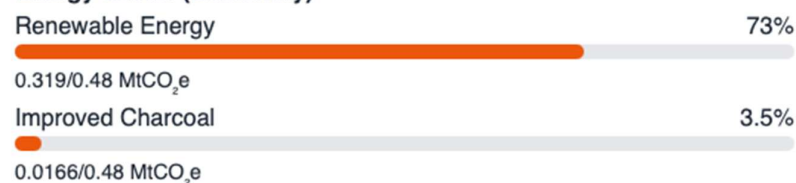
Figure ES. 0-1 Visual summary of Uganda's NDC 2.0 stocktake

• Mitigation Progress

AFOLU Sector



Energy Sector (Stationary)



Transport

Fuel efficiency improvements, alternative fuel switching, and Non-motorized transport: 0% progress toward mitigation targets

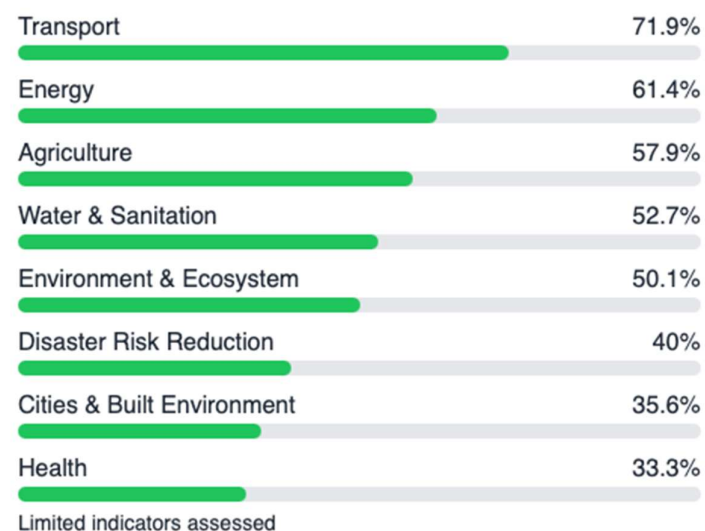
Waste



IPPU

Clinker substitution in cement: 0% progress - No reductions reported yet

• Adaptation Progress



Other Sectors With Limited/No Data:

Tourism, Manufacturing/Industry/Mining, Fisheries, Education, Forestry

• Means of Implementation

Financial Requirements

\$28.1B

Total Required by 2030

Domestic Resources \$4.1B (15%)

International Support Needed \$24B (85%)

ES.6 Priority Recommendations for NDC 3.0

The stocktake report highlights seven strategic priorities to guide the design and implementation of Uganda's third NDC (NDC 3.0) as follows:

- Strengthen data and transparency systems: Strengthen the national MRV and centralize climate data management system across all key sectors to address persistent data and reporting gaps (Section 10).
- Set a clear long-term emissions pathway: Define a national emissions peak year and align Uganda's commitments with a mid-century net-zero trajectory to enhance ambition and predictability (Section 3.1).
- Mainstream adaptation planning: Integrate the forthcoming National Adaptation Plan (NAP) into NDC 3.0, supported by sectoral monitoring and evaluation frameworks (Section 10).
- Clarify and empower institutions: Strengthen institutional mandates, especially local governments, and ensure adequate resourcing for implementation (Sections 5 and 10).
- Mobilize finance and carbon market opportunities: Leverage Article 6 and develop a pipeline of bankable projects in renewable energy, forestry, and resilient agriculture to close financing gaps (Sections 9 and 10).
- Increase domestic ambition: Expand the scope of unconditional commitments, demonstrating stronger national resolve and ownership of climate action (Section 3.1).
- Promote a just and inclusive transition: Embed principles of just transition, gender equity, and inclusivity across all NDC sectors to ensure fair and sustainable outcomes (Section 11).
- Pursue a greener pathway for oil and gas development to align with the national mitigation goals.

Uganda's NDC 2.0 has laid an important foundation, but this stocktake underscores that achieving its targets and enhancing ambition in NDC 3.0 will require stronger data systems, clearer institutional mandates, deeper adaptation integration, and mobilization of both domestic and international support. By acting on the strategic priorities outlined above, Uganda can position itself to deliver a more ambitious, credible, and implementable NDC 3.0; that advances national priorities while contributing meaningfully to global climate action.

1. Introduction

1.1 Stocktake and Alignment with the ToR

This stocktake is designed as a midpoint review of Uganda’s progress in implementing its Updated Nationally Determined Contribution (NDC 2.0) for the period 2020/21–2030/31. It provides a structured assessment of sectoral progress, institutional arrangements, and Uganda’s alignment with national development frameworks and international obligations under the Paris Agreement. Conducted within the five-year ambition cycle of the Paris Agreement, the stocktake evaluates achievements to date, identifies key gaps and barriers, and generates evidence-based recommendations to inform Uganda’s forthcoming NDC 3.0.

The assessment was guided by the Terms of Reference (TORs) agreed between the Government of Uganda and the African Development Bank under the Africa NDC Hub. The TORs emphasize three central dimensions:

1. Assessment of Progress– evaluating sectoral and cross-cutting implementation of NDC 2.0, including mitigation, adaptation, and means of implementation.
2. Institutional Analysis– reviewing the effectiveness of institutional arrangements, governance frameworks, and coordination mechanisms under the Climate Change Act (2021).
3. Strategic Alignment– ensuring Uganda’s NDC processes remain consistent with the objectives of the Paris Agreement, the Global Stocktake (GST), and domestic development goals such as Vision 2040 and the National Development Plans.

To demonstrate consistency and accountability, a TOR-to-Report mapping table was prepared, showing where each of the fourteen TORs (i–xiv) is addressed in this report.

Table 1.-1- Mapping of Terms of Reference (TORs) to Stocktake Report Sections-

TOR Ref.	TOR Description	Section(s) in Report Addressing TOR
i	Provide context on Uganda’s national circumstances relevant to NDC implementation	Section 2 (Introduction & Context: 2.2–2.7)
ii	Assess Uganda’s NDC 2.0 mitigation targets and ambition	Section 3.1 (NDC Targets); Section 6.2 (Pillar 1 – Ambition)
iii	Review institutional frameworks for NDC implementation and coordination	Section 3.2 (Institutional Framework); Section 5 (Stakeholder Engagement)
iv	Assess methodology and approaches used in NDC 2.0 design and implementation	Section 4 (Methodology)
v	Document stakeholder engagement processes in NDC implementation	Section 5 (Stakeholder Engagement)
vi	Evaluate alignment of Uganda’s NDC 2.0 with the SDGs and national development strategies	Section 6.1 (Alignment with SDGs); Section 11.8 (On alignment with MDGs/SDGs)

vii	Track progress on mitigation actions and GHG reductions	Section 7 (Tracking Progress on Mitigation)
viii	Track progress on adaptation actions and resilience building	Section 8 (Tracking Progress on Adaptation)
ix	Track support received and needed (finance, technology, capacity building)	Section 9 (Support Received and Needed)
x	Identify gaps, challenges, and barriers in NDC implementation	Section 10 (Assessment of Gaps and Barriers)
xi	Assess lessons learnt from NDC 2.0 implementation	Section 11 (Lessons Learnt and Recommendations)
xii	Evaluate inclusivity, equity, and transparency of processes (whole-of-society approach, gender, social justice)	Section 5 (Stakeholder Engagement); Section 11.7 & 11.9 (Whole-of-society, gender, social justice)
xiii	Provide actionable recommendations to strengthen ETF and BTR readiness, NDC 3.0, and financing strategies	Section 11 (Recommendations for NDC 3.0); Section 12 (Conclusion)
xiv	Ensure alignment with international reporting frameworks (ETF, GST, Decision 18/CMA.1, CTF tables)	Section 4.2 (NDC Assessment Framework); Section 6 (Six-pillar framework alignment); Section 12 (Conclusion – GST alignment)

Finally, the stocktake is designed to inform the next cycle of national climate targets. Its findings are not only intended to evaluate Uganda's progress but also to directly shape NDC 3.0, making it more ambitious, realistic, and responsive to national priorities while contributing effectively to global climate goals.

1.2 Reader's guide

This report is structured into twelve chapters that together provide a stocktake of Uganda's Updated NDC, assess progress to date, and generate recommendations for the NDC 3.0. The **Executive Summary** provides a high-level overview of the report's key findings. It outlines the purpose, scope, and methodology used for the assessment, and summarizes progress made on both mitigation and adaptation efforts. This section also highlights the key barriers and gaps encountered and concludes with a list of priority recommendations for the development of NDC 3.0. **Chapter 1** introduces the objectives and scope of the assignment and describes the structure of the report.

Chapter 2 outlines new developments in Uganda's national circumstances. It includes information on the country's government structure, population demographics, geographical features, and economic status. The chapter also describes Uganda's climate profile and details various economic sectors, concluding with an analysis of how these national circumstances have influenced the country's greenhouse gas emissions over time. **Chapter 3** presents Uganda's updated NDC targets and institutional framework. It details the quantitative and qualitative targets set for both mitigation and adaptation and describes the institutional framework established to facilitate the implementation and monitoring of these targets.

Chapter 4 explains the methodology. It outlines the sources of data and information, detailing how the assessment was aligned with international obligations and frameworks, such as Decision 18/CMA.1 and Decision 5/CMA.3. This section also explains the use of a six-pillar analytical framework to guide the assessment. Similarly, **Chapter 5** documents the stakeholder engagement process, summarizing consultations, governance arrangements, and coordination structures.

Chapters 6 to 9 comprise the core of the stocktake. **Chapter 6** assesses the content and quality of NDC 2.0 using the six-pillar framework presented in chapter 4. **Chapter 7 on Mitigation Progress** reviews Uganda's updated NDC and assesses the framework for monitoring progress. It evaluates the implementation of mitigation policies, measures, and actions across key sectors, including energy, transport, and AFOLU, highlighting achievements, gaps, and challenges in tracking emissions reductions. **Chapter 8 on Adaptation Progress** assesses progress against Uganda's stated adaptation goals and priority actions. It tracks implementation in sectors such as agriculture, health, water and sanitation, cities and the built environment, and forestry, identifying successes, bottlenecks, and areas where monitoring and reporting need strengthening. **Chapter 9 on Support Received and Needed** examines financial, technological, and capacity-building support for NDC implementation. Using ETF- and CTF-aligned frameworks, it tracks resources received and identifies gaps, highlighting priority areas for additional support and opportunities to mobilize international and private sector resources effectively.

Chapter 10 synthesizes key barriers and gaps, including MRV and data limitations, institutional and coordination challenges, financing constraints, technical and human capacity issues, policy and regulatory barriers, socioeconomic and behavioral challenges, and economy-wide coverage gaps. **Chapter 11** provides lessons and recommendations for NDC 3.0. It addresses ETF and BTR readiness, updating targets and baselines, strengthening adaptation and mitigation tracking, resilience and loss-and-damage frameworks, financing strategies and private sector engagement, sector-specific adaptation and mitigation pathways, institutional and governance strengthening, whole-of-society engagement, gender and social justice, and practical guidance on timelines and resource use. Finally, **Chapter 12** concludes by synthesizing Uganda's achievements, persistent barriers, and the opportunities for a more ambitious, inclusive, and implementable NDC 3.0.

2. National Circumstances

Uganda's national circumstances provide the context for understanding the country's greenhouse gas emissions profile, climate vulnerabilities, and institutional capacity for implementing its Updated NDC. As a landlocked East African country experiencing rapid population growth (~3% p.a) and ongoing development challenges, Uganda's emissions are dominated by Agriculture, Forestry, and Other Land use, while facing increasing pressure from urbanization and economic growth. The country's institutional framework has been significantly strengthened since 2021 through the National Climate Change Act and comprehensive NDC implementation planning.

Uganda submitted its first NDC in 2016 and its updated NDC in 2021, expanding sectoral coverage and specifying targets for mitigation and adaptation. The updated NDC reflected the country's increased ambition and alignment with national development goals such as Vision 2040 and the NDP III. Uganda is now in the process of preparing its third NDC (NDC 3.0), in which it is expected to further raise its level of ambition, informed by the outcomes of the ongoing NDC stock take exercise.

2.1 Government structure

Uganda operates a two-tiered government system comprising central government (Ministries, Departments, and Agencies) focused on policy formulation and standard setting, and local government responsible for planning and service delivery through a five-tiered structure from village (LC1) to district (LC5) levels.

The National Climate Change Act (2021) provides Uganda with a comprehensive legal framework for climate action, giving legal force to international agreements including the UNFCCC, Kyoto Protocol, and Paris Agreement. The Act establishes clear institutional roles, defines coordination mechanisms, and reinforces Uganda's international commitments with domestic legal backing. It also sets the basis for Uganda's participation in international cooperative approaches under Article 6 of the Paris Agreement, mandating environmental integrity, transparency in governance, and robust accounting to avoid double counting. Uganda's approach to Article 6 will be guided by sustainable development principles, ensuring that international cooperation complements rather than substitutes domestic climate action, while contributing to the achievement of the country's National Development Goals.

Central Climate Change Coordination

The Climate Change Department (CCD) within the Ministry of Water and Environment (MWE) serves as the lead coordinating body for climate change efforts and the national focal point for UNFCCC. The Commissioner of the CCD represents Uganda at international climate negotiations. Under the 2021 Climate Change Act, the CCD has been designated as the central "MRV-Control Center" with expanded mandates including:

- Coordinating, monitoring, and evaluating government climate programs and actions across all sectors
- Assessing adaptation and mitigation actions for synergies and co-benefits
- Promoting climate technology development, transfer, and innovation
- Serving as the national climate knowledge and information management center
- Promoting multi-stakeholder participation in climate planning and implementation
- Providing technical assistance to districts for developing Climate Change Action Plans
- Developing mechanisms for low-GHG/green growth pathways, MRV systems, resilience building, and climate finance mobilization
- Preparing and updating NDCs, national GHG inventories, and UNFCCC communications
- Coordinating with international climate mechanisms, including Article 6 of the Paris Agreement

High-Level Climate Governance

The National Climate Change Advisory Committee (NCCAC), established by the 2015 Policy and chaired by the Minister of Water and Environment, serves as a high-level, multi-sectoral stakeholder body providing independent technical advice on climate science, technologies, interventional programs, climate risk assessment, adaptive capacity enhancement, and target achievement. NCCAC membership includes representatives from line ministries, civil society organizations, NGOs, academia, and private sector.

The National Climate Change Policy Committee chaired by the Prime Minister coordinates implementation of the climate change policy and ensures information flow on resource allocation for the implementation of the policy. The committee constitutes ministers from various MDAs at national level.

NDC Implementation Coordination

Uganda has established a high level for coordinated NDC implementation comprising:

- Office of the Prime Minister (OPM): Coordinates different MDAs assigned to 18 National Development Programmes in addition to the National Climate Change Policy Committee as established in the 2015 national climate change policy.
- National Planning Authority (NPA): Ensures NDC priority actions are included in planning call circulars for MDAs and Local Governments, guides integration into plans and budgets.
- Ministry of Finance, Planning and Economic Development (MoFPED): Maintains the Development Partner database, sets procedures for accessing climate finance, creates private sector incentives, and leads climate finance mobilization.
- Climate Change Department (CCD/MWE): Acts as technical lead and MRV control center, defines priority NDC actions for integration into planning and budgetary systems.

Local Government Integration

Under the Climate Change Act (2021)², the Natural Resources Department serves as the district-level focal point for climate change, with District Environment and Natural Resources Committees designated for climate change coordination at district level. Lower local government climate change committees are appointed to address climate issues in their jurisdictions, creating a comprehensive multi-level governance structure.

2.2 Population profile

Uganda's population has grown dramatically from 36.34 million in 2014 to 45.9 million in 2024, representing a consistently high growth rate of 3% per annum. This rapid population growth has significant implications for GHG emissions, resource demand, and climate vulnerability. The population structure remains relatively youthful, with over 40% aged below 14 years, and maintains a high total fertility rate of 5.4 children per woman (UBOS & ICF, 2022).

Rural-Urban Demographics

Approximately 60% of the population remains rural and engaged primarily in rain-fed subsistence agriculture, making them highly vulnerable to climate variability and change³. However, urbanization is accelerating rapidly, with the urban population increasing from 13% in 2002 to 40% in 2023. Kampala, the largest urban

² [*National Climate Change Act, 2021.pdf*](#)

³ [*2023-Statistical-Abstract.pdf*](#)

center, hosts 1.9 million people and is the fastest-growing urban area, creating concentrated demand for energy, transport, and waste management services.

Development and Vulnerability Context

The Human Development Index improved modestly from 0.505 in 2018 to 0.582 in 2023⁴, moving to medium human development category. Critical vulnerabilities include:

- Only 53% of rural populations have access to basic drinking water services
- 68% of the population lacks access to basic sanitation (2020 baseline)
- Unemployment remains high at 12.6 % in 2024⁵, with approximately 2.5 million new jobs needed annually to absorb new labor market entrants

Refugee Population Impact

Uganda hosts one of Africa's largest refugee populations, estimated at over 1,780,000 people ⁶ by 2024 mainly from South Sudan, Democratic Republic of Congo, Somalia, and Burundi. This population adds significant pressure on natural resources, particularly forests (for fuelwood and charcoal), agricultural land, and water resources, directly impacting GHG emissions from land use change and energy consumption.

2.3 Geographical profile

Uganda is a landlocked East African country strategically located at the equator, between longitudes 29°34'E and 35°0'E, and latitudes 4°12'N and 1°29'S. The country covers approximately 241,550 km² (200,523 km² land area, 41,038 km² water bodies - representing 17% of total area). Uganda borders Kenya, Tanzania, Rwanda, Democratic Republic of Congo, and South Sudan.

Topographical Diversity

The country is largely situated on a plateau averaging 1,100 meters above sea level, sloping steadily downwards towards the South Sudan plains. The varied topography significantly influences climate patterns and includes:

Major Geographical Features:

- Lake Victoria: One of the world's largest freshwater lakes in the world (69,000 km²), shared with Kenya and Tanzania, and supports one of the biggest inland fisheries globally.
- Western Rift Valley: Part of the Great Rift Valley system, including Lakes Edward, George, and Albert Nile (lowest point 620m ASL)
- Mountain Ranges:
 - Rwenzori Massif (highest point at 5,111m ASL) - "Mountains of the Moon"
 - Mount Elgon (4,321m ASL) on the Kenya border
 - Imatong Mountain (3,029m ASL) and Moroto Mountains (3,085m ASL)
 - Mgahinga (4,127m ASL) in the Virunga range
- Extensive wetlands covering 26,308 km² (11% of land area), including 12 Ramsar Convention sites
- Five major river basins: Lake Victoria, Victoria Nile, Aswa River, Kidepo Valley, and Lake Kyoga

⁴ [*HDR25 Statistical Annex HDI Table.xlsx*](#)

⁵ [*National-Population-and-Housing-Census-2024-Final-Report-Volume-1-Main.pdf*](#)

⁶ [*UNHCR Uganda Fact Sheet - September - October 2024 - Uganda | ReliefWeb*](#)

Biodiversity and Ecosystem Services

The topographical diversity supports rich biodiversity, with Bwindi Impenetrable National Park and Mgahinga Gorilla National Park hosting more than half of the world's mountain gorilla population. The varied ecosystems provide critical services including carbon storage, water regulation, and climate buffering.

2.4 Economic Profile

Uganda is classified as a low-income developing country with significant development aspirations and growing economic challenges. Its economic trajectory demonstrates meaningful progress in key development indicators that create a foundation for enhanced climate mitigation efforts. With GDP growth reaching approximately \$60 billion in 2024 and per capita income rising from \$1,042 in FY 2021/22 to \$1,146 in 2024, the country is building economic capacity that can support green development initiatives. The poverty levels reduced from 21.4% in 2017 to 16.1% in 2024⁷ indicating improving living standards that can facilitate community participation in climate-friendly practices and reduce pressure on natural resources driven by subsistence needs. While the 12.6% unemployment rate remains a challenge, targeted climate mitigation investments in renewable energy infrastructure, sustainable agriculture, and green job creation can simultaneously address employment concerns while advancing Uganda's climate goals. This economic progress provides an opportunity to integrate climate considerations into development planning, ensuring that continued growth follows a low-carbon pathway that aligns with both poverty reduction objectives and global climate commitments.

Sectoral Economic Structure

Sectors with significant contribution to economic growth include the services sector, industry, Agriculture, forestry, & fishing, and Taxes on products.

Agriculture, forestry, & fishing aim to promote attainment of food and nutrition security, income generation, employment creation, and enhancing export earnings. It grew by 6.6% in 2024/2025 up from 5.6% in 2023/2024 driven by varying growth in food crops, cash crops, livestock, fishing and forestry with total 26.2 % contribution to Uganda's GDP up from 24.6% signifying increasing dependence on the sector (**Figure 2-1**).

Uganda aims to promote sustainable Industrialization for inclusive growth, employment and sustainable wealth creation through the industry sector⁸. The sector supports the national socio-economic development agenda through job creation, foreign exchange generation, and foreign direct investment attraction. It grew by 7% in 2024/2025 compared to the 5.5% growth 2023/2024 driven by growth in manufacturing, construction, water supply, mining and quarrying, and electricity. The sector contributed about 24.5 % to the national GDP 2024/2025, signifying the importance and role of the sector to national development⁹.

⁷ [Uganda-National-Household-Survey-Report-20232024.pdf](#)

⁸ [NDPIII.pdf](#)

⁹ [*AGDP-2024_25-PRELIMINARY-ESTIMATES.pdf](#)



Figure 2-1: GDP Growth Rates and Sectoral Performance for Fiscal Years 2020/21 - 2024/25 (UBOS, 2025)

Agricultural and climate vulnerability

Agriculture remains the backbone of Uganda's economy, directly or indirectly employing over 70% of the labor force and dominating rural livelihoods. The sector drives external trade through exports of coffee, fish, tea, cotton, maize, beans, and tobacco. However, the sector faces significant climate-related challenges including small-holder farming systems that constrain economies of scale, reliance on rain-fed agriculture which makes production highly vulnerable to climate variability such as frequent droughts and floods which impact productivity and food security. Moreover, low agricultural inputs (fertilizers, improved seeds) and post-harvest losses estimated at 37% due to inadequate storage and processing facilities¹⁰ limiting resilience and productivity of the agricultural sector in the face of climate change.

Livestock Sector Significance

The livestock sector contributes 4% to GDP but plays a crucial role in GHG emissions. Beef, milk, and other cattle products contribute approximately 73% of the gross value of all livestock output. The growing livestock population is a major source of methane emissions from enteric fermentation.

Development Vision and Climate Implications

Uganda is committed to becoming an upper middle-income country by 2040 as outlined in the Third National Development Plan (NDP III) 2020/21–2024/25¹¹ and complemented by the Fourth National Development Plan. This vision targets key sectors including:

¹⁰ *Reframing Food Futures: Making Foresight Transformative*

¹¹ budget.finance.go.ug/sites/default/files/NDPIII.pdf

- Manufacturing and agro-processing with potential for increased industrial emissions
- Tourism development - leveraging natural assets while maintaining ecosystem integrity
- Oil and gas development - confirmed reserves of 1.4-1.7 billion barrels with significant future emission implications
- Mineral extraction including important export minerals such as gold and cobalt

These present challenges and opportunities for Uganda to ensure a low carbon development pathway while ensuring precautionary exploitation of oil reserves.

2.5 Climate Profile

Uganda has a tropical savanna climate, with generally warm temperatures and a clearly defined wet and dry season (Figure 2-2). The country experiences moderate temperatures throughout the year, with the hottest month being January with mean temperatures of 18–28°C and the coldest month being July with mean temperatures of 17–25°C. Total annual average precipitation is 1,197 millimeters with mean monthly precipitation varying from 39.6 millimeters in January to 152.7 millimeters in April according to the World Bank Climate Change Knowledge Portal. Most parts of the country have two distinct rainfall seasons, one between March and May and the other between September and December, with dry seasons between July and August, and January and February. Future projections show that the temperature is expected to increase across all parts of the country under both the dry/hot¹² and wet/warm¹³ scenarios (World Bank Climate Change Knowledge Portal).

Under the dry/hot scenario, temperatures are expected to increase by 1.34°C by 2050 compared to the 1995-2020 average, while under the wet/warm scenario, temperatures are expected to increase by 0.85°C above the 1995-2020 average. Under the dry/hot scenario, rainfall is expected to increase by about 1.6 percent in most parts of the country by 2050 compared to the 1995-2020 average, while under the wet/warm scenario, it is predicted to increase by an average of about 12 percent over the same period (World Bank Climate Change Knowledge Portal).

¹² Dry/hot scenarios refer to scenarios around the 10th percentile of mean precipitation changes (dry) and the 90th percentile in mean temperature changes (hot)

¹³ Wet/warm scenarios refer to scenarios around the 90th percentile of mean precipitation changes (wet) and the 10th percentile in mean temperature changes (warm)

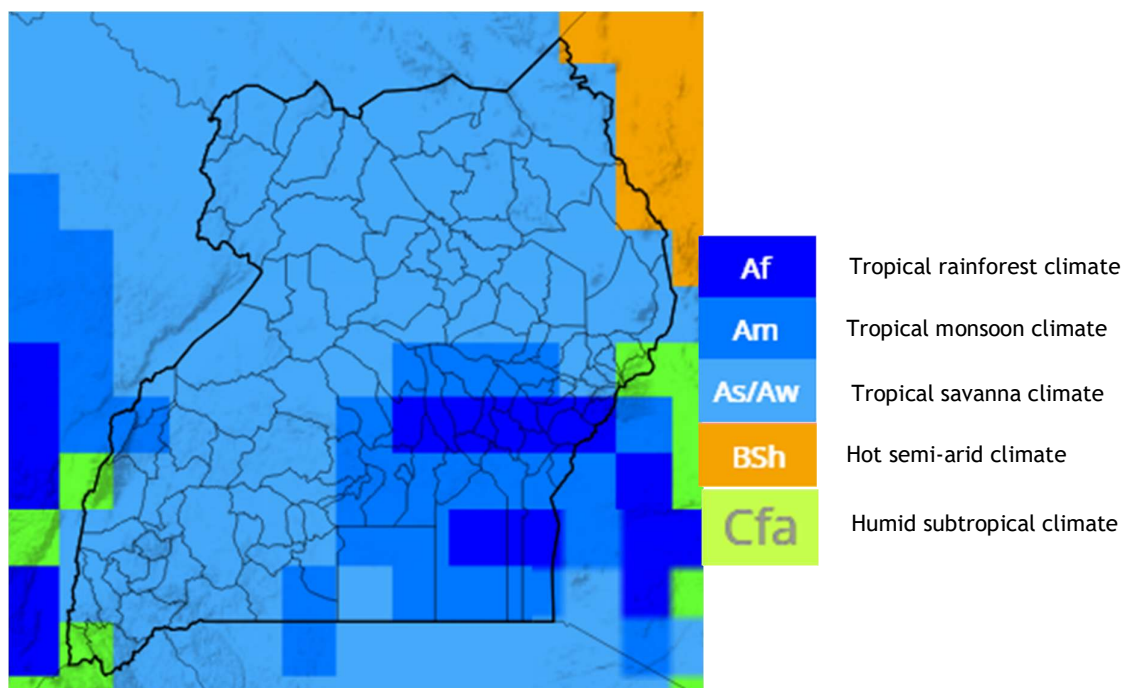


Figure 2-2. Koppen-Geiger Climate Classification Map for 1991-2020 climate period

(Source; World Bank Climate Change Knowledge Portal)

Climate Variability and Extremes

Uganda is experiencing heightened climate variability that creates complex feedback loops between environmental stress and greenhouse gas emissions while threatening sustainable development progress¹⁴. The increasing frequency of droughts is intensifying pressure on natural resources, driving deforestation as communities seek alternative livelihoods and forcing greater reliance on biomass energy sources, thereby increasing carbon emissions from land-use change and unsustainable harvesting practices. Simultaneously, more frequent flooding in urban areas and river basins is overwhelming inadequate infrastructure systems, disrupting economic activities that support low-carbon development investments. Gradual warming trends affecting highland ecosystems are compromising critical carbon storage capacity in montane forests while reducing agricultural productivity, potentially increasing food insecurity and driving expansion into previously protected carbon-rich landscapes¹⁵. The shifting seasonal rainfall patterns are disrupting traditional agricultural calendars, forcing farmers to adopt more emission-intensive practices such as increased fertilizer use, irrigation systems, and crop burning, while simultaneously reducing the resilience of natural carbon sinks. These interconnected climate impacts underscore the urgent need for adaptation measures that also deliver mitigation co-benefits, such as drought-resistant agroforestry systems, flood management through wetland restoration, and climate-smart agriculture practices that can break the cycle of climate vulnerability driving increased emissions.

¹⁴ [37189456_Uganda-NC3-1-Final TNC Uganda.pdf](#)

¹⁵ [Regional Fact Sheet Africa](#)

2.6 Sector details

Uganda's GHG emissions profile reflects its development stage, natural resource base, and economic structure. The emissions are dominated by land-based sectors¹⁶, with growing contributions from energy and transport:

Agriculture, Forestry and Other Land Use (AFOLU)

The AFOLU sector dominates Uganda's greenhouse gas emission profile¹⁷ with a 6% average annual increase between 1995-2019, accounting for 76% of total national emissions in 2019. Under the NDC BAU scenario, emissions are projected to reach 122.2 Mt CO₂e by 2030.

Within this sector, land use change and forestry emerges as the leading emission source, accounting for 60% of AFOLU emissions, driven by the dramatic decline in forest cover from 24% in 1990 to merely 10% in 2015, representing a catastrophic loss of over 760,000 hectares between 2000-2015 at an alarming 2% annual deforestation rate that directly undermines Uganda's natural carbon sequestration capacity. This forest degradation is driven by intersecting pressures including population growth demanding agricultural land expansion, biomass energy extraction for firewood and charcoal production serving 89% of primary energy needs, and settlement development accommodating rapid demographic increase.

Methane and nitrous oxide emissions from livestock constitute the second-largest contributor at 34% of AFOLU emissions, reflecting Uganda's significant pastoral and mixed farming systems where cattle, goats, and other ruminants produce enteric fermentation and manure management emissions that intensify with growing food demand from the expanding population.

Aggregate sources and non-CO₂ emissions from land account for 7% of the AFOLU sector, primarily from burning of woodlands and rangelands for land clearing and pasture management, as well as nitrous oxide emissions from managed soils where fertilizer application and agricultural practices release potent greenhouse gases. The agricultural component of AFOLU emissions reflects Uganda's economic structure where 70% of employment depends on farming, creating inevitable tensions between food security imperatives and emission reduction goals. However, this sector also presents Uganda's greatest mitigation opportunity, as sustainable forest management, agroforestry systems, climate-smart agricultural practices including improved livestock management and soil conservation, controlled burning alternatives, and wetland restoration could simultaneously reduce emissions across all three major sources, enhance carbon sequestration, improve agricultural productivity, and strengthen climate resilience, making AFOLU the cornerstone of any meaningful national mitigation strategy that aligns environmental sustainability with development needs.

Energy Sector

Uganda's energy sector demonstrates rapid emission growth with a 10% average annual increase between 1995-2019, contributing 9.1% of total national emissions, with total emissions increasing from 2.966 Mt CO₂e in 1995 to 10.17 Mt CO₂e in 2019¹⁸, representing a 10% annual increment driven by the complex interplay between energy poverty, biomass dependence, and growing modern energy demand accompanying

¹⁶ unfccc.int/sites/default/files/resource/64251087_Uganda-BUR2-1-UGANDA'S_SECOND_BIENNIAL_UPDATE_REPORT.pdf

¹⁷ unfccc.int/sites/default/files/resource/64251087_Uganda-BUR2-1-UGANDA'S_SECOND_BIENNIAL_UPDATE_REPORT.pdf

¹⁸ unfccc.int/sites/default/files/resource/64251087_Uganda-BUR2-1-UGANDA'S_SECOND_BIENNIAL_UPDATE_REPORT.pdf

economic development. Under the NDC BAU scenario, emissions are projected to reach 12.44 Mt CO₂e plus 9.6 Mt CO₂e in 2030.

The sector's emissions originate from four major subcategories with distinct profiles: the "other sectors" subcategory dominates at 45% of energy emissions (decreasing from 70% in 1995 to 54% in 2019), comprising household, commercial and institutional sectors, and off-road combustion activities in agriculture, forestry and fishing, where methane from burning biomass-based fuels (charcoal and wood) accounts for over 70% of emissions in CO₂e terms, while carbon dioxide emissions derive mainly from liquid fuel combustion in fishing, off-road forestry and agricultural activities, and limited LPG use for cooking. Transportation has emerged as the fastest-growing subcategory, increasing its share from 20% in 1995 to 27% in 2019, with emissions predominantly in the form of CO₂ (98%) and minor contributions from CH₄ and N₂O (1% each), making transport the single largest source of CO₂ emissions within the energy sector at 60% of total energy-related carbon dioxide. Energy industries account for 16% of energy sector emissions, with fuel combustion from thermal electricity generation experiencing significant fluctuations during Uganda's energy shortage period from 2011 to 2015, exemplified by sudden emission increases from 2009 following commissioning of thermal generators to compensate for reduced hydropower output due to declining Lake Victoria water levels, highlighting climate vulnerability in the energy system. Manufacturing industries and construction contribute 11% of energy emissions but demonstrate the most dramatic growth trajectory, with emissions increasing six-fold from 0.178 to 1.151 Mt CO₂e between 1995-2019, representing an average 22% annual increase driven by rising industrial fuel demand, particularly from the cement industry, signaling the emission intensification accompanying Uganda's industrialization aspirations.

Future developments: Oil and gas sector development (EACOP pipeline completion slated for 2025, oil refinery planned) will significantly impact future emissions

Waste Sector

Uganda's waste sector demonstrates a concerning 5.7% average annual emission increase between 1995-2019, contributing 14.5% of total national emissions in 2019, reflecting the systemic challenges of rapid urbanization outpacing waste management infrastructure development and institutional capacity. Total emissions increased from approximately 6.430 Mt CO₂e in 1995 to 16.241 Mt CO₂e in 2019 while under the NDC BAU scenario, waste emissions are projected to reach 3.19 Mt CO₂e by 2030.

Wastewater treatment emerges as the dominant emission source, rising from 3.352 Mt CO₂e in 1995 to 8.370 Mt CO₂e in 2019 with an average increment of 6% per annum. Solid waste disposal constitutes the second major emission source, increasing from 3.071 Mt CO₂e in 1995 to 7.612 Mt CO₂e in 2019 with an average annual increment of 5.9%, driven by growing waste generation from Uganda's 3% annual population growth and accelerating urbanization, with predominant reliance on open dumping and uncontrolled disposal practices creating significant methane emissions from decomposing organic waste in the absence of modern landfills with gas capture systems. Biological treatment of waste, including composting of solid waste and limited anaerobic biogas digestion of animal waste and wastewater for cooking energy and agricultural slurry production, represents an important mitigation activity with significantly lower emissions, currently accounting for only 2% of total waste sector emissions but presenting substantial expansion potential for reducing methane generation while creating renewable energy and valuable soil amendments. Incineration and open burning of waste, generating methane, nitrous oxide, and carbon dioxide emissions, increased modestly from 0.007 Mt CO₂e in 1995 to 0.011 Mt CO₂e in 2019 with approximately 2% annual growth, maintaining a minimal 0.1% contribution to overall waste sector emissions.

The waste sector's emission profile reveals critical mitigation opportunities through integrated approaches: expanding sanitation infrastructure and wastewater treatment facilities with methane capture could address the largest emission source while delivering public health benefits; implementing comprehensive solid waste

management systems with controlled disposal, recycling programs, and organic waste diversion could dramatically reduce landfill methane; scaling up biological treatment through community composting programs and biogas digesters could transform waste from an emission source into a resource that provides renewable energy and supports agricultural productivity; and establishing proper waste collection and disposal systems could eliminate open burning practices, collectively enabling Uganda to manage the waste challenges of urbanization while reducing emissions and creating employment opportunities in the emerging circular economy.

Industrial Processes and Product Use (IPPU)

The IPPU sector, while currently contributing less than 1% of Uganda's total emissions, exhibits the highest emission growth rate at 10.3% average annual increase between 1995-2019, signaling an emerging emission source that requires proactive mitigation planning as the country pursues industrialization. Emissions from IPPU are predominantly carbon dioxide, primarily originating from cement and lime production processes that release CO₂ through chemical reactions in addition to energy-related emissions. Cement production has emerged as the overwhelmingly dominant source within this sector, with CO₂ emissions rising continuously from 1995 to 2019 and contributing 74% of total IPPU emissions.

An increasingly significant component of IPPU emissions originates from hydrofluorocarbons (HFCs) used as substitutes for Ozone Depleting Substances, primarily in fire protection and refrigeration applications. HFC emissions have grown substantially from 0.0137 Mt CO₂e in 1995 to 0.0879 Mt CO₂e in 2015, reaching 0.112 Mt CO₂e in 2019, representing an average annual increase of 5.4% that reflects Uganda's growing demand for cooling and refrigeration services accompanying economic development and urbanization. The major sources of these emissions are mobile air conditioning and refrigeration systems, along with stationary air conditioning, collectively estimated at 0.105 Mt CO₂e, while fire protection agents contribute approximately 0.007 Mt CO₂e, with CO₂ emissions from lubricant use constituting less than 0.003% of total IPPU emissions. This growing HFC emission trajectory presents both challenges and opportunities, as the Kigali Amendment to the Montreal Protocol provides a framework for phasing down HFCs while Uganda's expanding cooling sector offers potential for leapfrogging to low-global warming potential alternatives, energy-efficient cooling technologies, and proper refrigerant management systems that can reduce emissions while meeting legitimate development needs for thermal comfort, food preservation, and healthcare cold chains.

2.7 Effect of national circumstances on GHG emissions and removals over time

Uganda's national circumstances create a complex web of drivers that fundamentally shape its GHG emission trajectory and removal capacity:

Population-Driven Emission Pressures

Uganda's high population growth rate of 3% per annum, coupled with a predominantly youthful age structure, generates escalating demographic pressures that directly intensify greenhouse gas emissions across multiple sectors while simultaneously constraining natural carbon removal capacity. The expanding population creates relentless demand for energy services, particularly biomass for cooking needs, driving systematic forest degradation as over 80% of energy requirements are met through solid biomass extraction¹⁹, resulting in significant emissions from deforestation and forest degradation. Agricultural land expansion to feed the growing population necessitates continuous conversion of carbon-rich forests and

¹⁹ *Uganda Energy Transition Plan*

wetlands into cropland and settlements, eliminating critical carbon sinks while releasing stored carbon through land-use change processes. The demographic transition intensifies demand for construction materials including timber, sand, and cement, further accelerating forest loss while increasing industrial emissions from cement production and transportation of building materials to serve expanding urban and rural settlements²⁰. Rapid urbanization driven by population growth creates escalating demand for transport services, increasing vehicle ownership and petroleum consumption that drives emissions growth in the transport sector, while simultaneously generating growing urban waste streams that overwhelm existing management capacity and create significant methane emissions from inadequate waste disposal systems. These population-driven pressures create a reinforcing cycle where demographic growth simultaneously increases emission sources while reducing the natural ecosystem capacity to absorb carbon, highlighting the critical need for sustainable development pathways that can accommodate population growth while minimizing environmental impact.

Economic Development-Climate Nexus

Uganda's position as a low-income country with ambitious upper middle-income aspirations by 2040²¹ creates fundamental tensions between immediate development needs and long-term climate sustainability that shape the nation's emission trajectory and mitigation potential. The economy's heavy dependence on agriculture, employing 70% of the population, ensures that the Agriculture, Forestry, and Other Land Use (AFOLU) sector remains the dominant source of greenhouse gas emissions while simultaneously making the entire economic system highly vulnerable to climate variability and extreme weather events that threaten both livelihoods and emission reduction efforts. Limited infrastructure development, exemplified by only 51.5% grid electricity access²², perpetuates widespread biomass dependence that drives systematic forest degradation, creating a development trap where energy poverty directly fuels environmental destruction and carbon emissions²³. Development financing constraints severely limit the country's capacity to invest in low-carbon alternatives such as renewable energy infrastructure, efficient transportation systems, and climate-smart agricultural technologies, forcing continued reliance on high-emission development pathways despite growing awareness of climate risks. The ambitious goal of achieving upper middle-income status by 2040 presents both opportunities and risks, as rapid industrial growth and transport expansion could dramatically increase emissions without adequate low-carbon planning and green growth strategies that integrate climate considerations into core development planning processes. This economic development-climate nexus highlights the critical need for innovative financing mechanisms and international support that can enable Uganda to pursue prosperity while avoiding the high-carbon development patterns that characterized earlier industrial transitions.

Natural Resource Degradation Cycles

Uganda's high annual deforestation rate of 2%²⁴ has established destructive reinforcing cycles that systematically erode the country's natural carbon storage capacity while perpetuating emission-intensive development patterns across multiple interconnected systems²⁵. The overwhelming reliance on biomass for 89% of primary energy needs creates direct and immediate pressure for forest conversion, particularly

²⁰ [*Updated NDC Uganda 2022 Final.pdf*](#)

²¹ [*vision20204011.pdf*](#)

²² [*Access to electricity \(% of population\) - Uganda | Data*](#)

²³ [*Uganda Energy Transition Plan*](#)

²⁴ [*unfccc.int/sites/default/files/resource/64251087_Uganda-BUR2-1-UGANDA'S SECOND BIENNIAL UPDATE REPORT.pdf*](#)

²⁵ [*Updated NDC Uganda 2022 Final.pdf*](#)

through unsustainable charcoal production that transforms carbon-sequestering forests into emission sources while simultaneously reducing the landscape's capacity to absorb future emissions. Population-driven agricultural expansion compounds these pressures by converting remaining forest carbon sinks into cropland, creating a dual impact of releasing stored carbon through deforestation while eliminating critical natural sinks that could otherwise offset emissions from other sectors²⁶. The resulting reduction in forest cover disrupts essential local climate regulation services, creating cascading feedback loops where diminished precipitation patterns and increased temperature variability reduce agricultural productivity, forcing farmers to compensate through further land conversion and forest encroachment in a desperate attempt to maintain food security. These climate-agriculture interactions establish a vicious cycle where environmental degradation cascades further degradation, as reduced ecosystem resilience makes communities increasingly vulnerable to climate shocks that drive additional unsustainable resource extraction. Institutional and financial constraints severely amplify these degradation cycles by limiting technical capacity for deploying low-carbon technologies that could reduce forest pressure, while inadequate waste management infrastructure in rapidly expanding urban areas generates mounting methane emissions, weak enforcement of targeted policies and regulations, and insufficient rural extension services prevent adoption of climate-smart practices that could enhance productivity on existing farmland, and financial sector limitations block access to green finance and appropriate technology that could enable sustainable resource management and forest conservation initiatives.

²⁶ *Uganda Energy Transition Plan*

3. NDC Targets and Institutional Framework

3.1 NDC Targets

Uganda's climate commitments have evolved progressively since the first NDC in 2016. The updated NDC submitted in 2021 ("NDC 2.0") expanded sectoral coverage to ensure it is economy-wide and, includes all IPCC main sectors including: energy, AFOLU, transport, waste, and IPPU, with an overall mitigation target of 24.7% reduction in national GHG emissions by 2030 relative to Business-as-Usual (BAU), equivalent to lowering projected emissions from 148.8 MtCO₂e to about 112.1 MtCO₂e. Of this reduction, 5.9% is unconditional (financed from domestic resources), while 18.8% is conditional on international support in the form of finance, capacity building, and technology transfer.

The updated NDC provides a clear breakdown of the relative and absolute contributions by sector:

- Agriculture, Forestry and Other Land Use (AFOLU): AFOLU remains the dominant contributor to Uganda's mitigation efforts, delivering 82.7% of the total mitigation potential. This translates to an absolute reduction of about 30 MtCO₂e by 2030, making AFOLU the highest priority sector for emission reductions.
- Transport (sub-sector of energy): The transport sector contributes 7.56% of the mitigation potential, equivalent to approximately 2.7 MtCO₂e. This places transport as the second most important source of mitigation, underscoring its critical role in Uganda's NDC implementation.
- Stationary Energy (electricity, biomass, industrial and residential energy use): Contributes 6.36% of the mitigation impact, about 2.3 MtCO₂e. While smaller than transport, the sector is key for long-term decarbonisation, given Uganda's renewable energy expansion pathway.
- Waste Management: Accounts for 3% of the mitigation contribution, equal to roughly 1.1 MtCO₂e, through measures such as improved waste treatment, landfill gas recovery, and biogas expansion.
- Industrial Processes and Product Use (IPPU): Provides the smallest share, 0.4%, equal to 0.15 MtCO₂e, reflecting the relatively low but growing industrial base.

AFOLU dominates Uganda's mitigation landscape, followed by, though at a much smaller scale, energy, especially transport. The strong weighting toward AFOLU highlights the urgency of addressing deforestation, land-use change, and agricultural emissions, while the growing importance of transport indicates the need to accelerate low-carbon mobility solutions.

Currently, Uganda is in the process of preparing its third NDC (NDC 3.0), which is expected to represent a further progression in ambition, in line with the Paris Agreement's requirement for successive NDCs to reflect enhanced ambition. The design of NDC 3.0 will be informed by:

- The outcomes of this NDC stocktake exercise provide evidence on achievements, gaps, and barriers in NDC 2.0 implementation.
- National development priorities under the Fourth National Development Plan (NDP IV) and Vision 2040, ensuring that climate targets are integrated with socioeconomic goals.
- Global stocktake outcomes under the Paris Agreement, which highlight the need for greater collective ambition to limit global warming to 1.5°C.

It is anticipated that NDC 3.0 will:

- Further increase the level of unconditional mitigation ambition through domestically financed measures, while scaling up conditional commitments linked to international support.
- Incorporate a clearer emissions trajectory, aligning with a mid-century net-zero pathway. This should explicitly distinguish between conditional and unconditional actions, policies, and measures, and may also specify a national emissions peak year.
- Expand coverage of sectoral targets, with stronger commitments in clean energy, transport electrification, forestry restoration, and waste management.
- Provide more detailed adaptation objectives, drawing on the outcomes of the ongoing National Adaptation Plan (NAP) process.
- Strengthen integration of just transition, gender, and inclusivity principles across all targets.

3.2 Institutional Framework

The stocktake adopted the Institutional Framework of NDC 2.0 Implementation and Resource Mobilization Plan (MWE, 2024). This framework informed the stocktake Stakeholder Engagement Plan and initial rounds of indicator data collection and validation exercises. The said framework included four core functions performed by legally mandated institutions/organizations. These functions are: coordination; execution of priority NDC adaptation and mitigation actions; resource mobilization; and monitoring, reporting and evaluation.

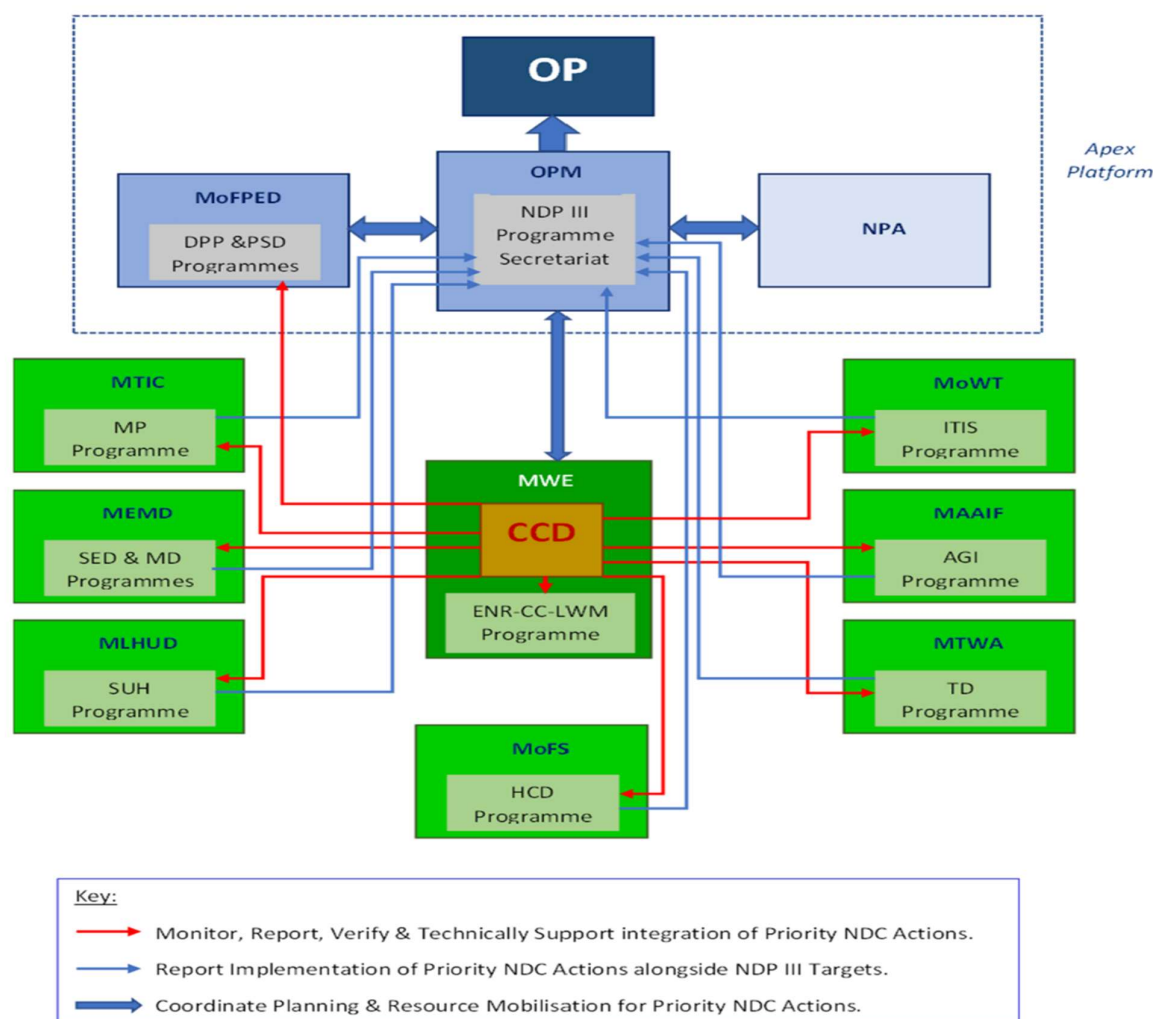


Figure 3-1: Institutional Framework for NDC 2.0 Coordination and Execution of Priority Adaptation and Mitigation Actions

Institution	Roles & Responsibilities	Mandating Policy/Legal Instrument
Office of the Prime Minister (OPM)	Coordinate the different MDAs assigned to the 18 National Development Programmes across which priority NDC Mitigation & Adaptation Actions are spread Liaise with NPA to form National Development Programme Technical & Steering Committees to follow-up implementation of NDC Actions Liaise with MoFPED & CCD to engage Development Partners interested in supporting NDC Implementation	NDP III; Uganda Partnership Policy (2013)
	Liaise with NPA & MoFPED to define priority NDC actions to be integrated in planning & budgetary systems of MDAs & LGs	National Climate Change Act (2021); National Environment Act (2019)

Institution	Roles & Responsibilities	Mandating Policy/Legal Instrument
Climate Change Department (CCD), MWE	Include NDC priorities in Uganda's Framework Strategy on Climate Change and National Climate Change Action Plan	National Climate Change Act (2021)
	Guide MDAs & District ENR Committees to include jurisdiction specific NDC priority actions in their Climate Change Action Plans	
	Coordinate NDC Implementation Plan execution	
	Collate, verify & disseminate information on NDC Implementation	
	Liaise with multi-stakeholders through NCCAC in developing NDCs	NDP III
	Communicate progress of NDC Implementation to the UNFCCC Secretariat	
	Follow up OPM & NPA to create National Development Programme Technical & Steering Committees to facilitate NDC implementation across multiple line ministries	
Ministry of Finance, Planning & Economic Development (MoFPED)	Liaise with OPM, MoFPED & NPA to engage Development Partners interested in NDC Implementation	Uganda Partnership Policy (2013); National Planning Authority (Development-Plans) Regulations (2018)
	Maintain database of Development Partners interested in supporting NDC priority mitigation & adaptation actions	Uganda Partnership Policy (2013)
National Planning Authority (NPA)	Include defined priority NDC actions on the Planning Call Circulars issued to MDAs & LG Planning Committees Guide MDAs & LG Planning Committees to integrate priority NDC actions in their plans & budgets	National Planning Authority Act (2002); National Planning Authority (Development-Plans) Regulations (2018); NDP III

Source: NDC Implementation and Resource Mobilization Plan (MWE, 2022)

However, the scope of this institutional framework is limited to NDC-relevant climate change actions implemented by MDAs and NGOs and Private Sector Actors in partnership with public entities, using resources that are appropriated by Parliament of Uganda, through the Ministry of Finance, Planning and Economic Development (MoFPED & OPM-DU, 2024). These resources termed as national budget support, include domestically resourced revenue through internal revenue of public institutions, taxes and domestic debt (Treasury Bill and Bonds); and externally from bilateral and multilateral sources and commercial banks in combinations of loans, grants and guarantees (MoFPED, 2024).

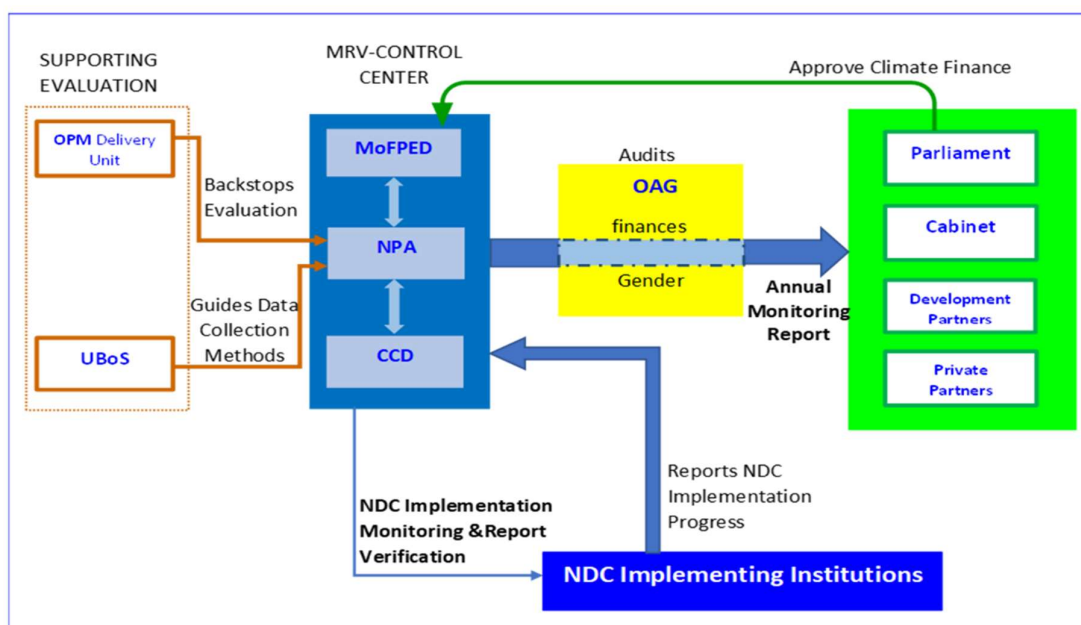


Figure 3-2: Institutional Framework for NDC 2.0 Resource Mobilization; and Monitoring and Evaluation

Institution	Roles & Responsibilities	Mandating Policy/Legal Instrument
National Planning Authority (NPA)	Prepare a national monitoring & evaluation framework to track the progress of NDC Implementation	NDP III; National Planning Authority Act (2002)
	Cooperate with OPM to evaluate performance of NDC Implementation against targets set out in this Plan	
	Liaise with Uganda Bureau of Statistics (UBoS) to identify within institutions implementing NDC the capacity needs limiting the collection of gender disaggregated data	
	Liaise with MoFPED & CCD to compile annual monitoring report on NDC Implementation & present this report before the Cabinet, Parliament, Development Partners & Private actors investing in implementing NDC priority actions	
Ministry of Finance, Planning & Economic Development (MoFPED)	Report on utilization of Climate Change Finance loans & grants before Parliament	Uganda Partnership Policy (2013)
Office of Auditor General	Ensure timely completion of annual audits of Climate Finance from Development Partners	Uganda Partnership Policy (2013)
	Support gender audits of NDC aligned projects	
Climate Change Department (CCD), MWE	Liaise with registered verifiers to approve information & reports in line with NDC priority Mitigation & Adaptation actions; and Climate Finance Flows	National Climate Change Act (2021)

Source: NDC Implementation and Resource Mobilization Strategy (MWE, 2022)

It is a challenge to get a complete inventory of all NDC-relevant activities, especially those carried out by non-state actors. This is primarily because there is no central database to track these projects. These activities often fall into several categories: Projects implemented directly by Development Partners; Projects carried out through NGOs and Civil Society Organizations (CSOs); and Projects run by public institutions (e.g., academia, national research organizations) using off-budget funding. Currently, there is no central register, like the Ministry of Finance, Planning and Economic Development's (MoFPED) audited project database, that captures this category of work.

Moreover, these actors are not obliged to be accountable to central authorities. In situations when non-state actors have been licensed to carry out activities, as in the case of waste management under the National Environment Management Regulations (waste management); and forest management under the National Forestry and Tree Planting Regulations, the actions of the licensees are captured under the licensing body, that is NEMA and NFA respectively, which requires various forms of reporting. This situation makes it, therefore, difficult to capture a legitimate institutional arrangement that governs the true extent of efforts in response to Climate Change. Additionally, as these actors, together with their activities are not "audited", it is difficult to verify the outcomes of their efforts. Or if they are included, there appears risk for double counting. Therefore, this indicates a gap in how to include climate change actions performed outside the scope of the national budget support.

4. Methodology

4.1 Introduction to Approaches and Sources of data and information

A mixed-method approach was employed in the NDC stocktake, aligned with international best practices but tailored to the country's context. The methodology included the gathering of information and data from national reports and sectoral data; from stakeholder consultations; and the application of a structured assessment framework, grounded in the UNFCCC Enhanced Transparency Framework (ETF). This integrated approach ensured that both quantitative data and qualitative insights (e.g. institutional experiences, barriers, and enablers) were captured in the analysis.

4.2 Sources of data and information

The primary sources of data and information for assessing mitigation and adaptation progress included official documents and input from experts in the given sectors, ensuring both qualitative and quantitative insights on progress were captured. Documents reviewed included:

- National policy documents: The Updated NDC (2021), results from the 2024 National Population and Housing Census, the Third National Development Plan (NDPIII), the NDPIII mid-term review, and the Fourth National Development Plan (NDPIV).
- Technical Working Group (TWG): Workshops convened by the Ministry of Water and Environment's Climate Change Department (MWE/CCD), which served as a platform for sector experts to provide updates and validate information.
- Sectoral inputs from MDAs: Customized data templates were developed and shared with designated personnel in relevant ministries, departments, and agencies for each sector and subsector covered in NDC 2.0. These inputs formed the backbone of sectoral progress assessments for both mitigation and adaptation.
- Follow-up interviews: Conducted where possible with sector focal points and subject matter experts to clarify or fill data gaps in the information shared.
- Development partner and international reports: Key documents from the African Development Bank (AfDB), the World Bank, UNDP and NDCP were used to complement national sources, especially where official data was incomplete.
- Climate reporting instruments: Uganda's Biennial Update Reports (BURs), the INDC, Uganda's updated NDC, the third national communication, and the REDD+ strategy and action plan, provided insights into GHG emissions and removals trends and long-term climate pathways.
- Additional documents and background studies used in the stocktake are listed in the Literature References of this report.

4.3 Alignment with International Frameworks (Decision 18/CMA.1 and Decision 5/CMA.3)

The NDC Assessment Framework developed for the stocktake exercise mirrors the Enhanced Transparency Framework (ETF) under the Paris Agreement, ensuring a comprehensive review of mitigation, adaptation, and means of implementation. in line with Decision 18/CMA.1 (Modalities, Procedures and Guidelines – MPGs), the assessment considered:

1. Information necessary to track progress in implementing and achieving the NDC (mitigation targets and policies, per Article 4).

2. Information related to climate change impacts and adaptation (per Article 7).
3. Information on financial, technology transfer, and capacity-building support needed and received (per Articles 9–11).

Uganda's MRV system is already evolving to align with the ETF requirements. The country has an ongoing GHG Inventory Improvement Plan, including the development of country-specific emission factors and updated documentation of methods and uncertainty. Efforts are also underway to establish a centralized digital portal that will allow MDAs to submit sectoral data directly, enhancing consistency, timeliness, and institutional ownership.

The Common Tabular Format (CTF) structure was applied in the NDC tracking framework developed for this stocktake, to ensure transparency and comparability with international standards. For mitigation, chapter III of decision 18/CMA.1 and guidance from decision 4/CMA.1 was adapted to compile indicators of progress such as sectoral GHG reductions and policy milestones, together with methodologies and accounting approaches consistent with UNFCCC guidance. Adaptation progress was tracked against Uganda's NDC goals using a framework analogous to chapter IV of decision 18/CMA.1, capturing both qualitative and quantitative outcomes across priority sectors. Support needed and received was assessed with reference to the NDC Implementation Plan budgets and finance data, consistent with CTF Tables under Annex III Decision 5/CMA.3 (finance, technology, and capacity-building). However, a fully ETF-integrated digital MRV system (GHG inventory) is not yet operational, and this remains an area for further strengthening in preparation for NDC 3.0 and the first Biennial Transparency Report (BTR).

4.4 Use of the six-pillar analytical overlay

In addition to the ETF-aligned review, the stocktake applied a six-pillar analytical framework to evaluate the NDC 2.0's overall robustness and implementation potential. The framework provides a six-pillar methodology for a structured lens on assessing qualitative aspects of the NDC 2.0 design and execution beyond the quantitative assessment. The pillar focus areas include:

Pillar 1 - Ambition: This pillar examines the strength of Uganda's climate targets and commitments. This includes the ambition level of the emission reduction targets, the coverage of sectors and greenhouse gases, the balance of unconditional against conditional targets, and the timeframe of achieving the targets. It essentially assesses if there is a progression in ambition over time, in line with Paris Agreement Article 4.3, relative to the INDC.

Pillar 2 - Quality: The pillar focuses on the robustness and clarity of the NDC content. It assesses the reliability of the data and assumptions used, the level of detail in the sectoral implementation roadmaps and in the proposed mitigation / adaptation measures, how well the plans align with other national policies, and the effectiveness of the systems used to monitor and report progress. A high-quality NDC would have transparent baselines, well-defined indicators, and integration with development plans.

Pillar 3 - Process: This pillar evaluates the inclusivity of the formulation and implementation process. It checks for transparency, the involvement of a wide range of stakeholders (a "whole-of-society" approach), the integration of gender considerations, and the incorporation of "just transition" principles, which ensure that the shift to a green economy is fair and equitable for everyone. This pillar reflects lessons on governance – whether the NDC process builds broad ownership and accountability.

Pillar 4 - Implementability: This pillar looks at the practical side of putting the plans into action- reviewing the institutional and operational readiness to implement the NDC actions. It assesses whether there are clear,

time-bound milestones, strong institutional frameworks with clear roles and accountability, and effective coordination between different government ministries, departments and agencies.

Pillar 5 - Investability: This pillar is all about the financial aspects. It evaluates whether the climate actions have detailed cost estimates, if there are clear sources of funding (both domestic and international), and if Uganda is developing a pipeline of "bankable" projects that are ready for investment. It also considers innovative financing mechanisms and private sector engagement as indicators of how readily implementable (and attractive to funders) the NDC actions are.

Pillar 6 - Credibility: This final pillar assesses how credible and globally aligned Uganda's NDCs are, say to its Long-term Climate Strategy (LTS) and global commitments. It checks if the plans incorporate insights from the Global Stocktake, whether they are linked to Uganda's long-term climate strategy, and the country's track record of implementing its previous NDCs. A credible NDC is realistic (grounded in past performance and capacities) yet forward-looking and aligned with international goals, thereby instilling confidence domestically and internationally.

By combining the ETF-aligned quantitative tracking with this six-pillar qualitative review, the methodology ensures a holistic stocktake. The approach illuminated not only what progress has been made (or not) on NDC targets, but also why— by identifying underlying factors like data reliability, institutional capacity, stakeholder engagement, and financing gaps. All analysis was done in close collaboration with Ugandan stakeholders and aligns with the Paris Agreement's broader objectives of improving transparency and facilitating the global stocktake process. This methodology sets a strong foundation for Uganda as it prepares to meet ETF reporting obligations (e.g. its first Biennial Transparency Report) and as it plans for the next NDC cycle.

4.5 Approach to assess Means of Implementation and Support

Means of implementation include climate finance, capacity building; and technology development and transfer. Uganda considered gender responsiveness as an imperative for NDC 2.0; and catered for it under means of implementation for NDC 2.0.

Climate Finance relevant for NDC 2.0 Implementation

Tracking the progress of climate finance for implementing NDC 2.0 requires information on climate finance needed, climate finance committed, and climate finance received. This Stocktake considered ***committed climate finance*** to be the approved costs/budgets of projects identified as aligned to NDC 2.0 priority actions in accordance with the said projects' financing documents. The disbursed share of the committed finance was considered as the ***received climate finance*** for the period of 2020 to 2025. The cost of the priority NDC Actions, as per the costed NDC 2.0 Implementation Plan (MWE, 2022), was considered as the ***targeted climate finance needed***. The difference between climate finance received and needed is what was used to track progress for climate finance in the context of NDC 2.0 Implementation. A positive value indicates achievement of climate finance target while a negative value indicates failure to achieve the set climate finance target.

Regarding climate finance needed for implementing NDC 2.0, 14% of the total NDC 2.0 adaptation cost was expected to be unconditional and 86% conditional; whilst for mitigation, 15% of the total cost was expected to be unconditional and 85% conditional (GoU, 2022). This Stocktake understood Unconditional climate finance to be that finance through Government of Uganda funding with monies mobilized mostly from domestic sources following the country's Public Finance Management policies and legal framework, national programme-based planning and budgeting procedure as indicated in the national budget and Public Investment Plan; and Conditional climate finance to mean any finance mobilized via external public and

private sources including bilateral and multilateral development partners, through both national budget support and off-budget support (GoU, 2022; MoFPED & DU-OPM, 2023; MoFPED, 2025). Off-budget support is channeled through projects that are either directly implemented bilaterally or multilaterally by the Development Partners; or through Non-Governmental Organizations, Civil Society Organizations or Private organizations but with funding from Development Partners. As such, the Stocktake expected climate finance channeled through national budget support to be likely a combination of domestic and external finance, that is Unconditional and Conditional, respectively; whilst financial support via off budget supported projects to be exclusively externally mobilized finance and therefore exclusively Conditional.

However, to reduce the risk of double counting conditional finance, the stocktake considered as domestic finance and therefore unconditional, all external finance that was channeled through national budget in the form of concessional loan instruments. This is because Concessional loans are monies which the Government of Uganda has borrowed with approval of the Parliament, to implement in the short term, its priority national development programmes with the intention of paying back the external loan source-both the interest on the loans (normally, five years after loan agreements are signed) and the principal in the long term (15-20 years or even up to 40 years for some infrastructure projects). Moreover, even Grants from external sources that are meant for national budget support are portions of concessional loans, according to Uganda's Public Debt Management policies; and reports published in 2024 by MoFPED on public debt, grants, guarantees and other financial liabilities. To control such discrepancies in future NDC's, Uganda might need to disaggregate its budgeted Conditional External Financial support as either portion expected through Debt instruments such as loans; and those from non-debt instruments, that is, for example purely Grants.

Against this background, data to track climate finance committed and received, relevant to NDC 2.0 implementation and channeled through the national budget, was collected from reports of 2020 till the second quarter of 2025 published by the Ministry of Finance, Planning and Economic Development (MoFPED) and the Delivery Unit of the Office of the Prime Minister (OPM-DU) on the performance of projects aligned to NDP III and supported through the National Budget from FY2018/19 to FY 2023/2024 and FY 2024/2025. This data was corroborated and checked for consistency with information solicited from Development Partners such as the NDC Partnership and World Bank. Only those projects whose objectives, output, outcomes and activities aligned to NDC 2.0 priority mitigation, adaptation and cross-cutting actions and activities (GoU, 2022; MWE, 2022) were included for the purpose of tracking of the climate finance component of NDC 2.0 Means of Implementation. This was irrespective of whether the data source indicated the projects as a climate project, had climate adaptation co-benefits for projects intended to reduce GHG emissions, or mitigation co-benefits for projects intended primarily to reduce vulnerability to impacts of climate-related hazards. This approach is aligned to international best-practice in tracking climate finance (EIB, 2022; 2023).

Data on climate finance committed and received, for off-budget bilaterally and multilaterally funded projects was obtained from the Development Aid portal, d-portal (<http://www.d-portal.org/faq.html#gen1>). This is a global web-based platform of all data from the International Aid Transparency Initiative (IATI) grouped by recipient country or organization, comparable with the latest Creditor Reporting System data of the Organization for Economic Co-operation and Development's Development Assistance Committee (OECD DAC CRS). This data dates from 1960 to "present year +2". IATI is a voluntary, multistakeholder initiative that seeks to improve the transparency of aid, development, and humanitarian resources to increase their effectiveness in tackling poverty. IATI data is sourced from the IATI Registry, and the CRS data is sourced from CRS tables and stored on Github. Newly published and re-import data files of the d-portal are updated every night. For the purposes of the Stock take, explore search filters of d-portal were used to generate the lists of off-budget projects considered in the Stock take analysis. The search filters included projects implemented in Uganda from 2020 to date (either completed or ongoing), projects aligned to climate change response, UNFCCC, Convention to combat Desertification, environmental protection, gender, youth, women, refugees

and refugee hosting communities. From the generated lists of projects from all NDC 2.0 priority adaptation and mitigation sectors, a final list was developed consisting of only those projects whose objectives, output, outcomes, and activities aligned to NDC 2.0 priority actions. This final list was corrected for projects that had already been identified as externally funded via the national budget.

Gender and Capacity-building

NDC 2.0 merely enumerated five priority gender actions; and seven capacity building needs that had to be addressed to guarantee implementation of priority NDC adaptation, mitigation, coordination and cross-cutting actions. However, the updated NDC did not include targets for the priority gender actions nor the stated capacity-building needs. Moreover, it is not clear whether these NDC gender and capacity-needs identification had been based on a prior gender gaps analysis and capacity-needs assessment specifically for NDC 2.0 implementation. Therefore, the approach adopted by the Stocktake assessed the status of both gender and capacity-building, as opposed to tracking their progress, *per se*.

It is worth noting that all the NDC 2.0 actions that were prioritized for both Gender and Capacity-building relate to Institutional Capacity-building.

This stocktake categorized capacity building as Institutional capacity-building and Innovation capacity-building. Institutional capacity-building encompasses the ability and competence of Ministries, Departments and State Agencies (MDAs) mandated to coordinate different facets of planning, monitoring and resource mobilization of climate action at the national and upper local government level, including policy and legislative frameworks. Innovation capacity-building focusses on actors that implement NDC actions and activity projects up to the community level.

Institutional capacity-building constitutes five thematic areas (Bakhtiari, et al.,2018); viz (i) Capacity to integrate NDC Priorities into sectoral and cross-sectoral programs and projects, to ensure that the latter do not undermine efforts to achieve the former, or vice versa; (ii)Resources to train relevant government agency staff (and possibly non-government agency staff too) with the view to increasing the technical and managerial skills of these individuals; (iii) Capacity to engage all relevant stakeholders, through consultations designed to elicit their input, so that it can be taken into consideration, thus increasing buy-in from stakeholders; (iv)Competence to conduct a regulatory framework revision, to streamline and complement existing laws and regulations, and strengthen related governmental processes and entities; and (v) Aptitude to monitor progress, and report on it, making best use of existing data collection mechanisms, and strengthening related capacities where needed.

With regards to Institutional capacity-building, nineteen indicators (19) were developed using the five thematic components of Institutional capacity building as criteria. These indicators were sub-categorized as: (i) NDC Coordination expected to be done through the Office of the Prime Minister (OPM), Climate Change Department, Ministry of Water and Environment (CCD), National Planning Authority (NPA), and the eighteen National Development Plan Program (NDP) Coordinators; (ii) Execution of NDC 2.0 priority actions through MDAs; (iii) NDC 2.0 Resource Mobilization through NPA, Ministry of Finance, Planning and Economic Development (MoFPED), Ministry of Justice and Constitutional Affairs (MoJCA), and CCD; (iv) NDC 2.0 Monitoring and Evaluation through Office of Auditor General, NPA, MoPED, CCD, and OPM. Due to data access challenges, it was not possible to gather information on these indicators within the time frame of the stocktake assignment. A number of these challenges revolved around administrative clearance for data access and could be catered for in advance within the next round of NDC, to facilitate the comprehensive assessment of Uganda's institutional capacity in line with NDC implementation. Off budget supported projects in line with Institutional capacity building and are implemented directly by Development Partners were identified and listed alongside other relevant projects under the respective priority NDC sectors.

On the other hand, data on innovation capacity-building was deduced from projects that were planned and implemented with support from Uganda's national budget. These projects' performance was monitored each financial year, from FY2019/2020 to FY2024/2025. This project performance information was verified and submitted by MoFPED to the Parliament of Uganda. The projects that were selected as relevant to NDC 2.0 were those whose scope and objectives were aligned with the priority NDC 2.0 adaptation and mitigation actions and activities. It is only these projects' components of capacity-building that were assessed in this stocktake. Projects that were categorized as "Climate Change Projects" by MoFPED and DPs but were not aligned with NDC 2.0 priority sectoral actions were not included and were considered outside the scope of this Stock take.

Technology Transfer and Development

The Stock take understood technology as a contributor to people's livelihoods and includes three of the five sustainable livelihood assets: physical assets (physical infrastructure, machinery and equipment); human assets (knowledge and skills; and capacity to organize and use all physical and human assets); natural assets (biological technology with which actors produce in agriculture advances in crop nutrition & protection) (Clements, et al., 2011). This direct association with livelihoods dictates the importance of having a people-focused/people-centered definition of technology. Moreover, in the context of sustainable economic development, climate change impacts the economy of a country through the interaction with livelihood assets within the country (WBG, 2025). Technologies were categorized as hard, soft, and organizational technologies. According to Clements and others (2011), hard technologies refer to tangible aspects such as manufactured objects, machinery, equipment and tools required to produce goods & services; while soft technologies refer to processes associated with the production and use of the hard technologies, including know-how-manuals & skills. Soft technologies encompass elements of awareness-raising including education & training as in the case of capacity building on implementing a technology. However, for the purpose of this Stocktake, capacity building to implement hard technology was categorized under the "Capacity-building Component" of Means of Implementation. It is only the know-how like manuals that remained as soft technology. Organizational technologies refer to the organogram involved in the adoption and diffusion process of new technology or where the technology will be used e.g. establishment of farmer groups and water use associations

The components of capacity building; technology transfer and development; and constraints/barriers and gaps encountered in the implementation of the identified NDC 2.0-relevant projects were adopted to inform the analysis of this Stock take. Additionally, the recommendations made during the public audit of these projects by MoFPED and OPM-DU were taken as the recommendations from key NDC 2.0 implementation stakeholders.

5. Stakeholder Engagement

5.1 Summary of consultations and technical workshops

The government ministries, departments, agencies, and key experts across relevant sectors were closely involved in developing the NDC Stocktake Assessment Report. Their input and validation were relevant for conducting a thorough assessment. Led by the Ministry of Water and Environment (MWE)/CCD and in collaboration with AfDB, stakeholder consultations were held to ensure an inclusive process. These consultations involved a technical workshop, technical meetings, and group discussions to collect input and build consensus. Meetings were also held with cooperating partners involved in the NDC 3.0 development process. These include AfDB, AFD, the World Bank, and UNDP. Further consultations were made with the teams involved in NDC 3.0 climate modeling, LTS development, and NDC 3.0 write-up. These collaborative exchanges were designed to ensure harmony in the different initiatives and for the project team to share early findings from the stocktake assessment to inform the next cycle of NDC.

The plan for technical consultations was designed to follow the format below:

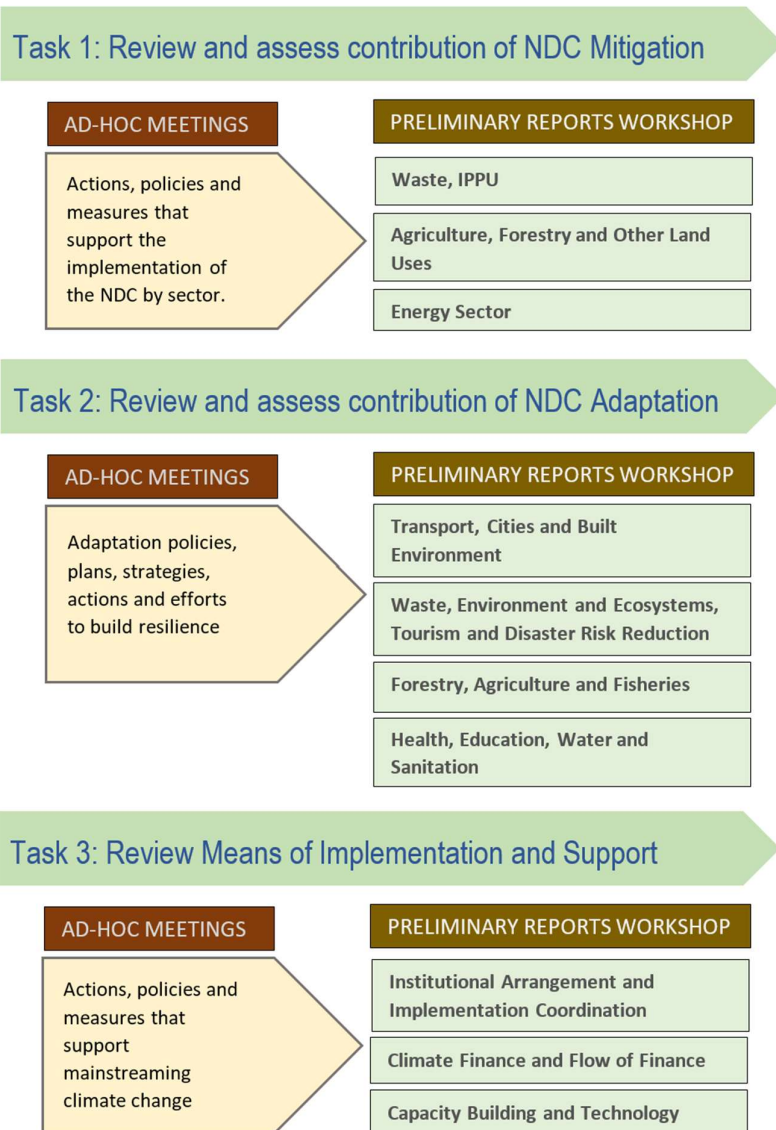


Figure 5-1: Technical consultation process for the NDC stocktake assessment

There were challenges in accessing all identified key stakeholders and subject matter experts.

The following table indicates the list of proposed stakeholders for the NDC Stocktake:

Table-5-1: List of stakeholders proposed to reach out for the NDC 2.0 Stocktake

Stakeholder Category	Specific Entities	Key Regions
Government Ministries, Departments, and Agencies (MDAs)	Ministry of Water and Environment (MWE) (Climate Change Department)	National
	Ministry of Agriculture, Animal Industry and Fisheries (MAAIF)	National, Key agricultural regions
	Ministry of Energy and Mineral Development (MEMD)	National, Areas with energy infrastructure
	Ministry of Finance, Planning and Economic Development (MoFPED)	National
	National Environment Management Authority (NEMA)	National
	Uganda Farmers Federation	Regional / District level
	District Local government	Regional
	Ministry of Health	National
	Department of Meteorology (DOM)	National
Private Sector	Private sector Foundation; Uganda Manufacturers Association (UMA)	Regional
	Renewable Energy Businesses	National
	Agricultural Businesses	Key agricultural regions
Civil Society Organizations (CSOs)	Climate Action Network Uganda (CAN-U); Green Africa Youth Organization (GAYO)	National, Vulnerable communities
	Environmental NGOs	National and regional
Local Communities	District Local Governments	Regional
	Traditional Leaders	Areas with traditional governance structures
Youth	Youth Climate organization; Youth Climate Council of Uganda; Uganda National Youth Council	Urban and rural areas
Women	Women's Rights Organizations	All regions, with a focus on marginalized communities
Academia	Universities and Research Institutions include Makerere University Climate change and Innovation Centre	National

During the stocktake, it was possible to reach out to the following entities and MDAs:

Office of the Prime Minister; Parliamentary Standing Committee on Climate Change; Ministry of Water and Environment; Ministry of Finance, Planning and Economic Development; Ministry of Works and Transport; Ministry of Health; Ministry of Agriculture, Animal Industry and Fisheries; Ministry of Energy and Mineral Development; Ministry of Gender, Labour and Social Development; Ministry of Tourism, Wildlife and Antiquities; Climate Change Department; National Planning Authority; Uganda Bureau of Statistics; Uganda Railway Cooperation; National Environment Management Authority; Kampala City Council Authority; Uganda Manufacturing Authority; National Water and Sewerage Cooperation; Uganda Tourism Board.

More stakeholders, especially from the Academia, Civil Society Organizations, and NGOs, were met during the technical workshop held in July 2025. This information is available from the lists of attendance to the 2 workshop reports that were submitted to CCD. However, a detailed list of stakeholders mapped for consultation is available in the annex to this report.

5.2 Governance and coordination

The governance and coordination challenges/opportunities encountered include:

Availability of information and data sources: The stock take was intended to be analytical and relied on the timely availability of complete information and datasets, including meta data. It was not possible to get all the information and data sets required. However, documents that were made available by key MDAs and the NDCP with supplementary information obtained from Uganda's submissions to the UNFCCC secretariat and other official government publications were used.

Establishment of the Technical Working Group: The role of the Technical Working Group was critical, especially as sector focal persons, and review of preliminary assessment results. One technical workshop was held where the TWG was established. The project team was able to access its members and received support during the workshop and follow-up emails. The TWG provided valuable input to facilitate tracking of NDC tracking of the NDC indicators including provision of activity data and sectoral reports where possible. Key data providers included the NPA, UBOS, and NDCP while other TWG members could not support the data collection process for various reasons, including the absence of information and not tracking the NDC indicator.

Participation in Technical Workshop: We believe that participation should be inclusive and of wider geographic coverage. The private sector, local government, and other NDC implementing partners attended the technical workshop and inception report validation workshop, where there was active participation.

Stakeholders' timings: We were aware that the project timing may be affected by the stakeholders' availability to both provide data and participate in the workshops. The project team was flexible to accommodate the key stakeholders' time availability. In addition, our deliverables depend on the client's and timely feedback on submissions. This was achieved within acceptable and foreseen timelines.

6. Assessment of NDC 2.0 Content and Gaps

Uganda's NDC 2.0 shows clear progress in aligning climate action with national priorities such as Vision 2040, the Third National Development Plan, and the Green Growth Development Strategy, while also situating it within the 2030 Agenda for Sustainable Development. The updated NDC highlights co-benefits for poverty reduction, food security, energy access, and sustainable cities, and promotes a whole-of-society approach that includes government, private sector, civil society, academia, and youth. However, while these linkages are broadly articulated, progress could be strengthened by explicitly mapping NDC actions to specific SDG targets and national development indicators to clarify and track synergies.

This section assesses the content of Uganda's Updated NDC (NDC 2.0) by checking its alignment with the Sustainable Development Goals and the six-pillar NDC Assessment framework. The 6-pillar assessment covers the ambition and clarity of the NDC targets, the feasibility of implementation plans, financial and institutional arrangements, costing of measures, and the inclusion of adaptation commitments. Uganda's Updated NDC (2021) reflects significant progress from the INDC in scope, ambition, and alignment with national and international priorities.

6.1 Alignment of NDC 2.0 with SDGs and national development

Uganda's NDC 2.0 explicitly links climate action to national strategies such as Vision 2040, the Third National Development Plan (NDP III), and the Green Growth Development Strategy. It also positions climate commitments within the broader framework of the 2030 Agenda for Sustainable Development, including co-benefits for poverty reduction, food security, energy access, and sustainable cities. For example, the updated NDC highlights how its climate actions support resilience of communities and ecosystems, which dovetails with SDG targets on poverty alleviation, food security, and sustainable cities.

The Updated NDC also underscores a whole-of-society approach involving government, private sector, academia, civil society, youth and others in NDC implementation. This broad inclusion corresponds to SDG 17 (partnerships) and ensures climate objectives are mainstreamed into Uganda's socio-economic goals. While the NDC mentions these linkages in general terms, it could be strengthened by explicitly mapping specific NDC actions to individual SDG targets or national development indicators. Doing so in the future would clarify explicit synergies with other SDGs e.g. affordable clean energy (SDG 7) and green jobs (SDG 8).

6.2 Ambition Level of Targets (Pillar 1 – Ambition)

Uganda's NDC 2.0 demonstrates a clear progression in ambition. The updated target of 24.7% emission reductions below BAU by 2030 (5.9% unconditional, 18.8% conditional) is higher than the 22% reduction target in the INDC. This increase, though modest, is significant given Uganda's status as a low-emitting country with pressing development needs.

Ambition has also broadened in scope. NDC 2.0 covers all major emitting sectors: AFOLU, Energy (stationary and transport), Waste, and IPPU. Interim milestones for 2025 were introduced for several measures, which signal near-term ambition and provide touchpoints for tracking progress.

Uganda's conditional target remains highly dependent on international finance and technology transfer. While this reflects the principle of common but differentiated responsibilities (CBDR), it also highlights vulnerability: without predictable support, the conditional ambition may remain aspirational. Future NDC

iterations could further raise ambition in unconditional contributions, especially in no-regrets areas like renewables, efficiency, and sustainable land management.

6.3 Implementation Feasibility and Clarity (Pillar 2 - Quality)

The updated NDC provides a much clearer roadmap for action than the previous one. It delineates specific mitigation and adaptation measures for each sector, each with associated indicators and targets for 2025 and 2030. For instance, the NDC lists interventions like expanding renewable energy capacity, promoting forest natural restoration initiatives to halt the current high deforestation rates (~2%) etc., along with numeric targets (e.g. MW of renewables, hectares reforested). This level of detail is a significant improvement, making the NDC more actionable and measurable. The NDC 2.0 was developed in tandem with Uganda's Long-Term Low Emissions Development Strategy (LTS) which was under development back then, ensuring short-term actions are aligned with long-term pathways. It also references alignment with existing national policies (for example, linking adaptation actions to Uganda's National Disaster Risk Management policy and others), which helps integrate NDC implementation into the broader policy framework.

Uganda's NDC demonstrates enhanced specificity compared to its predecessor but would benefit from strengthening its implementation framework to ensure effective delivery of commitments. The NDC presents comprehensive measures and targets yet falls short of more detailed implementation pathways with clearly defined milestones throughout the 2021-2030 period, including specific timelines, designated responsible institutions, and interim checkpoints beyond the current 2025/2030 endpoints to facilitate systematic progress tracking and institutional accountability. Certain targets, such as achieving 50% clean energy access for cooking by 2025 from a 2015 baseline of 15%, represent ambitious commitments that would benefit from accompanying implementation strategies detailing financing mechanisms, distribution systems for clean cookstoves, regulatory frameworks, and capacity-building requirements to demonstrate feasibility and enhance credibility.

Transparency in methodological approaches, including baseline scenario assumptions, emissions factors, and calculation methodologies, could be strengthened by integrating technical details of NDC 2.0 from annexes into the main document to improve accessibility and facilitate understanding by diverse stakeholders. The NDC would benefit from a more robust description of its monitoring, reporting, and verification framework, specifically outlining domestic measurement and reporting procedures, institutional arrangements, and quality assurance mechanisms to track implementation progress effectively. Enhancing these elements would align the NDC more closely with international best practices while providing a clearer implementation blueprint that supports Uganda's climate ambitions through systematic, accountable, and transparent action.

6.4 Pillar 3 - Process

The process of updating the NDC 2.0 was inclusive and participatory, involving a multi-sectoral technical committee, extensive consultation with ministries, civil society, private sector, academia, and development partners. Special attention was paid to engaging women, youth, and vulnerable groups, aligning with principles of just transition and equity.

This participatory approach improved ownership and legitimacy, and the NDC reflects broad consensus on Uganda's priorities. However, the document could have more explicitly described how stakeholder inputs shaped the final commitments, to provide transparency and traceability of the consultation outcomes.

Overall, Uganda's NDC update process demonstrates a strong commitment to inclusivity, but future processes should emphasize documenting and communicating stakeholder contributions as part of good governance.

6.5 Pillar 4 - Implementability

The NDC document situates its implementation within Uganda's existing institutional framework for climate change. It notes that the Climate Change Department (CCD) under the Ministry of Water and Environment will coordinate the NDC actions, and it references the involvement of various ministries and agencies in a whole-of-government approach. The NDC update process itself was overseen by a multi-sectoral technical committee (as acknowledged in the document's annexes/acknowledgements), indicating an inclusive approach to planning. The NDC also came after the passing of the Climate Change Act, 2021, which provides a legal basis for climate action coordination through entities like the National Climate Change Advisory Committee (NCCAC). This gives the NDC an improved institutional anchor compared to the first NDC.

6.6 Pillar 5 Investability

Uganda's NDC stands out for providing concrete cost estimates for its climate actions. As noted, the total implementation cost through 2030 is estimated at \$28.1 billion, with breakdowns by sector and by source of funding (domestic vs international). The document lists indicative costs for major initiatives, which improves transparency and helps with resource mobilization. For example, significant costs are associated with adaptation priorities (over \$17 billion needed)- reflecting the emphasis on adaptation, while mitigation actions are estimated at around \$10 billion. Uganda committed to mobilizing USD 4.1 billion domestically (15% of total), with the remainder conditional on international support.

By providing a clear costing baseline, Uganda has made its NDC more investable. Future updates should build on this by detailing finance mobilization strategies and identifying bankable projects, enhancing the NDC's attractiveness to both donors and private investors.

6.7 Pillar 6-Credibility

Uganda's NDC 2.0 enhances credibility through its comprehensiveness and evidence base. Adaptation coverage expanded from 7 sectors to 13, including new areas such as fisheries, ecosystems, urban systems, and tourism. Each sector now has targets and indicators (with 2015 baselines and 2025/2030 goals), making adaptation commitments measurable.

The NDC is also informed by a national Climate Risk and Vulnerability Assessment, ensuring actions are aligned with Uganda's most pressing risks. Integration with long-term national goals and consistency with Global Stocktake (GST) requirements further reinforce credibility.

However, credibility will ultimately depend on implementation and reporting. Uganda should establish mechanisms to regularly publish progress updates, link interim results with global reporting cycles, and transparently revise assumptions as needed.

7. Tracking Progress on Mitigation

Uganda is making tangible progress in AFOLU and stationary energy sectors, particularly in renewable energy deployment and early forest interventions. However, high-impact sectors like transport and IPPU remain largely unquantified or unimplemented, and most mitigation measures are behind schedule for achieving NDC 2.0 targets for 2030. Accelerated implementation, stronger MRV, and securing conditional financial support are needed to meet pledged reductions.

AFOLU Sector: Early progress shows 1.89 MtCO₂e total emissions removals (~6% of the 2030 target) with natural forest restoration (1.68 MtCO₂e), Commercial small-holder and community bioenergy woodlots (0.13 MtCO₂e vs. 2.9 MtCO₂e, ~4% of the 2030 target) and Commercial small-holder and community poles & timber plantations with coffee agroforestry (0.068 MtCO₂e vs. 0.37 MtCO₂e, ~18% of the 2030 target)

- Most climate-smart agriculture interventions remain unimplemented and/or not reported, emissions reductions are far below the NDC 2.0 2030 target of 3.1 MtCO₂e due to lack of specific activity data to allow for estimation. However, Uganda has launched the Climate Smart Agricultural Transformation project²⁷ with support from the World Bank whose implementation across 69 districts will enhance reporting and implementation of climate smart agriculture interventions.
- Non GHG targets: Wetlands coverage has increased from 8.9% in 2020 to 9.32% by the end 2024. No verifiable data was provided to track changes in national deforestation rates while forest cover increased from 12.5% in 2020 to 12.6% in 2024 according to FSSD which is below the 15 % target by 2025.

Energy Sector (Stationary): Overall progress at 15% with Renewable energy generation being largely on track, achieving ~0.350 MtCO₂e (~73% of 0.48 MtCO₂e target for energy generation), but improved charcoal production is still at pilot stage (0.0166 MtCO₂e, 3.5%) and industrial efficiency/fuel switching shows minimal progress.

- Non-GHG targets: Electricity generation capacity has increased from 1,276MW to 2047MW, with an Increased renewable electricity share in the grid, standing at 98% according to ERA; and electricity access has increased from 24% to 58% by end of FY 2024/2025.

Transport Sector: Despite infrastructure gains (rail and NMT), no quantified emissions reductions have been reported, meaning 0% of NDC 2.0 2030 mitigation targets (3.06 MtCO₂e combined) .

- Non-GHG targets: 5.45km of NMT constructed in Kampala, 400km of MGR rehabilitated, 29km of operational passenger rail in GKMA, and 1, 127 cleaner vehicles by end of FY 2024/2025.

Waste Sector: Methane recovery and solid waste composting achieve ~18% of the 2030 targets by FY 2024/2025 (0.197 MtCO₂e), but the sector's overall impact is small (<2% of total mitigation potential).

²⁷ <https://www.agriculture.go.ug/ucsatp/>

- **Non-GHG: 148,658 tonnes/yr of waste diverted from landfills in 11 urban centres (excluding Kampala), waste collected from 50-60% of Kampala waste collected (up from 40%) and 1 wastewater plant upgraded to recover and utilize methane for electricity generation by end of FY 2024/2025.**

IPPU Sector: No measurable reductions reported; interventions remain at planning/pilot stages.

7.1 Uganda's Updated NDC

Uganda's updated NDC (2021) expanded sectoral coverage to AFOLU, energy, transport, waste, and IPPU, and committed to a 24.7% reduction in emissions by 2030 relative to BAU. It builds on the 2015 INDC while aligning with national development goals.

The NDC set a single-year target for 2030 with the annual net GHG emissions in MtCO₂e below the BAU GHG emissions in 2030. The base year was 2015, with emissions estimated at 90.1 MtCO₂e and the implementation period running from 2021 to 2030.

In terms of coverage, the NDC is economy-wide²⁸, covering energy, IPPU, AFOLU, and waste sectors, as well as the four main gases (CO₂, CH₄, N₂O, and HFCs), applying IPCC AR5 GWP metrics.

Uganda currently hosts Clean Development Mechanism (CDM) projects under the Kyoto Protocol, primarily in the renewable energy, efficiency and forestry sectors. The approval of CDM projects that have applied for transitioning to Article 6.4 mechanism of the Paris Agreement is under consideration by CCD. Looking ahead, Uganda expects to host Article 6.4 projects and potentially engage in cooperative approaches under Article 6.2, capitalizing on its forest restoration and renewable energy potential. However, currently, Uganda is not participating in any Article 6 activities under the Paris Agreement, and therefore no corresponding adjustments have been undertaken in this stock take report.

7.2 Framework for tracking progress towards Uganda's NDC

In line with Decision 18/CMA.1 (paras. 65–78) and Decision 5/CMA.3, Uganda applies a single indicator to track progress in implementing and achieving its updated NDC. Information is provided below, with supplementary details structured in accordance with CTF summary Tables 1–4.

7.2.1 Indicator for tracking progress of the NDC

Uganda tracks its NDC using annual total net greenhouse gas (GHG) emissions, expressed in million tonnes of carbon dioxide equivalent (MtCO₂e). This indicator is consistent with the scope of the NDC, uses the same unit and metric as the base year (2015) and target year (2030), and covers all relevant sectors and gases. The indicator was identified in Annex 1 of the updated NDC.

²⁸ An NDC is considered **economy-wide** if it covers all major IPCC inventory sectors (energy, IPPU, AFOLU, and waste) and includes the main greenhouse gases (CO₂, CH₄, N₂O, and fluorinated gases). In addition, the chosen indicator—typically total net national emissions—must be consistent with the scope and methodologies of the national GHG inventory. Excluding entire sectors, gases, or using an inconsistent indicator means the NDC is not fully economy-wide

7.2.2 Consistency with NDC Target

The selected indicator is directly aligned with Uganda's NDC target, which is defined as an economy-wide BAU scenario target. It encompasses all sectors and gases included in the most recent national GHG inventory, excluding GHG emissions from natural disturbances.

7.2.3 Definitions and Scope

The indicator of “total net GHG emissions” corresponds to the annual total emissions and removals across all covered sectors and gases, reported in MtCO₂e. This definition is consistent with Uganda’s national GHG inventory practice.

7.2.4 Accounting Approach

Progress towards the NDC is accounted for using anthropogenic emissions and removals consistent with Article 4, paragraphs 13–17 of the Paris Agreement, and annex II of Decision 4/CMA.1. Uganda’s inventories are compiled in line with the 2006 IPCC Guidelines and apply 100-year Global Warming Potentials (GWPs) from the IPCC’s Fifth Assessment Report. For this stock take, quantitative tracking of the indicator across the full implementation period (2021–2030) is constrained. Uganda’s most recent GHG inventory update covers up to 2019 and was submitted in August 2025 as part of the Second BUR²⁹. This highlights inherent challenges in near-real-time reporting of GHG and activity data. Methodological details consistent with Paris Agreement guidance are provided in CTF Table 3.

However, Uganda in its second BUR presents a comprehensive National Inventory Improvement Plan (NIIP) with sector specific actions that would result in improvement of country specific carbon accounting methodologies including emission factors and activity data collection for key emissions and removals sources. Further improvement of the integrated MRV tool managed by the CCD under MWE needs to be undertaken including capacity building of the activity data providers up to subnational level, quality control specialists and a complementary set of regulations and guidelines need to be developed to strengthen the existing legal and institutional frameworks e.g NCCA specifically by instituting mandates for activity data provision across sectors on a regular basis including from the MDAs, CSOs, development partners, industries and other private sector entities to facilitate and enhance compilation of the national GHG inventory in a transparent, accurate, consistent, complete, and comparable manner in alignment with the ETF reporting requirements.

7.2.5 Most recent information for the indicator

According to Uganda’s Second Biennial Update Report (BUR2), the most recent GHG inventory covers the period 1995–2019. Net national emissions increased from 43.9 MtCO₂e in 1995 to 112.1 MtCO₂e in 2019, reflecting an average annual growth rate of 6.2%. Compared to Uganda’s First BUR, which reported total national emissions of about 77 MtCO₂e in 2015, the BUR2 update shows a continued upward trend, with emissions rising by approximately 35 MtCO₂e over the subsequent five years.

In 2019, Agriculture, Forestry and Other Land Use (AFOLU) remained the dominant source, contributing about 76% of total emissions, followed by waste at 14.5%, energy at 8%, and IPPU at less than 0.5%. These results were reported using the 2006 IPCC Guidelines with 100-year Global Warming Potentials from the IPCC Fifth Assessment Report. No inventory data are yet available for 2020 to 2025, which constrains quantitative tracking of NDC implementation during the current period.

²⁹ https://unfccc.int/sites/default/files/resource/64251087_Uganda-BUR2-1-UGANDA%E2%80%99S%20SECOND%20BIENNIAL%20UPDATE%20REPORT.pdf

7.2.6 Methodological Consistency

In terms of coverage, scope and methodological approaches, the NDC targets and indicator are consistent with the 2015 National Inventory Report which formed the basis for NDC accounting, and with subsequent inventory updates. Both target and indicator are economy-wide, covering AFOLU, energy, IPPU, and waste, and the four gases (CO₂, CH₄, N₂O, HFCs). Uganda has maintained methodological consistency by applying 2006 IPCC Guidelines and AR5 GWP values. Land sector and includes 4 gases i.e. CO₂, CH₄, N₂O, and HFCs, GWP metrics from the IPCC's 5th Assessment Report have been used. Further information regarding methodological consistency is provided in the structured summary [table 3](#) (Annex 1: Methodologies and Accounting Approaches).

7.3 Progress of Mitigation Policies, Measures and Actions

Uganda in her updated NDC commits to implementation of prioritized mitigation measures across key mitigation sectors including in the AFOLU, energy (stationary), transport, waste, and IPPU. Most of the prioritized measures have been reported in other official submissions to the UNFCCC secretariat including but not limited to the First BUR³⁰, the green schools NAMAs³¹, waste management NAMA³², and the REDD+ strategy and action plan for mitigation actions mainly in AFOLU³³. Uganda committed to implement other new mitigation actions not previously reported in the IPPU, transport, and energy (stationary).

Uganda has shortlisted mitigation actions and measures focusing on those that have significant impact on GHG emissions and removals, prioritizing them for implementation under the updated NDC.

The summary tables in this section are extracted from [Annex 3 – table 5: Comprehensive NDC Assessment Framework](#). While this section presents a structured summary of mitigation actions and results, the annex contains the full assessment framework with more detailed information, including additional parameters, sources of data, implementing agencies, implementation period and sectoral breakdowns not captured in the summary tables below.

7.3.1 Agriculture, Forestry and Other Land Use (AFOLU)

AFOLU

The implementation of the mitigation actions in the AFOLU sector is guided by commitments mainly in the REDD+ strategic options in the REDD+ strategy aimed at reversing current wood and biomass extraction into sustainable abatement activities. The strategic options were subdivided into sub options which were prioritized as mitigation measures in the sector to reverse deforestation rates (~2%), land degradation and increase forest cover from about 12.5% in 2020 to 21% in 2025 and 21% in 2030. By 2024, FSSD reported a forest cover increase to 12.6%.

The activity data used to compute emissions reductions from the AFOLU sector were primarily obtained from the NDC partnerships Country Engagement Update Reports for the implementation of the updated NDC and districts forestry data compilations from the Forest Sector Support Department (FSSD) in the Ministry of Water and Environment which oversees and coordinates implementation of the REDD+ strategy and action plan, and hosts the national REDD+ focal point. However, there was limited disaggregation of activity data to specific interventions and therefore emissions reductions were aggregated for several measures where activity data disaggregation was not possible. This further exemplifies capacity gaps in data collection,

³⁰ [Uganda. Biennial update report \(BUR\). BUR 1. | UNFCCC](#)

³¹ [NAMA Report Uganda FNL.pdf](#)

³² [NAMA Integrated Waste Management](#)

³³ https://redd.unfccc.int/media/final_-_uganda_redd_strategy_and_action_plan-october_2017.pdf

monitoring and reporting of the REDD+ strategy hence there is need to strengthen monitoring and evaluation of the REDD+ strategic interventions as per its already established indicators in the REDD+ M&E framework and/or refinement to allow for effective regular bottom-up tracking of the carbon and non-carbon outcomes from implementation of the interventions.

The progress of the prioritized interventions under the updated NDC in the AFOLU sector are described below.

Climate Smart Agriculture

Climate Smart Agriculture as a strategic intervention is aimed at reducing agricultural expansion to forested land through sustainable intensification on already cultivated land hence achieving mitigation benefits. The mitigation benefits are realized via avoided deforestation since higher carbon per hectare is sequestered by forests compared to croplands. Its implementation under the updated NDC was estimated to provide 3.1 MtCO₂e mitigation potential. There exist isolated interventions reported under this strategic intervention with support from development partners e.g. through the SNV - Climate Resilient Agribusiness for Tomorrow (CRAFT) project and USAID provided targeted training on Climate-Smart Agriculture (CSA) practices and Good Agronomic Practices (GAPs) across different agricultural value chains. However, reporting on this intervention remains limited, constraining effective estimation of emissions reductions.

In addition, Uganda through the Ministry of Agriculture, Animal Industry and Fisheries launched a 6-year Uganda Climate Smart Agricultural Transformation project³⁴ with funding from the World Bank. The project will be implemented in 69 districts and is expected to contribute to implementation of the expected outcomes of this strategic option.

The climate smart agriculture strategic option covers 3 sub-options including sustainable land use management and agroforestry, rainwater harvesting and irrigation, and green house cultivation of vegetables as described below.

Sustainable Land Management use management and agroforestry

Sustainable land management practices are promoted as some of the most cost-effective interventions in the REDD+ strategy aiming for adoption by all rural farming households. These include agroforestry practices such as plantation of fruit trees, fodder trees, fuelwood and construction wood species to cover sustainably for all kinds of tree product needs for the households, hence reducing burden on natural and forested areas. Moreover, additional benefits from utilization of non-timber forest products such as through apiculture can be realized, providing additional sources of income. By adopting such practices, Uganda aims to reduce GHG emissions through enhanced carbon sequestration in soils, decreased reliance on forests and degradation of forests for timber and firewood, and improved overall farm resilience, contributing to the country's climate change mitigation efforts.

It is estimated that implementation of these interventions would remove 1.2 MtCO₂e from the atmosphere by 2030. Estimation of achieved emissions reductions from the implementation of these practices was constrained by limited disaggregation of activity data to capture the area under these practices. However, Uganda finalized the National Agroforestry strategy³⁵ which is envisaged to scale up agroforestry with improved monitoring and evaluation of agroforestry outcomes. The tables below summarize the sustainable land management strategic sub-option.

³⁴ <https://www.agriculture.go.ug/ucsatp/>

³⁵ THE UGANDA NATIONAL AGROFORESTRY STRATEGY 2024/25 - 2033/34

-Table 7-1: Sustainable land management and agroforestry

Name of action	Description	Objective	Type of instrument	Status of implementation	Sector (s) affected	Gases affected	Implementing entity	Start year of implementation
Sustainable Land Use Management and Agroforestry	Land use management practices, 40 million-tree campaign and REDD+ strategic interventions	Reduce GHG emissions through sustainable land management	Other	Implemented	AFOLU	CO ₂ , CH ₄ , N ₂ O	MAAIF; NARO; NaFORI; MoLG(DP,DFS, DNR)	2024

-Table 7-2: Estimates of GHG emissions reductions from sustainable land use management and agroforestry

Estimates of GHG emission reductions (Mt CO ₂ e)		
Achieved		Expected
Year	Emissions reductions	
2021	NE	NE ³⁶
2022	NE	NE
2023	NE	NE
2024	NE	NE
2025	NE	NE

Rainwater harvesting and irrigation

This is the second sub-option under the climate smart agriculture strategic intervention mainly aiming for storage of rainwater to allow for its utilization and/or to prolong the two crop cultivation seasons in Uganda especially for times when rainfall is not sufficient in crop cultivation season. Moreover, Northern Uganda has only one long rainy season hence collecting rainwater from house roofs and piping it into a closed storage tank in the ground from where drip irrigation water can be distributed on the crop fields and the vegetable

³⁶ Emission reductions are “not estimated, NE” due to limited disaggregation of activity data to allow for estimation of GHG reductions associated to this specific measure.

gardens or for providing drinking water for livestock will boost crop yields and productivity in farm operations. The interventions complement and align with the National Irrigation Policy (2017), targeting 50% of irrigation potential by 2040. It is estimated that implementation of these interventions will provide 1.6 MtCO₂e abatement by 2030 from avoided deforestation to create new farmland.

In support of this intervention 20 and 42 solar powered irrigation systems were provided under World Bank led Uganda Intergovernmental Fiscal Transfer Program (UgIFT) and GIZ, (SEFFA) respectively in 2023. Additionally, MAAIF and National Agriculture Advisory Services (NAADS) with support from the UgIFT installed four (4) sets of solar micro-irrigation at four beneficiary sites in Mityana, Nwoya, Terego and Nakaseke DLGs in 2024.

Name of action	Description	Objective	Type of instrument	Status of implementation	Sector (s) affected	Gases affected	Implementing entity	Start year of implementation
Rainwater harvesting and irrigation	Contributes to National Irrigation Policy (2017), targeting 50% of irrigation potential by 2040	Mitigate water shortages from drought hazards	Other	Implemented	AFOLU	CO ₂ , CH ₄ , N ₂ O	MAAIF, Districts (DP & DSD)	2023

Estimates of GHG emission reductions (Mt CO ₂ e) ³⁷		
Achieved		Expected
Year	Emissions reductions	

Green house cultivation of vegetables

This is the third sub-option for the climate smart agriculture strategic intervention in the REDD+ strategy. It is also aimed at reducing expansion of agricultural land into forested areas through intensification of agricultural production per unit area, mainly in crop production especially in rural/urban and sub-urban areas. It is estimated that implementation of this intervention will cumulatively result in 0.3 MtCO₂e abatement by 2030. In addition to the limited activity data on baseline values on greenhouse cultivation and

³⁷ Estimates of Achieved and Expected emissions reductions are “not estimated, NE” as actions are not direct mitigation measures in themselves. These measures incentivize farmers and boost all year-round production limiting deforestation.

no activity data from the Country Engagement Update Reports of the NDC-Partnership on this intervention was reported hence emissions reductions were not estimated (NE).

-Table 7-3: Greenhouse cultivation of vegetables

Name of action	Description	Objective	Type of instrument	Status of implementation	Sector (s) affected	Gases affected	Implementing entity	Implementation period
Greenhouse cultivation of Vegetables	Reduces emissions through soil carbon build-up and reduced deforestation	Reduce GHG emissions in rural/semi-urban households	Other	Planned	AFOLU	CO ₂ , CH ₄ , N ₂ O	MAAIF, Districts (DP & DSD)	2024-2026

Estimates of GHG emission reductions (Mt CO ₂ e) ³⁸	
Achieved	Expected
NE	NE

Sustainable Fuelwood and commercial charcoal production

Sustainable fuelwood and commercial charcoal production is the second strategic option under the REDD+ strategy aimed at addressing energy poverty in the context of climate change by promoting sustainable fuelwood and charcoal production in Uganda. It provides opportunities to reduce emissions while fostering significant sustainable development benefits through increasing access to clean energy. Also, this allows for diversification of livelihood options improves the resilience of households with other benefits on women's and children's daily workload (climate-care nexus). By ensuring sustained production of fuelwood and charcoal from sustainable sources, household energy supply is improved with health and nutritional impacts. Community woodlots and tree planting in farm forestry provide livelihood and income benefits in the form of food, fibre, energy and higher charcoal volumes with an improved pricing structure will allow better financial returns to the producers of charcoal. Moreover, tree planting contributes to landscape restoration in degraded areas.

It is estimated that implementation of these interventions will result in a cumulative emission reduction of 3.7 MtCO₂e by 2030. By 2025, implementation of these interventions contributed to an estimated total emissions removal of 0.2 MtCO₂e with 0.131 MtCO₂e from commercial small-holder and community bioenergy woodlots, and 0.068 MtCO₂e from commercial small-holder and community poles and timber

³⁸ Estimates of Achieved and Expected emissions reductions are "not estimated, NE" due to lack of specific activity data to facilitate computation of the emissions reductions

plantations (with coffee agroforestry) while emissions reductions from Improved charcoal kilns linked to bioenergy woodlots were estimated in the energy (stationary) sector.

The strategic option constitutes 3 sub-options as summarized below.

-Table 7-4: Commercial small-holder and community bioenergy woodlots

Name of action	Description	Objective	Type of instrument	Status of implementation	Sector (s) affected	Gases affected	Implementing entity	Start year of Implementation
Commercial small-holder and community bioenergy woodlots	Supports sustainable wood fuel production in tree-based bioenergy woodlots	Support sustainable wood fuel production	Other	Implemented	AFOLU	CO ₂ , CH ₄ , N ₂ O	MEMD, Districts (DP, DFS/DNR & DSD), Private Landowners	2023

Estimates of GHG emission reductions (Mt CO ₂ e)		
Achieved		Expected ³⁹
Year	Emissions reductions	
2021	NE	NE
2022	NE ⁴⁰	NE
2023	0.077	NE
2024	0.054	NE
2025	NE	NE
Total	0.131	NE

-Table 7-5: Commercial small-holder and community poles and timber plantations (with coffee agroforestry)

³⁹ Expected emissions reductions NE as in the updated NDC. Only a 2030 target expected emission reduction was available in the updated NDC (2.9 MtCO₂e).

⁴⁰ Achieved Emissions reductions only estimated for years with reported activity data and not estimated “NE” where there was no reported activity data.

Name of action	Description	Objective	Type of instrument	Status of implementation	Sector (s) affected	Gases affected	implementing entity	Start year of Implementation
Commercial small-holder and community poles & timber plantations with coffee agroforestry	Contributes to GHG reduction through woodlots and farm forestry practices	Reduce GHG emissions through sustainable agro-forestry practices	Other	Implemented	AFOLU	CO ₂ , CH ₄ , N ₂ O	MEMD, Districts (DP, DFS/DNR & DSD), Private Landowners	2024

Estimates of GHG emission reductions (Mt CO ₂ e)		
Achieved		Expected ⁴¹
Year	Emissions reductions	
2021	NE	NE
2022	NE	NE
2023	NE ⁴²	NE
2024	0.068	NE
2025	NE	NE
Total	0.068	NE

⁴¹ Expected emissions reductions NE as in the updated NDC. Only a 2030 target expected emission reduction was available in the updated NDC (0.37 MtCO₂e)

⁴² Achieved Emissions reductions only estimated for years with reported activity data and not estimated “NE” where there was no reported activity data

Table 7-6: Improved charcoal kilns linked to bioenergy woodlots-

Name of action	Description	Objective	Type of instrument	Status of implementation	Sector (s) affected	Gases affected	Implementing entity	Start year of implementation
Improve charcoal kilns	Aims to increase uptake of efficient charcoal production technologies by improving efficiency from 12% in 2020 to 75% by 2030	Balance supply and demand for biomass energy in alignment with Uganda's biomass energy strategy	Other	Implemented	AFOLU	CO ₂ , CH ₄ , N ₂ O	MEMD, Districts (DFS/DNR & DSD), Private Sector	2024-2026

Estimates of GHG emission reductions (Mt CO ₂ e) ⁴³	
Achieved	Expected
IE	NE

Large Scale Commercial Timber Plantations

Large scale commercial timber plantations intervention is the third strategic option in the REDD+ strategy. It is aimed at promoting responsible investments in large-scale commercial transmission pole and timber plantations thereby contributing to Inclusive Green Economy (IGE)⁴⁴ and is aligned with Uganda's green growth strategy. It is estimated that full implementation of these interventions will reduce Uganda's GHG

⁴³ Estimates of achieved and expected emissions reductions by 2025 were not reported as these emissions reductions were accounted for in the energy (stationary) sector and including the reductions here would result in double counting. However, the updated NDC estimates full implementation of this measure to reduce Uganda's emissions by 3.37 MtCO₂e by 2030

⁴⁴ https://redd.unfccc.int/media/final_-_uganda_redd_strategy_and_action_plan-october_2017.pdf

emissions by 10 MtCO₂e by 2030. However, due to difficulty in disaggregation of activity data to specific interventions, aggregate emissions reductions for this intervention and its constituent sub-options were estimated under the restoration of natural forests in the landscape strategic intervention.

The strategic intervention has 2 sub-options i.e commercial transmission pole and timber plantations, and commercial pole and sawlog plantations as described below.

Commercial transmission pole and timber plantations

This is a sub-option under the commercial scale timber plantations aimed reducing wood demand from natural forests by increasing sustainable production of wood products e.g fuelwood, charcoal, small poles, transmission poles and sawn timber. Implementation of this intervention is expected to reduce Uganda's GHG emissions by 5 MtCO₂e by 2030.

Table 7-7 provides a summary of commercial transmission pole and timber plantations and information on the emissions reductions from this intervention.

Table 7-8: Commercial transmission pole and timber plantations-

Name of action	Description	Objective	Type of instrument	Status of implementation	Sector (s) affected	Gases affected	Implementing entity	Start year of implementation
Commercial transmission pole and timber plantations	Reduces wood demand from natural forests	Provide sustainably sourced wood from plantations for poles and timber	Other	Implemented	AFOLU	CO ₂ , CH ₄ , N ₂ O	NFA, Districts (DFS/DNR & DSD), Private Landowners	2023

Estimates of GHG emission reductions (Mt CO₂e)⁴⁵

Achieved	Expected
IE	NE

Commercial pole and sawlog plantations

Similarly, this intervention is also aimed at reducing wood demand from natural forests by increasing supply of sustainable sourced wood products including small pieces of wood as small poles, fuelwood or charcoal. By implementing this intervention, co-benefits including improvement of rural livelihoods by creating employment through fuelwood, charcoal, pole and sawn timber production business is expected. Additionally, this will reduce erosion on large areas and support biodiversity restoration. It is estimated that implementation of this intervention will reduce Uganda's GHG emission by 5 MtCO₂e by 2030.

Table 7-9 summarizes the commercial pole and sawlog plantations.

⁴⁵ Estimates of achieved and expected emissions reductions by 2025 were not reported as these emissions reductions under the restoration of natural forests in the landscape intervention due to difficulty in disaggregating activity data and including the reductions here would result in double counting.

Table 7-9: Commercial pole and sawlog plantations

Name of action	Description	Objective	Type of instrument	Status of implementation	Sector (s) affected	Gases affected	Implementing entity	Implementation period
Commercial pole and sawlog plantations	Enhances sustainable production of wood products	Enhance sustainable wood product production (poles, fuelwood, charcoal)	Other	Implemented	AFOLU, Energy	CO ₂ , CH ₄ , N ₂ O	NFA, Districts (DFS/DNR & DSD), Private Land Owners	2024-2026

Estimates of GHG emission reductions (Mt CO ₂ e) ⁴⁶	
Achieved	Expected
IE	IE

Restoration of natural forests in the landscape

This is the fourth strategic intervention in the REDD+ strategy aimed at restoring and/or rehabilitating natural forests within the context of climate-smart landscapes thus enhancing the carbon sequestration capacity and the delivery of other forest-related goods and services. By implementing this strategic intervention, other co-benefits will be realized such as enhanced support for biodiversity, mitigating natural disasters, and reduction of soil erosion. The strategic option constitutes 2 sub-options i.e. natural forest regeneration and restoration of degraded protected natural forest with an estimated mitigation potential of 0.6 MtCO₂e by 2030.

The implementation of this intervention removed approximately 1.68 MtCO₂e from Uganda's atmosphere between 2021 and 2024. However, due to difficulty in disaggregation of activity data to specific interventions, these removals also constitute removals from the large-scale commercial timber plantations intervention with its sub-options. Therefore, improved monitoring including ground truthing of activity data will be required to improve accuracy and allocation of activity data to associated removals for these interventions.

Error! Reference source not found.11 summarizes the natural forest regeneration sub-option with its respective emissions reductions while table 7.13 summarizes emissions reductions from restoration of degraded protected natural forest.

-Table 7-10: Natural Forest regeneration

⁴⁶ Estimates of achieved and expected emissions reductions by 2025 were not reported as these emissions reductions under the restoration of natural forests in the landscape intervention due to difficulty in disaggregating activity data and including the reductions here would result in double counting.

Name of action	Description	Objective	Type of instrument	Status of implementation	Sector (s) affected	Gases affected	Implementing entity	Start year of implementation
Restoration of degraded protected forests	Restores natural forests in reserves and protected areas	Restore forests through indigenous tree species planting >100,000 ha restored	Other	Implemented	AFOLU	CO2	NFA, UWA, Districts (DFS/DNR & DSD), Private large landowners	2021

Name of action	Description	Objective	Type of instrument	Status of implementation	Sector (s) affected	Gases affected	Implementing entity	Start year of implementation
Natural forest regeneration	Aims to rehabilitate 100,000 hectares of forest areas	Rehabilitate forests through collaborative management	Other	Implemented	AFOLU	CO ₂	NFA, UWA, Districts (DFS/DNR & DSD), Private large landowners	2021

-Table 7.11 Summarizes natural forest regeneration natural forests sup-option with respective emissions reductions.

Table 7-11:Natural forest regeneration-

Estimates of GHG emission reductions (Mt CO ₂ e) ⁴⁷	
Achieved	Expected
IE	IE

Table 7-1212: Restoration of degraded protected natural forest

⁴⁷ Estimates of achieved and expected emissions reductions by 2025 were not reported as these emissions reductions under the restoration of degraded natural forests in the landscape intervention due to difficulty in disaggregating activity data and including the reductions here would result in double counting.

Estimates of GHG emission reductions (Mt CO2e)		
Achieved		Expected
Year	Emissions reductions	
2021	1.05	NE
2022	0.50	NE
2023	0.04	NE
2024	0.09	NE

2025	NE ⁴⁸	NE
Total	1.68	NE

Energy efficiency

The energy efficiency strategic intervention in the REDD+ strategy aims at improving and addressing traditional cooking methods. It constitutes measures to increase energy efficient fuelwood and charcoal stoves and complements the Uganda biomass energy strategy, the renewable energy policies (2007 and

⁴⁸ Achieved Emissions reductions only estimated for years with reported activity data and not estimated “NE” where there was no reported activity data

2023) and the energy transition plan⁴⁹ by promoting clean cooking solutions and energy efficient biomass use. By implementing this intervention, Uganda benefits from reduced energy poverty by increasing access to clean, safe and efficient energy with associated health benefits from reduced indoor air pollution. Additionally, gender related benefits from reduced care⁵⁰ needs allowing for more time for productive activities. Full implementation of this intervention is estimated to reduce Uganda's emissions by 6.8 MtCO₂e by 2030. **Error! Reference source not found.**7.13 summarizes energy efficient wood and charcoal stoves intervention.

Table 7-1313: Energy efficient fuelwood and charcoal stoves

Name of action	Description	Objective	Type of instrument	Status of implementation	Sector (s) affected	Gases affected	Implementing entity	Implementation period
Energy Efficient fuelwood and charcoal stoves	Promotes clean cooking solutions and efficient technologies	Promote efficient biomass energy use among households and institutions	Other	Implemented	AFOLU, Energy	CH ₄ , N ₂ O, CO ₂	MEMD, FSSD, Districts (DFS/DNR & DSD)	2021-2026

Estimates of GHG emission reductions (Mt CO ₂ e) ⁵¹	
Achieved	Expected
IE	NE

Livestock management

This is the 7th strategic intervention in the REDD+ strategy aimed at promoting sustainable livestock management especially in the cattle corridor. It constitutes interventions on promoting improved cattle breeds, feeds, and improvement of water availability through constructing water dams and valley dams. Additionally, other measures include establishment of fodder agroforestry plantations for grazing and stall feeding. By implementing these measures, livestock productivity is increased from improved breeds, diversification of products for example from fodder agroforestry provides additional sources of income and

⁴⁹ *Uganda Energy Transition Plan*

⁵⁰ *The climate-care nexus: Addressing the linkages between climate change and women's and girls' unpaid care, domestic, and communal work | Publications | UN Women – Headquarters*

⁵¹ Estimates of the achieved and expected emissions reductions are accounted for under this intervention and are included in the energy (stationary) sector hence the notation key "IE" has been used as accounting for these emissions reductions from the intervention here would lead to double counting as emissions from energy use would already account for these emissions reductions.

other ecosystem services such as improvement in soil fertility and increased carbon pool. It is estimated that implementation of these interventions will result in reduction of Uganda's emissions by 2.9 Mt CO₂e by 2030.

Whereas implementation of interventions under this strategic intervention are still in the nascent stages, government of Uganda through MWE and Conservation International initiated the **Fostering Water Security and Catchment Resilience in Uganda's Cattle Corridor (LDCF Uganda Cattle Corridor)** project to promote sustainable and climate-resilient land and water management in the Cattle Corridor of Uganda with support from the Global Environment Facility (GEF) through the Least Developed Countries Fund (LDCF), with a grant of USD 7,598,165 and a projected co-finance of USD 54,774,000.⁵²

Table 7.14 summarizes the livestock management intervention in the cattle corridor.

-Table 7-1414: Livestock management in the cattle corridor

Name of action	Description	Objective	Type of instrument	Status of implementation	Sector (s) affected	Gases affected	Implementing entity	Start year of implementation
Livestock management in cattle corridor	Promote improved cattle breeds and feeds, and water availability for livestock	Improve livestock breeding, feeding, and management through constructing water dams, valley tanks, and establish fodder agroforestry plantations for zero grazing and stall-feeding	Other	Implemented	AFOLU, Energy	CH ₄ , N ₂ O, CO ₂	MAAIF/DAR, NGBC, districts Commercial livestock farmers	2025

Estimates of GHG emission reductions (Mt CO ₂ e) ⁵³	
Achieved	Expected
NE	NE

⁵² *Fostering Water Security and Catchment Resilience in Uganda's Cattle Corridor (LDCF Uganda Cattle Corridor)* - Africa Innovations Institute

⁵³ Estimates of the achieved and expected emissions reductions are not estimated due to data limitations.

Wetlands and Peatland management

This measure aims to increase wetland coverage from 8.9% in 2020 to 9.57 % in 2025 and approximately 12 % by 2030. It constitutes wetland management practices such as demarcation, gazettement, and restoration of degraded wetlands. By implementing these practices ecological and socioeconomic functions of wetlands will be enhanced such as flood control, wildlife habitats and biological diversity protection, improvement in carbon storage, and sustaining livelihoods including artisanal fisheries, traditional crafts and water sources for domestic use ⁵⁴. It is estimated that implementation of these interventions would reduce Uganda's GHG emissions by 0.4 Mt CO₂e. Wetlands coverage has increased from 8.9% in 2020 to 9.32% by the end 2024.

Activity data disaggregation, for example with specification on form of wetland restoration and wetland type limited estimation of emissions reductions from these interventions. Targeted surveys and studies to estimate mitigation potential of these interventions including establishing Uganda specific wetland emissions factors will enhance tracking of mitigation benefits of this intervention.

-Table 7-1515: Wetlands and Peatland management

Name of action	Description	Objective	Type of instrument	Status of implementation	Sector (s) affected	Gases affected	Implementing entity	Implementation period
Wetland and Peatland management	Aims to increase wetland coverage from 8.9% (2020) to 9.57% (2025) to 12% (2030)	Implement wetland management practices	Other	Adopted	AFOLU, Ecosystem	CH ₄ , N ₂ O, CO ₂	MWE(WMD); NEMA; MoLG; Dev't Partners; NGOs	2021-2026

Estimates of GHG emission reductions (Mt CO ₂ e) ⁵⁵	
Achieved	Expected
NE	NE

Overall, despite challenges in reporting and availability of activity data for estimation of emissions reductions in AFOLU, early registered achieved emissions reductions include 1.89 MtCO₂e total emissions reductions from AFOLU (~6% of the 2030 sectoral emission reduction target). Natural forest restoration contributed 1.68 MtCO₂e, commercial small holder and community bioenergy woodlots contributed 0.13 MtCO₂e, and community poles and timber plantations with coffee agroforestry contributed 0.068 MtCO₂e to the total estimated achieved emissions reductions.

⁵⁴ [national wetland policies - uganda.pdf](#)

⁵⁵ Estimates of the achieved and expected emissions reductions are not estimated due to data limitations.

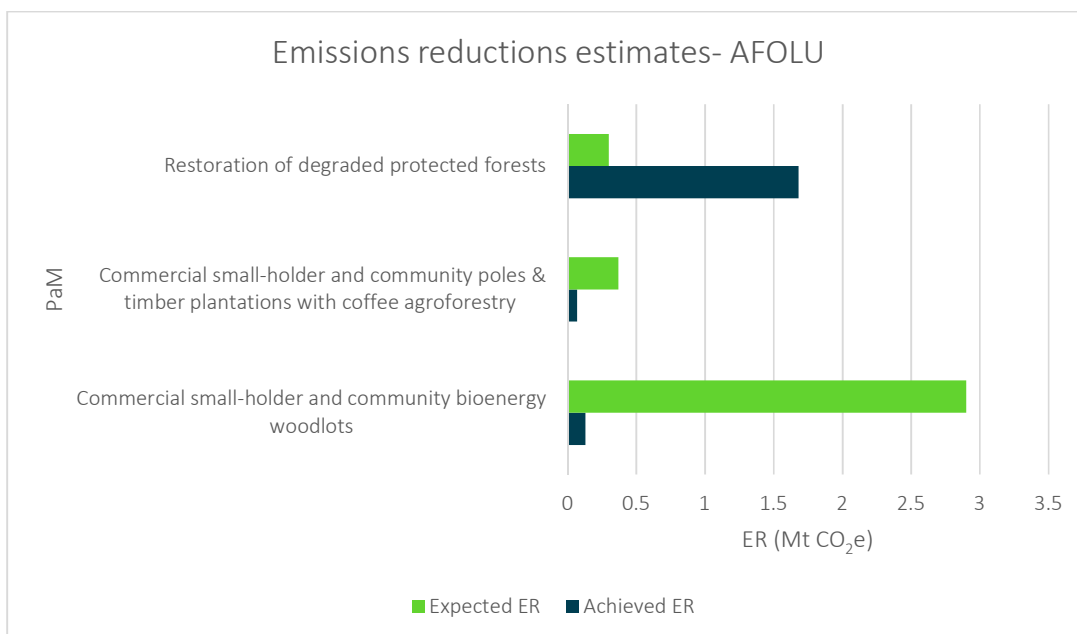


Figure 7-1 Some of the estimated GHG emission reductions in AFOLU by 2025

7.3.2 Energy Sector

Stationary Energy

Renewable energy generation has achieved substantial progress during the NDC implementation period. Net renewable electricity from new hydro and solar PV plants increased to 2.33 million MWh/yr by end of 2024, resulting in a reduction of 0.350MtCO₂e. This already corresponds to ≈72.8% of the 2030 target (0.480 MtCO₂e), making renewable power the dominant contributor to GHG removals within stationary energy. The results highlight that Uganda’s renewable roll-out is largely on track, though earlier modelling underestimated the pace and scale of deployment.

Challenges: No challenge experienced in obtaining data for this mitigation action, as data was readily available in the right format and over the course the entire implementation period⁵⁶.

Transmission and distribution (T&D) losses reduction shows mixed results. By 2025, distribution losses had fallen by 3.4%, while transmission losses worsened slightly (–1%). The combined effect is reduction of 0.000218 MtCO₂e by 2024. Compared with the very small NDC target of 0.00001 MtCO₂e by 2030, this corresponds to an apparent ≈2175% progress. This underscores a significant mismatch between baseline assumptions and observed performance, particularly since the current estimates are based on actual electricity generation and distribution data. While the absolute emissions impact is small, the relative progress suggests that Uganda has already surpassed the modelled NDC target for this intervention.

Challenges: No major data availability challenges were encountered. Required figures were readily obtained from UMEME annual reports and statistical abstracts, combined with energy transmission and distribution values from MEMD. However, results fluctuated across years, with some periods showing an increase in

⁵⁶ [Generation_Statistics_Q1_2025_20250706.xlsm](#)

emissions rather than reductions. This suggests that factors such as grid expansion, system efficiency improvements, and operational conditions introduce variability into outcomes.

Mitigation Energy: Mirama-Kabale Transmission Project (1409), Source: MoFPED, 2025



Commissioned Kabale substation and switchyard that was completed

Completed Kaale-Mirama Transmission line, Rubanda District

Figure 7-2 Construction of transmission lines

Improved charcoal production has lagged. By 2025, about 102,600 tonnes of charcoal were produced using an estimate of about 190 pilot Casamance improved kilns, yielding 0.0166 MtCO₂e reductions by 2024. The Casamance kilns have an estimated efficiency of 30% compared to the baseline efficiency of 10%. Against a target of 0.48 MtCO₂e by 2030, this intervention resulted in about 3.5% progress. The contribution remains modest and below expectations, largely because this is still at pilot stage and a full-scale roll-out will be required if significant progress is to be registered.

Challenges: No evidence of wider-scale adoption of improved charcoal kilns could be obtained. Estimates are based solely on pilot installations, and no additional rollout data was available, hence the poor progress.

Industrial efficiency and fuel switching: Progress could not be quantified due to lack of activity data. No information was available from the Ministry of Trade and Industry on the status of implementation, and preliminary research did not yield evidence of progress on clinker substitution in the cement industry.

Challenges: The absence of monitoring and reporting frameworks for this sub-sector highlights persistent gaps in Uganda’s MRV systems. Stronger follow-up will be needed under NDC 3.0 to confirm whether these measures are being implemented and to ensure that data is captured systematically

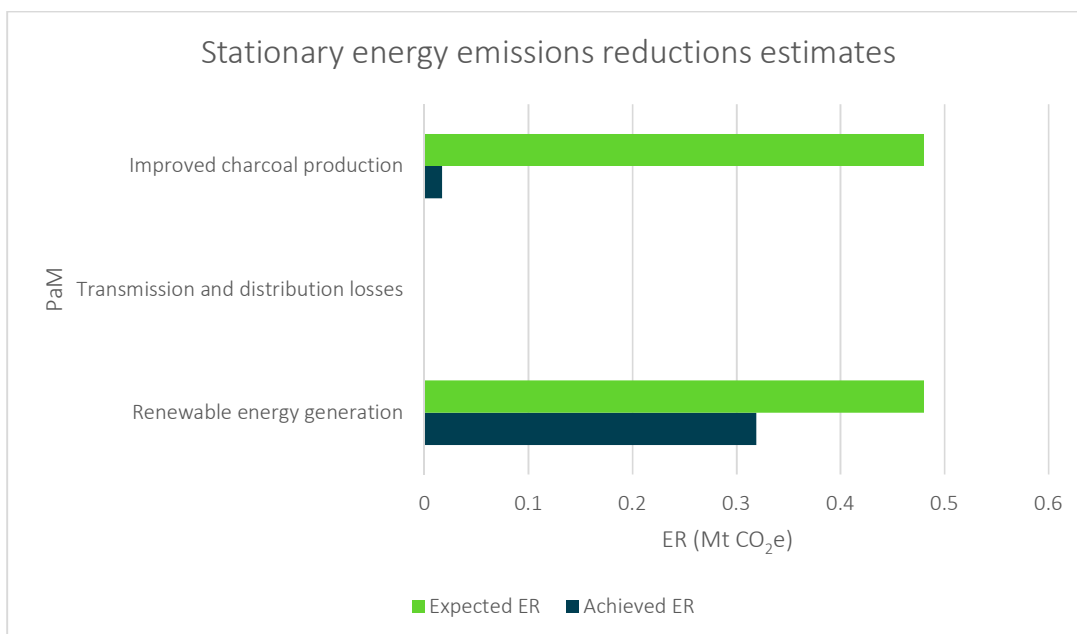
The **Overall progress for the stationary energy** target stands at 15% of the 2030 target, amounting to 0.35 Mt CO₂e of the overall NDC 2030 target of 2.34 MtCO₂e for the subsector.

-Table 7-16: Stationary Energy Progress Table

Mitigation policy or measure	Indicator used (units)	2020 value	2025/ latest value	Estimated Impact (MtCO ₂ e)	Progress (%) of 2030 target	Data Source and Assumptions
Overall Stationary Energy Progress (target =2.34 MtCO ₂ e)				0.35	14.9%	
Energy Generation (MW/yr)				0.35	72.8%	
Renewable energy generation	Net renewable energy from new non-carbon standard projects (MWh/yr)	0	2331747	0.32	100%	Source: MEMD Assumptions: CDM approved GEF for Uganda used and the actual generation data
Reduction in transmission and distribution losses	% annua; reduction in distribution and transmission losses (%/yr)	2% distribution, -1% transmission	3.4% distribution, -1% transmission	0.0002	100%	Source: UMEME, UETCL and MEMD ERs obtained from Total energy efficiency improvements over the implementation period x the baseline grid emission factor (0.137 tCO ₂ e/MWh)
Improved efficiency of charcoal production NE	Tonnes of Charcoal produced (tonnes/yr)	0	102600	0.0166	3.46%	Source: FAO, NDC partnerships reports ERs based on 190 pilot Casamance kilns with 30% efficiency compared to 10% baseline
	Share of Charcoal produced in improved kilns (% or dimensionless (proportion))	0	0.2%			
Energy Use MWh/yr				NE	0%	
Industrial Energy efficiency	Energy saved from energy efficiency interventions (GWh/yr)	NA	NA	NE	NE	NA (MEMD contacted)
Industrial Fuel switching	Quantity of fossil fuel substituted (by type) may add more rows (litres/yr)	NA	NA	NE	NE	NA (MEMD contacted)

Increased electricity access for households	Electricity consumed for cooking (kwh/yr)	NA	NA	NE	NE	NA (MEMD contacted)
Lighting energy efficiency in households	Number of efficient bulbs distributed and in use (units/yr)	NA	NA	NE	NE	NA (MEMD contacted)
Cooking mitigation measures	Baseline non-renewable biomass consumption (tonnes/yr)	NA	NA	NE	NE	NA (MEMD contacted)
Commercial/Institutional Energy Use				NE	0%	
Improved cookstove efficiency	Increase in average thermal efficiency of end-use appliances or systems (% or GJ input/GJ output)	NA	NA	NE	NE	NA (MEMD contacted)
Switching to cleaner fuels	Baseline fuel replaced by low-emission alternatives (litres/yr or GJ/ or m3)	NA	NA	NE	NE	NA (MEMD contacted)

In summary, as indicated in the Table 7.15 above, Stationary energy contributes a significant share of Uganda's mitigation potential, at roughly one-third of AFOLU's expected contribution, making it the second most important sector in absolute terms. Importantly, it is also the sector with the highest volume of measurable reductions to date, largely driven by renewable energy generation.



Transport Energy

Transport remains Uganda's second most ambitious mitigation sector, but one where progress is hardest to track. Measures such as fuel efficiency improvements are expected to yield 1.86 MtCO₂e of reductions each year by 2030, yet there are no measured results to date. Similarly, alternative fuel switching is targeted at 0.54 MtCO₂e per year and non-motorized transport (NMT) at 0.66 MtCO₂e annually, but again no data is available to assess actual implementation.

A significant number of infrastructure projects have moved forward. For instance, the Greater Kampala Metropolitan Area (GKMA) now has 29 km of operational rail, while the Meter Gauge Railway (MGR) rehabilitation has delivered 400 km of completed track, of which 140km are already operational. Despite these advances, there are no quantified annual emission reductions because the necessary activity data and emission factors are missing.

Other PAMs: Fuel efficiency (~1.86 MtCO₂e), fuel switching (~0.54 MtCO₂e), and NMT (~0.66 MtCO₂e) remain in planning or early implementation stages with no reported progress values.

Challenges:

Tracking progress in the transport sector faces several constraints. First, activity data on fuel consumption by vehicle type, modal share, and passenger- and freight-kilometres is unavailable, preventing quantification of emission reductions from fuel efficiency, switching to alternative fuels, and NMT interventions. Second, while infrastructure investments such as rail rehabilitation and BRT corridors are progressing, they lack a robust MRV systems to translate operational outputs into GHG outcomes. Third, institutional coordination between transport planning authorities, city councils, and CCD remains weak, limiting integration of climate metrics into transport planning. Finally, financing for large-scale modal shifts, particularly electric mobility and low-carbon freight, remains insufficient, leaving many planned policies and measures at concept or pilot stage.

Table 7-1617: Transport Progress Table

Mitigation policy or measure	Indicator used (units)	2020 value	2025/ latest value	Estimated Impact (MtCO ₂ e)	Progress (%) of 2030 target	Data Source and Assumptions
Road transport fuel efficiency	cummulative number of cleaner light-duty vehicles (count)	NA	NA	NE	NE	NA
Alternative fuel switch	Volume of alternative cleaner fuel 1 (litres)	NA	NA	NE	NE	NA
	Volume of alternative cleaner fuel 2 (litres)	NA	NA	NE	NE	NA
	Volume of alternative cleaner fuel 3 (litres)	NA	NA	NE	NE	NA
Development of Non-Motorized Transport (NMT) infrastructure	Total length of NMT infrastructure completed (km)	1.95	5.45	NE	NE	Source: KCCA Report No estimations as requisite emission factors and activity data not availed
MGR – Meter Gauge Railway rehabilitation for freight transit	Total length of MGR completed (km)	0	400	NE	NE	Source: URC Assumptions: No ER estimations as requisite emission factors and activity data not available
Efficient operation of public transportation	Improvement in fuel consumption resulting from improvement in operation	NA	NA	NE	NE	NA
Residential trip avoidance through town planning and transit-oriented development (TOD)	Residential trips avoided per person km (pkm)	NA	NA	NE	NE	NA
BRT-Bus Rapid Transit	Kilometers of BRT corridor completed (km)	0	0	0	0%	Source: MoWT
GKMA passenger service	Kilometers of rail in GKMA completed & operational (km)	0	29	NE	NE	Source: URC ERs not estimated as requisite EFs and

						activity data not available
Metro rail	Length of Metro Rail	0	0	0	0%	Source: KCCA, URC
LRT - Light Rail Transit	CO2e emissions saved per tonne-kilometer (tkm) of freight moved by light rail	NE	0	0	0%	Source: MoWT ERs not estimated as requisite EFs and activity data not available
SGR - Standard Gauge Railway	Length of SGR network constructed	0	0	0	0	Source: MoWT
Electric road vehicles	NE	NE	NE	NE	NE	NA
Electrification of rail	NE	NE	NE	NE	NE	NA
Road improvement projects	NE	NE	NE	NE	NE	NA
Oil pipeline	NE	NE	NE	NE	NE	NA
Utilisation of water transport	NE	NE	NE	NE	NE	NA
E-commuting	NE	NE	NE	NE	NE	NA

The above results illustrate a persistent challenge; while the transport sector carries a mitigation potential of over 3 MtCO₂e annually under NDC 2.0, none of it is being tracked. Without robust MRV systems, Uganda risks having its most high-impact sectors remaining invisible in official progress assessments.

7.3.3 Waste

The waste sector provides one of the clearest cases of measurable progress, though its overall contribution to Uganda's NDC remains relatively small. By 2025, wastewater treatment at the Nakivubo WWTP has captured an estimated 164,927 m³ of methane annually, equivalent to 0.003 MtCO₂e per year. This represents about 18% of the 2030 target of 1.1 MtCO₂e for wastewater methane recovery.

Solid waste composting across 11 towns and cities currently diverts around 148,658 tonnes of organic waste from landfill each year, generating approximately 0.194 MtCO₂e in annual reductions. Composting remains one of the sector's most reliable interventions; however, sustainability of these sites depends on adequate financing and operational support.

School bio-latrines, designed to capture biogas from human waste for cooking, are expected to contribute an additional 0.0006 MtCO₂e annually by 2030, but no recent activity data is available to confirm actual reductions. Similarly, information on agro-waste management (e.g. sugarcane residues) is missing, making it difficult to quantify associated mitigation benefits.

In total, the waste sector delivers around 0.197 MtCO₂e emissions reductions by 2025, equivalent to ≈18% of its 2030 target. This makes waste one of the best-monitored sectors; however, because it accounts for less than 2% of Uganda’s overall mitigation potential, its broader significance is limited.

Waste management(and adaptation solutions for water and sanitation) , Source: MoFPED & OPM-DU, 2024



Casting of concrete for Nalukolongo faecal sludge’s anaerobic pond side walls;

Figure 7-3 Waste management solutions

Challenges:

The waste sector faces persistent constraints despite early progress. Composting plants, once sustained by CER revenues from CDM projects, have struggled financially since the end of crediting in 2019, leaving many underfunded and operating far below capacity and facing a risk of closure. MRV systems remain weak, with fragmented or inconsistent data, while innovative solutions such as bio-latrines (Green Schools NAMA) have not moved beyond pilot stage, thus greatly impacting the overall progress registered under the waste sector.

Table 7-1718: Waste Progress Table

Mitigation policy or measure	Indicator used (units)	2020 value	2025/latest value	Estimated Impact (MtCO ₂ e)	Progress (%) of 2030 target	Data Source and Assumptions
Waste Sector -Main PAMs				0.20	17.9%	
Efficient Solid waste and wastewater management in Planned Green Cities	Volume of methane recovered from wastewater treatment and used for energy or heat generation (m ³)	0	164,927	0.003	17.9%	Source: NWSC-Nakivubo WWTP Assumptions: IPCC default methane EF used, GWP of 28 (AR 5 value). Density of methane at 20C and 1atm as 0.656 kg/m ³

	Tonnes of solid waste diverted from landfills or SWDS and treated aerobically in Solid Waste Management Sites (tonnes)	0	148,658	0.19		Source: NEMA Assumptions: NEMA value for 1 small plant obtained (Kabale) and multiplied by 11 assuming other plants comparable size. The value is further multiplied by a factor of 0.80 to cater for uncertainty
NAMA - Schools bio-latrines	Volume of biogas utilized for cooking (m3/yr)	NA	NA	NE		NA (contacted MEMD)
Additional Actions						
Sugarcane waste management	Tonnes of bagasse used for energy generation	NA	NA	NE	NE	NA (contacted USMA)

7.3.4 Industrial Processes and Product Use (IPPU)

Mitigation in the industrial processes and product use (IPPU) sector is still largely conceptual. The clinker substitution programme in cement production, aimed at reducing the share of clinker with alternative materials, is projected to deliver 0.1 MtCO₂e annually by 2030. However, as of 2025, no results have been reported. Other interventions, such as energy efficiency in steel production or the adoption of lower-emission fuels, remain in planning and pilot stages with no measurable emission reductions. This leaves IPPU as a sector with modest but real annual potential, yet one where progress cannot be quantified due to data gaps.

Challenges: No data was available from the Ministry of Industry, Trade and Cooperatives (MTIC) on the status of implementation of this mitigation action. A preliminary search did not yield any results to show any progress on clinker substitution within the cement industry. This action will therefore need to follow up to ensure it's fully implemented under NDC 3.0.

Table 7-1819: IPPU Progress Table

Mitigation policy or measure	Indicator used (units)	2020 value	2025/latest value	Estimated Impact (MtCO ₂ e)	Progress (%) of 2030 target	Data source and assumptions
Clinker substitution in	Clinker substitution rate [Change in clinker factor (total clinker	NA	NA	NE	NE	NA

cement production	used in tons/total cement produced in tons) as a result of substitution of clinker with low-carbon alternatives (fly ash, slag, etc.)) (% or dimensionless factor)					
	Total cement produced (Tonnes/yr)	NA	NA	NE	NE	NA
Circular economy management of refrigerants	Amount of refrigerants recovered/destroyed (tonnes/yr)	Not Implemented		0	0%	NA

7.3.5 Conclusion

While Uganda has made some measurable progress in selected areas—most notably in AFOLU interventions such as early forest restoration and in waste management through composting and methane recovery—the overall trajectory of NDC implementation is largely not on track to achieve the 2030 mitigation targets. Progress in high-impact sectors such as transport, IPPU, and industrial energy efficiency remains minimal or unquantified, and in several cases implementation has yet to move beyond planning or pilot stages.

A critical challenge is the persistent lack of timely and complete data across sectors. The most recent GHG inventory only covers up to 2019, leaving a major gap in assessing progress during the current NDC implementation period. Without updated inventories, reliable sectoral activity data, and systematic MRV, it is not possible to generate a full and accurate picture of Uganda’s progress toward its NDC targets.

In sum, while some interventions demonstrate promising early results, the combination of slow implementation and incomplete data indicates that Uganda’s NDC is not currently on track, and urgent action is needed to accelerate delivery, strengthen MRV systems, and close data gaps in order to credibly assess and achieve the 2030 commitments.

8 Tracking Progress on Adaptation

Uganda's adaptation progress under NDC 2.0 is uneven: forestry and health have already met or exceeded 2025 targets, and sectors like water, sanitation, transport, energy, and agriculture show moderate progress though often short of their goals. By contrast, cities, fisheries, tourism, education, and manufacturing remain far behind, largely due to weak data and baselines.

8.1 Overview of Uganda's adaptation goals and priority areas

The priority sectors for adaptation in the updated NDC include; Environment and Ecosystems, Water and Sanitation, Agriculture, Forestry, Fisheries, Energy, Transport, Manufacturing, Industrial Processes and Mining, Cities and the Built Environment, Tourism, Education, Health and Disaster Risk Reduction. Thirteen (13) pillars were identified to transform Uganda into a climate- resilient and low-carbon society by 2050 that is prosperous and inclusive.

1. Promote climate-resilient and low-carbon agricultural development
2. Promote and apply land management practices that support sustainable and productive use
3. Promote climate-resilient water supply systems, increase water supply capacity and use efficiency.
4. Promote sustainable management of ecosystems and the use of nature-based solutions, including through community engagement
5. Strengthen climate information services through improved data collection and sharing infrastructure
6. Develop and promote a clean and resilient energy system
7. Promote climate resilient and low-carbon urban planning and development
8. Increase availability and promote access to finance for climate-resilience, low-carbon investments and climate impacts recovery
9. Promote education and training on climate change science, and sharing of indigenous knowledge
10. Mainstream and institutionalize climate change responses in policies, plans, programmes and budgets at all levels of governance
11. Promote a multimodal shift to low carbon mobility and create climate-resilient transport infrastructure to support economic growth
12. Promote a resource-efficient circular economy
13. Further develop a resilient and “fit for future” health systems

8.2 Progress against planned actions

Environment and Ecosystems

The Environment and ecosystem sector has four adaptation actions centered around safeguarding natural capital base and enhancing ecosystem resilience. Ten indicators were identified for tracking the progress of these actions with five of them having baseline data. Present information was provided on three (60%) indicators that were used for tracking their progress.

Wetlands coverage has increased from the baseline value of 8.9% to 9.32% by the end of 2024/2025 compared to the NDC target of 9.57% by 2025. This represents a percentage progress of 62.7% during the NDC implementation period. The area of wetlands restored has increased from the baseline value of 16,906ha to 35,000ha compared to the NDC target of 55,906ha by 2025. This represents a percentage

progress of 46.3% during the NDC implementation period. The other indicator with present information is KM of wetland boundaries demarcated that has increased from the baseline value of 700km to 2100km compared to the NDC target of 4100km by 2025. This represents a percentage progress of only 41.6%. The progress made in the wetlands subsector is attributed to the implementation of the GCF wetlands restoration project in the ministry of water and environment. The overall progress of the sector based on the assessed indicators was 50.1% during this 5-year NDC implementation period, representing a near average progress for the sector.

The indicators of area degraded riverbanks and lakeshores restored and maintained, area of degraded hilly and mountainous areas restored and level of implementation (%) of the Rangeland Management and Pastoralist Policy had baseline values but no present data was obtained. The other five indicators did not have both baseline data and present information. Some indicators such as area of Peatlands restored, level of implementation of climate actions in National Biodiversity Strategy and Action Plan II & III, share of biodiversity contribution to national wealth and number of people benefiting from Ecosystem-Based Adaptation (EbA) interventions/ projects baselines were to be determined by the respective sectors during the NDC2.0 implementation period, but no information on the baseline was determined from the documents we reviewed.

Water and Sanitation

The water and sanitation sector has four adaptation actions attuned mainly to water resource and wastewater management for resilience building in the sector. Thirteen indicators were identified for tracking the progress of these actions with twelve of them having baseline data. Present information was provided on eight (66.6%) indicators that were used for tracking their progress.

National (rural) water access increased from the baseline value of 68% to 69% by the end 2024/2025 compared to the NDC target of 76% by 2025; representing a percentage progress of 12.5% in the NDC implementation period. On the other hand, National (urban) water access increased from the baseline value of 71% to 72.8% compared to the NDC target of 90% by 2025; representing a percentage progress of 8.5%. Water for Production storage capacity increased from the baseline value of 42 MCM to 52.6MC compared to the NDC target of 62 MCM by 2025; representing a percentage progress of 72.7%. In terms of sanitation infrastructure, Population with access to basic sanitation increased from a baseline value of 18% to 43.6% compared to the NDC target of 25% by 2025; representing a percentage progress of 365%. However, data received on sewer service coverage, the other indicator for sanitation, shows that it has remained at 23% as the baseline value in the NDC implementation period compared to 31% that was NDC target for the year 2025. Compliance with national water standards increased from the baseline value of 61% to 80% compared to the NDC target of 70% by 2025; representing a percentage progress over 100% from the target. Water permit compliance for surface water abstraction increased from the baseline value of 76% to 78.6% compared to the NDC target of 82% by 2025; representing a percentage progress of 43.3%. Water permit compliance for ground water abstraction increased from the baseline value of 76% to 78.7% compared to the NDC target of 81% by 2025; representing a percentage progress of 54% during the NDC implementation period. Catchment management plans increased from the baseline value of 17 to 19 compared to the NDC target of 21; representing a 50% progress. The overall progress of the sector based on the assessed indicators was 52.7% during this 5-year NDC implementation period.

It was not possible to get data on the number of Solar/wind powered water supply systems constructed in NDC implementation period and the indicator of the number of rainwater harvesting tanks constructed didn't have baseline data nor present information.

Transport

The transport sector has two NDC adaptation actions that capture mainstreaming climate change in multimodal transport policy, legislation, and infrastructure development. Five indicators were identified for tracking the progress of these actions with four of them having baseline data. Present information was provided on all five (100%) indicators that were used for tracking their progress.

The Km of paved national roads increased from a baseline value 4971km to 6312km by the end 2024/2025 compared to the NDC target of 7600km by 2025. This represents a percentage progress of 51%. The Km of urban paved roads increased from the baseline value of 1253km to 1836km by the end of 2024 compared to the NDC target of 1848km by 2025; representing a progress of 97.9%. This significant increase in paved urban roads is attributed to the Uganda Cities and Municipalities Infrastructure Development program that is supported by the World Bank. The Permanent railway road has increased from the baseline value of 262km to 391km by the end of 2024 compared to the NDC target of 482km by 2025; representing 58.6% progress. The Proportion of national road reserves with green infrastructure has increased from a baseline value of 1% to 24% by the end 2024 compared to the NDC target of 30% by 2025; representing a progress of 80%.

The other indicator was Number of revised transport regulations, guidelines and design codes integrating climate risks which had no baseline information. However, information received from the Ministry of works and transport shows that 6 transport regulations, guidelines, and design codes integrating climate risks have been revised between 2021 and 2025. These included; the National Transport and Logistic Policy, 2021, the National Integrated Transport Master Plan 2021-2040, the General Specification for Roads and Bridges, 2025 (Final Stages), the Urban Road Manual, 2022 and the MoWT Strategic Plan 2020/2021-2024/25.

The overall progress of the sector based on the assessed indicators was 71.9% during this 5 year NDC implementation period representing a very good progress for the sector.

Adaptation Transport: Kampala Flyover and Road Upgrading Project (1319), Source: MoFPED, 2025



Improved Clock Tower Junction with pedestrian Bridge & Flyover



Improved Nsambya junction with the Underpass in the Background

Figure 8-1 Examples of improved solutions in the transport sector for non-motorized transport and pedestrian safety

Energy

The energy sector has a total of four NDC priority adaptation actions that mainly prioritized access and utilization of electricity from sustainable resources and energy efficiency. 8 indicators were provided in the

NDC, and they all have baseline data. Results from the stock take show that seven (87.5%) had current information for tracking their progress with only one indicator (Renewable off-grid energy solutions) missing current information data for its tracking.

Of the indicators that were tracked, the proportion of the population with access to electricity increased from the baseline value 24% to 58% by end of 2024/2025 compared to the NDC target of 60% by 2025. This represents 94.4% progress of this indicator. Electricity generation capacity increased from the baseline value 1276.2MW to 2047MW compared to the NDC target of 3500MW; representing a 34.7% progress in the implementation period. Per capita electricity consumption increased from the baseline value 100kwh to 218kwh compared to the NDC target of 145kwh; representing more than 100% progress in the implementation period. High voltage transmission lines increased from the baseline value 2354km to 4518.7km compared to the NDC target of 4354km; representing more than 100% progress in the implementation period. The share of clean energy for cooking has increased from the baseline value of 15% to 25% by the end of 2024 compared to the NDC target of 50% by 2025; representing a 28.6% progress. The share of biomass energy used for cooking has decreased from the baseline value of 88% to 75% compared to the NDC target of 50% by 2025; representing a 34.2% progress in the implementation period. The proportion of households and institutions using efficient cooking technologies increased from the baseline value of 1% to 3.8% compared to the NDC target of 10% by 2025; representing 38% progress. High voltage transmission lines increased from the baseline value of 2354km to 4010km compared to the NDC target of 4354km by 2025; representing a percentage progress of 82.8%.

The overall progress of the sector based on the assessed indicators was 61.4% during this 5-year NDC implementation period, which is relatively good progress.

Agriculture

The agricultural sector has a total of six NDC priority adaptation actions that mainly focus on value addition and post-harvest handling, largescale irrigation; and climate smart agriculture including agro-ecology. Three of the actions did not have indicators for their tracking and nor baseline data to base on during the assessment. The results of the stock take assessment show that three of the four actions that had indicators (75%) had both baseline and current information, allowing for tracking their progress during the implementation period. The fourth indicator (Proportion of farmers practicing sustainable land management practices) did not have quantitative present data, but some analysis of its progress is provided qualitatively in this stock take.

The share of post-harvest losses has decreased from the base line value of 37% to 18.2% by the end of 2024/2025 compared to 12% target value for the year 2025. This represents 75.2% progress of this indicator. Area under irrigation has increased from the base line value of 19,776ha to 23,141ha compared to 28,934ha target value for the year 2025; representing a 36.7% progress. Another action that has data is the Promotion and development of climate resilient crop varieties. Data from the National Research Agriculture Organization (NARO) show that 21 new varieties were released for seven crops in the period between 2021 and 2025. Of these varieties 13 (61.9%) can be classified as climate resilient since they are resilient to certain diseases and pests that are climate dependent. For the indicator of proportion of farmers practicing sustainable land management practices that didn't have quantitative data, some information was obtained on capacity building initiatives where over 1,800 farmer groups and 15,000 individuals, focusing on sustainable production, enterprise development, and climate-smart agronomic practices by the Ministry of Agriculture, Animal, Industry and Fisheries (MAAIF) were trained. The climate smart agriculture transformation project in the same ministry has prioritized research and promotion of climate smart agriculture practices in 56 districts.

The overall progress of the sector based on the assessed indicators was 57.9% during this 5-year NDC implementation period.

Adaptation Agriculture: Local Economic Growth Support (1509), Source: MoFPED, 2025



Works at Rwakibira Valley Dam, Gomba District



Completed milk collection centre at Buseeta, Kibuka District



Completed Magoma Market, Nakaseke District



Disc Plough Tractor for Ongongoja Fruit Growers' Cooperative Society Ltd Oil Processing Project, Katakwi District

Figure 8-2 Examples of progress in the agriculture sector

Fisheries

The Fisheries sector has two NDC adaptation actions that focus on climate resilient capture fisheries and an ecosystem approach to aquaculture management. Promotion of climate resilient capture fisheries has an indicator of volume of fish stock that has a baseline value. Current production data shows that capture fisheries production increased to 728,369 metric tons⁵⁷ by 2023. The baseline value was 567,000 metric tons per year and compared with 1,200,000 metric tons per year at the 2025 target. This represents a percentage progress of 25.5% during this period. The other action of promotion of ecosystem approach to aquaculture management has no indicator or baseline data, and there was no current data to track its progress. Some qualitative information on the sector included the establishment of aquaculture training hubs, development

⁵⁷ [*National-Status-of-the-Environment-Report-NSOER-2024.pdf*](#)

of technical regulations, and training of over 230 farmers in integrated systems like rice-fish farming by MAAIF.

Forestry

The forestry sector has four NDC adaptation actions revolving around improved forest management and equitable forest product use. All four actions have indicators for their tracking and present information with only one having baseline data. Present information was provided on 3(75%) indicators that were used for tracking progress in the sector.

Under the action of promotion of afforestation and reforestation, area under planted forests has increased from the baseline value 107,608ha to 338555ha by the end of 2024 compared to the NDC target of 307,000ha by 2025. This represents more than 100% progress from the 2025 target. For the indicator of Area under Forest landscape restored, present information shows that about 109,879ha of forest has been restored by 2024/25 financial year compared to the NDC 2025 target of 200,000ha. However, there was no baseline information for this indicator that was to be determined during the NDC implementation. For the indicator of area under agroforestry landscape, forest landscape restored, The MWE under the department of forestry assessment and monitoring has developed the National Agroforestry strategy 2024/25 to 2033/34 that aims at strengthening and promoting coordinated development of agroforestry in Uganda. One of the strategy objectives is to enhance the promotion, adoption, and scaling up of agroforestry approaches and technologies in Uganda which aligns well with the adaptation targets of the NDC. The indicator of the number of people using non-timber forest products (NTFP) didn't have baseline data that was supposed to be determined during the NDC2.0 implementation.

Disaster Risk Reduction

The disaster risk reduction sector is accorded five adaptation actions that underscore the importance of mainstreaming disaster risk reduction in development planning. Five indicators were identified for tracking the progress of these actions with three of them having baseline data. Present information was provided on all three (100%) indicators that were used for tracking progress in the sector.

Accuracy of meteorological information increased from the base line value of 70% to 77% by the end of 2024/2025 compared to the NDC target of 90% by 2025 and this represents a 35% progress during this NDC implementation period. However, no information was provided on how these accuracy percentages were calculated scientifically. Automation of Weather and Climate Network increased from 30% to 67% compared to the NDC target of 82% by 2025; representing a percentage progress of 71.1% during the NDC implementation period. On the indicator of Proportion of Local governments with climate action plans, information received shows that between 2021 and 2025, only two districts (1.4%) of Buikwe and Namayingo had developed their district action plans. Compared to the 10% NDC 2025 target, this represents 14% progress in the NDC implementation period. The two actions of promotion of local, indigenous and traditional knowledge (ITK) and practices in disaster risk reduction and strengthening policy linkage and synergies on climate change, migration and disaster risk reduction had tracking indicators, but no baseline data and current information to track their progress.

The overall progress of the sector based on the assessed indicators was 40% during this 5-year NDC implementation period, representing average progress in the sector.

Cities and the Built Environment

Cities and the built environment sector has four adaptation actions aimed at bolstering sustainable urbanization that includes green spaces, efficient mobility and improved waste management. Ten indicators were identified for tracking the progress of these actions with eight of them having baseline data. Present information was provided on four (50%) indicators that were used for tracking progress in the sector.

The proportion of urban dwellers living in slums and informal settlements decreased from 58% to 54% by 2024/2025 compared to the NDC target of 40% by 2025; representing 22.2% percentage progress. Length of drainage channels constructed/improved in GKMA increased from 9km to 23.6km compared to the NDC target of 30km by 2025; representing a percentage progress of 69.5%. The other two indicators related to promotion of efficient mobility in cities where urban paved roads increased from 1348km to 1836km compared to the NDC target of 7740km by 2025; representing a percentage progress of only 7.6%. However, this indicator should be harmonized with a similar indicator in the transport sector that has different baseline values. KM of NMT (cycle/Pedestrian) lane in GKMA increased from 1.95km to 5.45km by the end of 2024 compared to the NDC target of 25km by 2025; representing a percentage progress of 15.2%.

The overall progress of the sector based on the assessed indicators was 35.6% during this 5 year NDC implementation period representing a less than average progress in the sector

Health

There are four adaptation actions of the health sector that are mainly about developing National Adaptation Plans for the health sector. Six indicators were identified for tracking the progress of these actions with all of them having baseline data as zero from the start of NDC2.0. Present information was provided on only two (33.3%) indicators that were used for tracking their progress.

The number of climate risk and vulnerability assessments conducted for the health sector was one which was conducted during the Health National Adaptation Plan (H-NAP). The NDC target was also one implying 100% progress on this indicator during the NDC implementation. The other indicator related to the development of Health National Adaptation Plan (H-NAP) which was developed in 2024 also indicates 100% progress of the target. However, it was not possible to access current data on the number of districts with developed climate health profiles, level of implementation of HNAP, level of linkage between the emergency medical call system with the National Disaster response call system and the number of transformed National and Regional Referral Hospitals (Climate smart hospitals).

Adaptation Health: Rehabilitation and Construction of General Hospitals-Refurbishinh and Equipping of Busolwe General Hospital Project (1243), Source: MoFPED & OPM-DU, 2024



Construction of ramp on existing wards, and compound leveling

Figure 8-3 Rehabilitation and construction of General Hospitals

Manufacturing, Industrial Processes and Mining

The manufacturing, industrial processes and mining sector has three NDC adaptation actions that involve scaling-up resource efficient technologies, research and development and promoting circular economy in industrial and manufacturing value chains. Three indicators were identified for tracking the progress of these actions with only one of them having baseline data while the other two did not have either baseline data or present information.

It was not possible to access present information on all the indicators and therefore could not track the level of NDC adaptation action implementation in this sector.

Education

The Education sector has three adaptation actions that are envisaged to contribute towards raising awareness of and developing knowledge for climate change resilience. Three indicators were identified for tracking the progress of these actions with two of them having baseline data. Present information was provided on only one (33.3%) indicator that was used for tracking their progress.

The indicator with current data is the level of integration of climate change in the national curriculum. Data from the National Curriculum Development Centre (NCDC) shows that climate change has been integrated in the lower secondary revised curriculum through the geography subject curriculum that is compulsory for all students. The Upper secondary curriculum has also been revised with some aspects of climate change integrated Geography and Biology curricula. However, these are optional subjects taken by a proportion of students and therefore not reaching all the learners. We propose inclusion of climate change related topics

in the General paper, a subject that is done by all the students in Advanced level. The integration of climate change in teaching and learning in tertiary institutions is still not harmonized with different universities offering specific courses that relate to climate change such as Meteorology and Environmental Sciences.

For the action of improving education and awareness raising climate change, no present data on the number of people trained on climate change disaggregated by gender was provided. Data was also not provided for the action of Promoting natural and cultural/heritage conservation where the number of knowledge systems integrating local, indigenous and traditional knowledge and practices was used as an indicator.

Tourism

The tourism sector has three adaptation actions ordered towards safeguarding nature-based tourism attraction sites. Five indicators were identified for tracking the progress of these actions with only one of them having baseline data while the other four did not have either baseline data or present information.

Regarding the number of climate risk and vulnerability assessments conducted for the tourism sector, the ministry has planned for this assessment in the financial year of 2025/2026. Other indicators in the sector had neither baseline data nor present data and were therefore not tracked. Other indicators were proposed to improve tracking and reporting for the next NDC.

9 Support Received and Needed

Uganda's NDC 2.0 indicates a requirement of USD 17.7 billion for adaptation, including USD 2.5 billion unconditional and USD 15.2 billion conditional on international support, alongside other needs for capacity building, technology development and transfer. To date, progress on mobilizing finance has been mixed: national budget support and external projects have covered multiple adaptation sectors (agriculture, health, forestry, water, transport, energy, and cities), while three of the five mitigation sectors (AFOLU, energy, transport) have attracted resources, leaving waste and IPPU unfunded. Technology transfer has advanced through renewable energy generation, resilient infrastructure, climate-smart agriculture, aquaculture innovations, and digital systems, while capacity-building has strengthened farmer groups, local governments, utilities, and sectoral institutions. Still, several sectors (Environment & Ecosystems, Tourism, Disaster Risk Reduction) remain largely unfunded and critical technology action plans have yet to be implemented. While there is evidence of functional capacity-building programs and technology deployment, the gap between NDC 2.0 financial needs and actual mobilized resources, as well as incomplete sectoral coverage, indicates that Uganda is only partially on track to meet the means-of-implementation targets for 2030.

The implementation of Uganda's updated Nationally Determined Contributions (NDC 2.0) under the Paris Agreement required significant and sustained investments, particularly in the form of climate finance to a tune of US\$ 28.1 Billion (MWE, 2022; MoFPED, 2025). As outlined in the NDC 2.0 Implementation Plan (MWE, 2022), the means of implementation for these commitments include climate finance, capacity building, and technology development and transfer. This section presents the findings of a climate finance stocktake relevant to the implementation of NDC 2.0, based on data collected from multiple sources.

NDC 2.0 relevant climate finance in Uganda was attached to a variety of projects that were financed either through national budget support or off-budget support. National Budget support consisted of Government of Uganda funding mobilized domestically in the national economy; and external funding from a diverse external public and private organizations through instruments such as loans, grants, guarantees and debt swaps. External sources of financing for national budget support include bilateral and multilateral Development Partners and commercial banks. Off-budget support was channeled through projects that were either implemented directly by bilateral Development Partners; or through Non-Governmental Organizations; Civil Society Organizations and Private Companies funded by Development Partners

This Stocktake understood Unconditional climate finance to be that finance through Government of Uganda funding with monies mobilized mostly from domestic sources following the country's Public Finance Management policies and legal framework, national programme-based planning and budgeting procedure as indicated in the national budget and Public Investment Plan; and Conditional climate finance to mean any finance mobilized via external public and private sources including bilateral and multilateral development partners, through both national budget support and off-budget support (GoU, 2022; MoFPED & DU-OPM, 2023; MoFPED, 2025). As such, the Stocktake expected climate finance channeled through national budget support to be likely a combination of domestic and external finance, that is Unconditional and Conditional, respectively; whilst financial support via off budget supported projects to be exclusively externally mobilized finance and therefore exclusively Conditional.

However, to reduce the risk of double counting Conditional finance, the Stocktake considered domestic finance and therefore unconditional, all external finance that was channeled through the national budget in the form of concessional loan instruments. This is because Concessional loans are monies which the Government of Uganda has borrowed with approval of the Parliament, to implement in the short term, its priority national development programmes with the intention of paying back the external loan source-both the interest on the loans (normally, five years after loan agreements are signed) and the principal in the long term (15-20 years or even up to 40 years for some infrastructure projects). Moreover, even Grants from external sources that are meant for national budget support are portions of concessional loans, according to Uganda's Public Debt Management policies; and reports published in 2024 by MoFPED on public debt, grants, guarantees and other financial liabilities. To control such discrepancies in future NDC's, Uganda might need to disaggregate its budgeted Conditional External Financial support as either portion expected through Debt instruments such as loans; and those from non-debt instruments, that is, for example purely Grants.

This Stocktake used the following definitions for climate finance-related terms:

Climate Finance Needed is the cost of the priority NDC Actions, as per the costed NDC 2.0 Implementation Plan (MWE, 2022).

Climate Finance Committed is the approved costs/budgets of projects identified as aligned to NDC 2.0 priority actions in accordance with the said projects' financing documents for projects financed through national budget support; and project information from IATI d-portal for projects financed through off-budget support.

Climate Finance Received is the disbursed share of the committed finance for the period of 2020 to 2025, for both national budget supported projects; and off-budget supported projects.

Climate Finance Progress is the difference between Climate Finance Received and Climate Finance Needed. A **positive** value indicates **achievement** of climate finance target while a **negative** value indicates **failure** to achieve the set climate finance target.

9.1 Tracking Financial support using ETF aligned framework

In the sections that follow, the progress of climate finance for each NDC 2.0 priority Adaptation and Mitigation Sectors; and Coordination for NDC 2.0 implementation is presented. The Stock take findings are presented as climate finance channeled through:

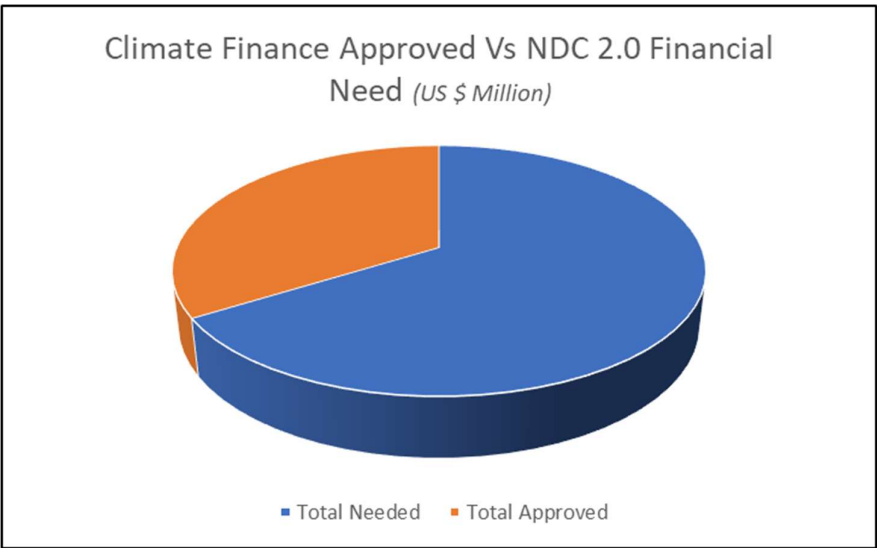
National Budget Support: The findings are presented as **Climate Finance Status** – detailing finance needed (conditional/unconditional), approved, received and progress; and **Funding Status**-providing details of funders, project implementers, project status; and financing instrument used

Off-Budget Support: The findings present exclusive Conditional Finance- detailing Climate Finance Committed and Climate Finance Received for the period of 2020 to 2025; and summaries of major sources for this form of External Off-Budget Finance

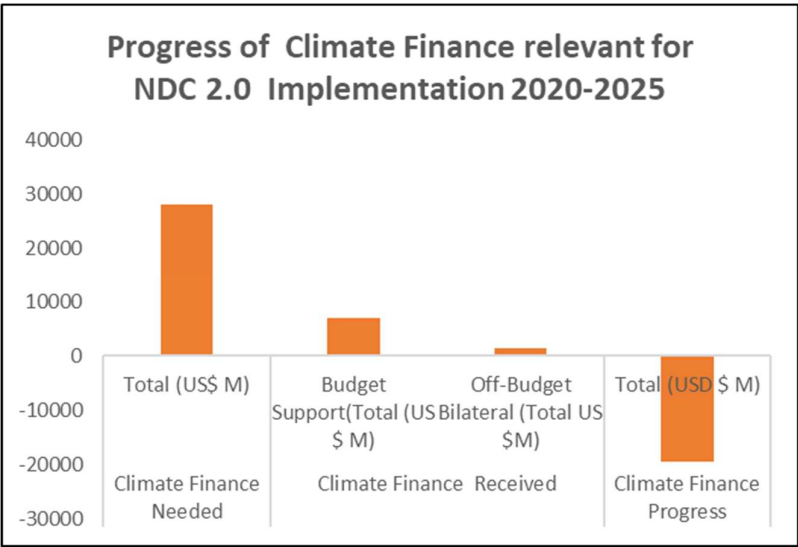
The details of the National Budget supported projects and Off-Budget supported projects that were identified and selected as aligned to NDC 2.0 priority actions are included in CTF reporting format in Annex 3: Sheet name_ MOI_Climate Finance Support Received.

By assessing funding status through this lens, the report provides a more accurate and policy-relevant view of how current investments align with climate objectives and where further financial efforts are required to close the implementation gap. This systematic approach not only supports national planning but also enhances transparency and accountability in tracking the flow and use of climate finance in Uganda.

Overall, the climate finance that was approved and committed for actions aligned with the implementation of NDC 2.0 fell short of the total climate finance needed.



Furthermore, the actual disbursed amounts—representing the climate finance received—were also below the required levels.



This resulted in an overall funding gap of approximately US \$ 19.5 Billion for the effective implementation of NDC 2.0. The overall climate finance progress stands at 30.3% of which National Budget Support contributes close to 82.3% whilst Off-Budget Support-exclusively Conditional was 17.7%

Status of NDC 2.0 Sectoral Climate Finance Progress				
NDC 2.0 Priority Sector	Climate Finance Needed	Climate Finance Received		Climate Finance Progress
	Total (US\$ M)	Budget Support (Total US \$ M)	Off-Budget Bilateral (Total US \$ M)	Total (USD \$ M)
Agriculture	3311.1	450.485	580.53	-2280.09
Fisheries	195.8	9.38	8.95	-177.47
Education	48.3	0	0	-48.3
Health	931.5	12.106	15.15	-904.244
Forestry	5693.5	20.95	4.35	-5668.2
Water & Sanitation	2729.3	862.96	220.21	-1646.13
Environment & Ecosystems	64.7	0	28.67	-36.03
Tourism	87	0	8.4	-78.6
Disaster Risk Reduction	62.85	0	89.59	26.74
Manufacturing, IP& Mining	97.14	249.92	77.15	229.93
Transport (Adaptation)	1031.22	2473.496	59.41	1501.686
Energy (Adaptation)	491.34	307.5948	6.18	-177.565
Cities & Build Environment	2925.18	327.1	4.11	-2593.97
Transport (Mitigation)	652.35	346.93	10.14	-295.28
Energy (Mitigation)	1327.5	1625.21	147.35	445.06
AFOLU	7592.7	2.33	87.25	-7503.12
Waste (Mitigation)	694.2	0	2.8	-691.4
IPPU (Mitigation)	56.34	0	0	-56.34
Coordination	106.7	324.08	160.62	378
Total	28098.72	7012.542	1510.86	-19575.3

Only the priority NDC 2.0 Actions under the Education Sector (Adaptation) and IPPU Sector (Mitigation) appear to be completely unfunded via both National budget and Off-Budget Support. The NDC Actions for the Adaptation Sectors of Environment and Ecosystems, Tourism and Disaster Risk Reduction; and for Waste Sector in Mitigation that were not financed via the National Budget, were financed, albeit meagerly through Off-Budget support. This Stock take finding is consistent with the acknowledge fact that activities related to Uganda's Environment and Natural Resources are usually under funded by public domestic finance, even when the economy is heavily dependent activities that are nature-based.

9.1.1 Coordination of NDC 2.0 Implementation

National Budget Support for Coordination of NDC 2.0 Implementation

Table 9-1: Support Needed and Received for NDC 2.0 Coordination

NDC-aligned Climate Finance Needed and Received for NDC 2.0 Coordination							
Climate Finance Needed			Climate Finance Approved			Climate Finance Actually Received	Climate Finance Progress
Conditional	Unconditional	Total (US\$ M)	Conditional	Unconditional	Total (US\$ M)	Total (US \$ M)	Total (USD \$ M)
99.75	6.98	106.7	500	0	500	324.08	217.38

For the implementation of NDC 2.0 coordination efforts, a total of US\$106.7 million in climate finance was identified as needed: US\$99.75 million conditional on external support and US\$6.98 million to be mobilized domestically. In contrast, US\$500 million in conditional finance has been approved, significantly exceeding the stated need. However, only US\$324.08 million has been received, with US\$217.38 million showing measurable progress.

This mismatch between approved and needed finance may stem from under-costing of coordination efforts during the initial planning stages, leading to an underestimated budget that does not reflect the true scale or complexity of implementation needs. It also suggests a possible disconnect between what has been requested and what development partners are willing or able to support.

Additionally, the complete absence of approved or received funds for the unconditional portion highlights a critical gap in domestic resource mobilization, which could limit national ownership and resilience of the NDC process.

To ensure successful implementation of NDC 2.0, it is crucial to revisit and potentially revise cost estimates, strengthen alignment between funding approvals and actual needs, improve disbursement processes, and enhance efforts to secure both domestic and international support.

Table 9-2: Funding Status for Coordination of NDC 2.0 Implementation

Resource Mobilization for Uganda's NDC 2.0 Implementation	Funding Status of Projects relevant to NDC 2.0 Actions							
	NDP Project Title (Code)	NDP III Programme Category for Project	Project Funder	Nature of Instrument	Project Implementer	Project Status	Project Duration Start Year	End Year
Procedures to guide MDAs to maximize NDC aligned projects in their sector plans & Mid-Term Expenditure Framework & Budget	Uganda Intergovernmental Fiscal Transfers Programme (UgIFT)	Human Capital Development	World Bank	Loan & Grant	MoFPED; MoH; MWE; DLG; MAAIF; MoES	Restructured (Ongoing)	2020	2024

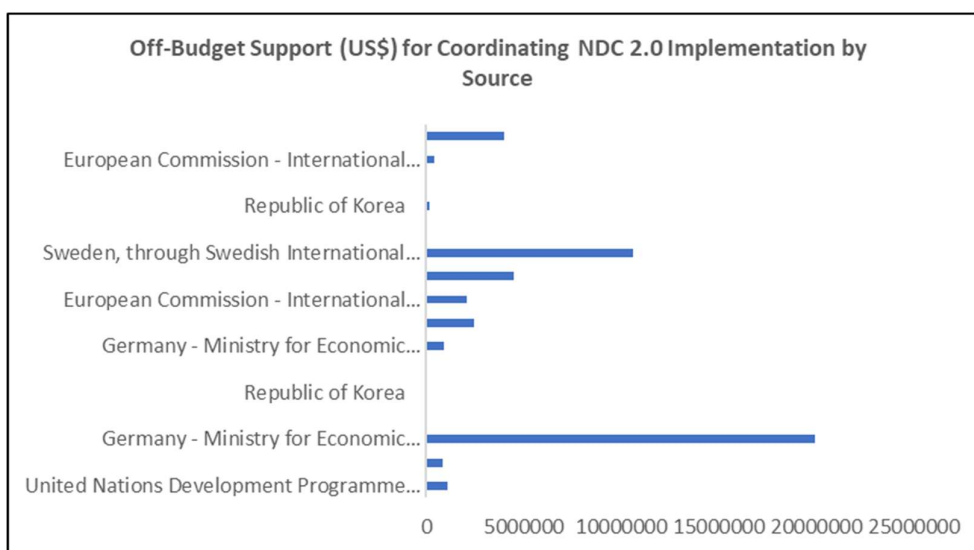
One of the key projects supporting Uganda's NDC 2.0 implementation is the Uganda Intergovernmental Fiscal Transfers Programme (UgIFT), funded by the World Bank through a combination of loan and grant instruments. Implemented by multiple ministries and agencies, including MoFPED, MoH, MWE, MAAIF, MoES, and DLGs, the project aims to guide MDAs integrating NDC-aligned actions into sector plans and budgets. Although currently under restructuring, the project has been active since 2020 and is set to conclude in 2024, playing a critical role in aligning fiscal planning with climate goals under the Human Capital Development programme.

Off-Budget Support for Coordination of NDC 2.0 Implementation

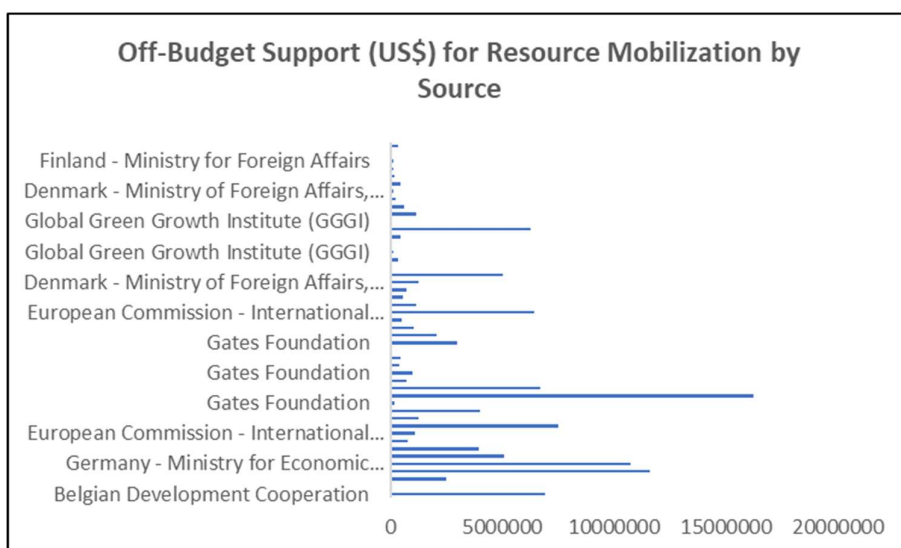
US \$99.7 M was budgeted in NDC 2.0 as Conditional to external support. The results from the Stock take analysis reveal that exclusive Conditional finance for NDC 2.0 exceeded the target by 161%

Off-Budget Bilateral Support for NDC 2.0 Coordination		
NDC 2.0 Coordination Component	Climate Finance Committed (US\$ M)	Climate Finance Received (US\$)
High-Level National Coordination	0	0
NDC Implementation	47.6	47.1
Resource Mobilization	387.5	113.6
Monitoring & Evaluation	0	0
Total	435.1	160.7

The German-Ministry for Economic Cooperation and Development was a key source of off-budget financial support for activities related with coordinating NDC 2.0 Implementation.



The Gates Foundation was a main source for Off-Budget support for the Resource Mobilization component of NDC 2.0 Coordination.



9.1.2 Support for NDC 2.0 Sectoral Adaptation Actions

Overall, the climate finance that was approved and committed for actions aligned with the implementation of NDC 2.0 was less than the total climate finance needed. Similarly, the actual disbursed amounts—representing climate finance received—were also below the required levels, resulting in an overall funding gap for NDC 2.0 implementation. Notably, there are some sectors without NDP III projects explicitly aligned with NDC 2.0 priority actions. However, components of these sectors are embedded within projects under other sectors, and these have been identified as subsectors in the Climate Tagging Framework (CTF) table in Annex 3: *MOI_Climate Finance Received*. Under the Adaptation component, only ten (10) of the thirteen (13) priority sectors had NDC 2.0 actions and activities that could be matched with the selected projects supported through the national budget. The sectors whose NDC 2.0 priority actions and activities appear not to be directly funded through National budget support include Environment and Ecosystems, Tourism, and Disaster Risk Reduction. However, these sectors are funded through Off-budget support.

The following sections provide a detailed breakdown of climate finance status by priority sector, highlighting specific funding gaps and alignment with Budget-supported NDP III projects resulting into implementation of NDC 2.0 adaptation actions.

Agriculture NDC 2.0 Adaptation Actions

National Budget Support for Agriculture NDC 2.0 Adaptation Action

Table 9-3: Support Needed and Received for Agriculture NDC 2.0 Adaptation Action

NDC-aligned Climate Finance Needed & Received for Agriculture NDC 2.0 Adaptation Actions								
NDC 2.0 Priority Sector	Climate Finance Needed			Climate Finance Approved			Climate Finance Actually Received	Climate Finance Progress
	Conditional	Unconditional	Total (US\$ M)	Conditional	Unconditional	Total (US\$ M)	Total (US \$ M)	Total (USD \$ M)
Agriculture	3177	133.9	3311.1	1382.296	432.315	1814.611	450.485	-2860.62

In Uganda's NDC 2.0, the agriculture sector identified a total climate finance need of US\$3.31 billion (US\$3.18B conditional, US\$134M unconditional). However, only US\$1.81 billion has been approved, and just US\$450.49 million has actually been received.

This reflects a significant financing gap of over US\$2.86 billion, highlighting slow progress in mobilizing and disbursing climate finance for agriculture—Uganda's largest NDC priority sector. The shortfall signals the urgent need for enhanced alignment between commitments and disbursements to meet climate adaptation goals in the sector.

Table 9-4: Funding Status for Agriculture NDC 2.0 Adaptation Actions

Agriculture NDC 2.0 Adaptation Action	Funding Status of Projects relevant to NDC 2.0 Actions							
	NDP Project Title (Code)	NDP III Programme Category for Project	Project Funder	Nature of Instrument	Project Implementer	Project Status	Duration	
							Start Year	End Year
Expanding value addition, efficient harvesting, post-harvest handling and storage and access to markets, including micro- finance	Project for Restoration of Livelihoods in the Northern Region (1381)	Agro- Industrialization	GoU; Adaptation for Smallholder Agriculture Programme (ASAP); IFAD; Project Beneficiaries	GoU Counterpart Funding; Grant; Loan; Beneficiary Contribution	MoLG-Agago, Amuru, Gulu, Lamwo, Nwoya, Omoro, Pader & Adjumani	Completed	2015	2024
	Rural Development and Food Security in Northern Uganda (RUDSEC) Project (1760)	Regional Development	Kfw; GoU Counterpart financing	Grant; GoU Counter financing	MoLG	Ongoing	2023	2028
	Uganda Climate Smart Agricultural Transformation Project (1786)	Agro- Industrialization	International Development Association (IDA)(WBG); GoU	Loan	MAAIF	Ongoing	2022	2029
	Agriculture Cluster Development Project (1263)	Agro- Industrialization	International Development Association (IDA)(WBG)	Loan	MAAIF	Completed	2017	2024
	National Oil Palm Project (1508)	Agro- Industrialization	IFAD; Bidco Uganda Limited; GoU	IFAD Loan & Grant; GoU Counterpart; Private sector Contributions; Loan reflows & farmer contributions	Bidco Uganda Limited and Farmers	Ongoing	2019	2029
Undertake comprehensive management needs assessment of existing large- scale irrigation schemes	Irrigation Scheme Development in Central and Eastern Uganda (1323)	Agro- Industrialization	JICA; GoU	Grant; Counterpart Funding	MAAIF	Ongoing on extension	2018	2026
	Local Economic Growth Support (1509)	Regional Development	Islamic Development Bank; GoU & Community	Loan; Counterpart funding; Beneficiary Contribution	Ministry of Local Government	Ongoing (Near Completion)	2019	2025
	Development of Solar Powered Water Supply and Irrigation Systems Project (1666)	NRE-CCLWRM	United Kingdom Export Finance (UKEF) 85%; GoU (15%)	Loan	MWE; Nexus Green	Ongoing (Near Completion)	2021	2025
	Irrigation for Climate Resilience Project (1661)	Agro- Industrialization	International Development Association (IDA)(WBG)	Loan	MWE, MAAIF, NFA	Ongoing	2020	2026
Scaling-up Climate Smart Agriculture (CSA) Practices including agro- ecology	Agriculture Value Chain Development Programme (1444)	Agro- Industrialization	African Development Fund	Loan	MAAIF	Ongoing (Near Completion December 2025)	2018	2025
	Developing a Market Oriented and Environmentally Sustainable Beef Meat Industry (1493)	Agro- Industrialization	European Union	Grant	MAAIF	Near Completion April 2025)	2018	2025
	China-Uganda South-South Cooperation Project Phase 3 (1663)	Agro- Industrialization	Government of the People's Republic of China	Grant	FAO (Project Facilitator) MAAIF	Ongoing (Near Completion (June 2025)	2022	2025
	Enhancing National Food Security through Increase Rice Production Project (1493)	Agro- Industrialization	Islamic Development Bank; GoU & Pearl Rice Company and Busowa Traders and Farmers' Cooperative Society Company Limited	Loan	Pearl Rice Company and Busowa Traders and Farmers' Cooperative Society Company Limited	Completed	2016	2023
	Promotion of Rice Development Project Phase II (1709)	Agro- Industrialization	Japan International Cooperation Agency (JICA)	Grant		2nd Phase completed; Potential of 3rd Phase	2019	2024
	National Oil Seeds Project (1772)	Agro- Industrialization	IFAD; OPEC Fund for International Development (OFID); GoU; Heifer International; Kuehne Foundation; Beneficiaries & Private Sector	Loan	MAAIF; MoLG	Ongoing	2021	2028
	Farm Income Enrichment and Forestry Conservation Project Phase II (1417)	NRE-CCLWRM	African Development Bank; Nordic Development Fund; GoU	Loan	MWE; MAAIF	Completed	2016	2024

Uganda's NDC 2.0 adaptation actions in agriculture are being financed through a diverse mix of financial instruments, including loans, grants, government counterpart funding, and beneficiary contributions. Key financiers include IFAD, the World Bank (IDA), the African Development Bank, the European Union, JICA, KfW, and the Islamic Development Bank, among others.

Most projects fall under the Agro-Industrialization programme of the NDP III, with others aligned to Regional Development and Natural Resources, Environment, Climate Change, Land and Water Resource Management (NRE-CCLWRM). Instruments vary by project and funder—for example:

- The Uganda Climate Smart Agricultural Transformation Project (IDA loan) supports long-term resilience under Agro-Industrialization.

- The Local Economic Growth Support Project (IsDB loan + beneficiary contributions) falls under Regional Development.
- The Farm Income Enrichment and Forestry Conservation Project Phase II was funded via loans from the AfDB and NDF under the NRE-CCLWRM programme.

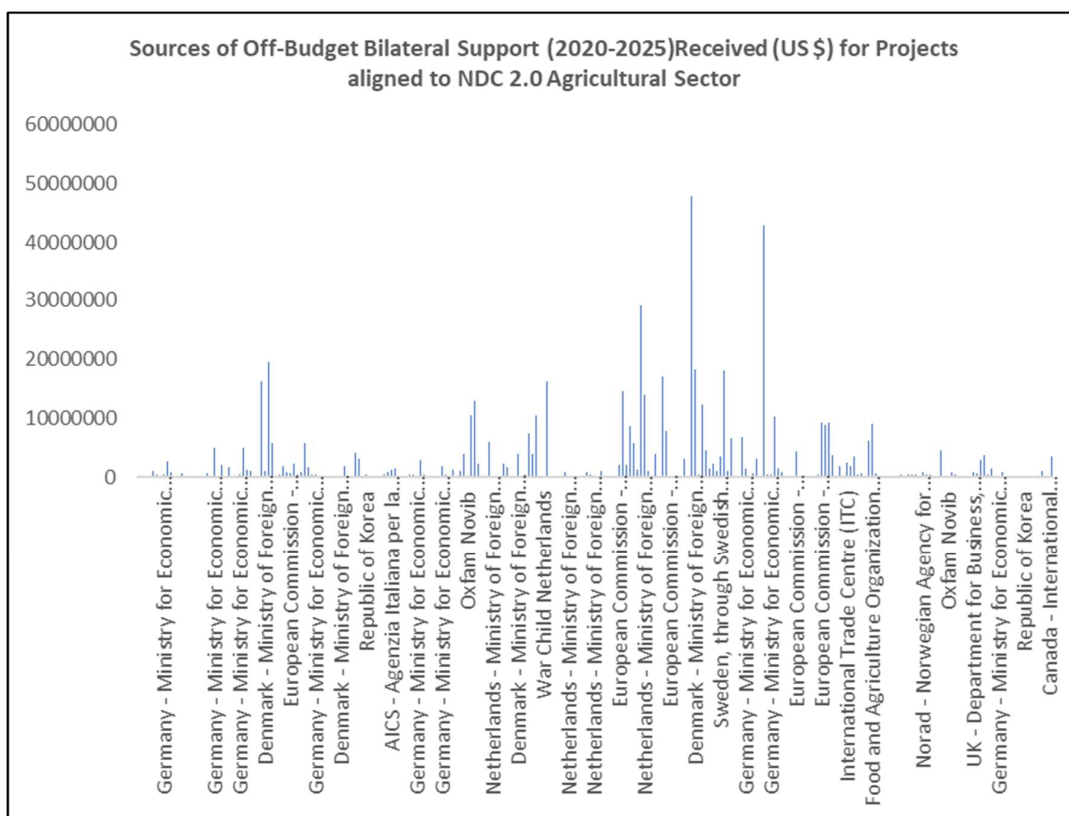
This blended finance approach reflects a growing commitment to use innovative, multi-stakeholder financing models to advance climate-resilient agriculture and fulfill Uganda's NDC goals.

Off-Budget Support for Agriculture NDC 2.0 Adaptation Action

US \$3.17 Billion was budgeted as the target for Conditional finance of priority actions under the Agriculture Adaptation sector. In the period of 2020 to 2025, only 26.9% of exclusive Conditional Climate Finance has been committed for the implementation of priority NDC 2.0 actions for the Agriculture Sector. Only 15.6% of the targeted Conditional Finance for Agriculture has been received.

Off-Budget Bilateral Support for Agriculture NDC 2.0 Adaptation	
Climate Finance Committed (US\$ M)	Climate Finance Received (US\$ M)
998.4	580.5

The Denmark-Ministry of Foreign Affairs was a key source of Off-budget support for NDC 2.0 Actions of the Agriculture sector under Adaptation. This was followed by the Germany-Ministry of Economic Cooperation and Development; and the Netherlands-Ministry of Foreign Affairs



Fisheries NDC 2.0 Adaptation Actions

National Budget Support for Fisheries NDC 2.0 Adaptation Action

Table 9-5: Support Needed and Received for Fisheries NDC 2.0 Adaptation Actions

NDC-aligned Climate Finance Needed & Received for Fisheries NDC 2.0 Adaptation Actions							
Climate Finance Needed			Climate Finance Approved			Climate Finance Actually Received	Climate Finance Progress
Conditional	Unconditional	Total (US\$ M)	Conditional	Unconditional	Total (US\$ M)	Total (US \$ M)	Total (USD \$ M)
137.06	58.74	195.8	1.624	9.963	11.587	9.38	-186.42

Uganda's NDC 2.0 Fisheries sector requires US\$195.8 million in climate finance—US\$137.06 million conditional and US\$58.74 million unconditional. To date, only US\$11.59 million has been approved, and just US\$9.38 million received.

This leaves a substantial financing gap of approximately US\$186.4 million, underscoring the critical underfunding of climate adaptation actions in the fisheries sector. The slow progress highlights the need for greater investment and targeted support to build climate resilience in this vulnerable sector.

Table 9-6: Funding Status for Fisheries NDC 2.0 Adaptation Actions

Fisheries NDC 2.0 Adaptation Action	Funding Status of Projects relevant to NDC 2.0 Actions							
	NDP Project Title (Code)	NDP III Programme Category for Project	Project Funder	Nature of Instrument	Project Implementer	Project Status	Project Duration Start Year	End Year
Promote climate resilient capture fisheries	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Promote ecosystem approach to aquaculture management	Promoting Environmentally Sustainable Commercial Aquaculture (1494)	Agro-Industrialization	European Development Fund (EDF)	Grant ; Project Beneficiaries	MAAIF; NARO	Completed	2017	2024
	China-Uganda South-South Cooperation Project Phase 3 (1663)	Agro-Industrialization	Government of the People's Republic of China	Grant	FAO (Project Facilitator) MAAIF	Ongoing (Near Completion (June 2025)	2022	2025

Climate adaptation actions in Uganda's fisheries sector under NDC 2.0 have primarily been financed through grants and beneficiary contributions, with funding sourced from bilateral partners under the Agro-Industrialization programme of the NDP III.

The financing instruments used reflect a reliance on concessional resources, with no evidence of loan financing or private sector blended finance to date. While these instruments have supported progress in aquaculture, capture fisheries remain unsupported, indicating a gap in climate finance allocation.

As Uganda moves toward NDC 3.0, there is an opportunity to diversify financing instruments, strengthen sector-specific resource mobilization, and ensure that both aquaculture and capture fisheries are strategically included. Expanding the range of instruments and sources will be essential for scaling up resilience across the entire fisheries value chain.

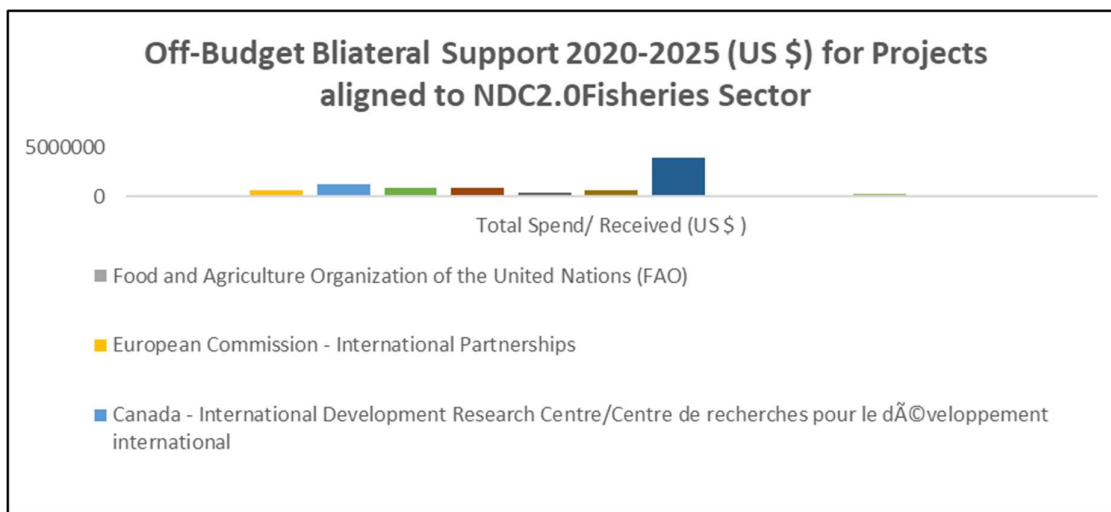
Off- Budget Support for Fisheries NDC 2.0 Adaptation Action

US\$ 137.06 Million was the budget for exclusive Conditional Finance for the priority NDC 2.0 Adaptation actions in the Fisheries Sector. There is an 84.8% funding gap between the Committed exclusive Conditional Finance and the exclusive Conditional Finance needed for implementing priority actions for the Fisheries

sector. Additionally, only 6.4% of exclusive Conditional Finance needed was received during the period from 2020 to 2025.

Off-Budget Bilateral Support for Fisheries NDC 2.0 Adaptation	
Climate Finance Committed (US\$ M)	Climate Finance Received (US\$ M)
20.9	8.9

The key sources for off-budget financial support, by volumes, for priority actions in the Fisheries sector were the Canada-International Development Research Centre, European Commission-International Partnerships; and the Food and Agriculture Organization of the United Nations (FAO)



Education NDC 2.0 Adaptation Actions

National Budget Support for Education NDC 2.0 Adaptation Action

Table 9-7: Support Needed and Received for Education NDC 2.0 Adaptation Actions

NDC-aligned Climate Finance Needed & Received for Education NDC 2.0 Adaptation Actions							
Climate Finance Needed			Climate Finance Approved			Climate Finance Actually Received	Climate Finance Progress
Conditional	Unconditional	Total (US\$ M)	Conditional	Unconditional	Total (US\$ M)	Total (US \$ M)	Total (USD \$ M)
36.23	12.08	48.3	150	21.6	171.6	0	-48.3

Uganda's education sector identified a need for US\$48.3 million in climate finance to support adaptation actions under NDC 2.0, with US\$36.23 million conditional on external support and US\$12.08 million expected from domestic sources.

To date, US\$171.6 million has been approved—well above the identified need. However, none of this funding has been received, resulting in zero disbursement and a full financing gap of US\$48.3 million in practical terms.

This highlights a critical disconnect between finance approvals and actual flows and underscores the urgency to address disbursement delays and implementation bottlenecks. For the education sector to contribute meaningfully to climate resilience, climate finance must move beyond commitments to timely and targeted delivery.

Table 9-8: Funding Status for Education NDC 2.0 Adaptation Actions

Education NDC 2.0 Adaptation Action	Funding Status of Projects relevant to NDC 2.0 Actions							
	NDP Project Title (Code)	NDP III Programme Category for Project	Project Funder	Nature of Instrument	Project Implementer	Project Status	Project	
							Start Year	End Year
Integrate climate change into the national curriculum	Uganda Secondary Education Expansion Project (1665)	Human Capital Development	International Development Agency-WB; GoU	Loan with Grant component; GoU Counterpart financing	Ministry of Education & Sports	Ongoing	2022	2025
Improve education and awareness raising on climate change	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Develop knowledge systems for scaling up adaptation	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Under Uganda's NDC 2.0, the education sector has begun accessing climate finance through a single notable initiative focused on climate-proofed infrastructure for education facilities. The Uganda Secondary Education Expansion Project, aligned with the Human Capital Development programme of the NDP III, is currently ongoing (2022–2025).

The project is funded by the International Development Association (World Bank) and the Government of Uganda, using a combination of a loan with a grant component and counterpart financing. It is implemented by the Ministry of Education and Sports and supports the construction and improvement of school infrastructure that is resilient to climate impacts.

However, no dedicated projects are currently addressing other NDC 2.0 education priorities, such as climate change awareness or knowledge systems for adaptation.

As Uganda develops NDC 3.0, this experience offers a useful entry point to scale up investments in resilient education systems, while also broadening the scope to include climate literacy and capacity-building. There is a clear opportunity to engage a wider range of financing instruments and development partners to fill these critical gaps.

Off- Budget Support for Education NDC 2.0 Adaptation Action

For the period of 2020 to 2025, there appears to be a net 100% funding gap from exclusive Conditional Climate Finance sources for the priority actions of the Education Sector, as there's no record for committed nor received climate finance for projects aligned to priority NDC 2.0 actions and activities in the Education Sector

Health NDC 2.0 Adaptation Actions

National Budget Support for Health NDC 2.0 Adaptation Action

Table 9-9: Support Needed and Received for Health NDC 2.0 Adaptation Actions

NDC-aligned Climate Finance Needed & Received for Health NDC 2.0 Adaptation Actions							
Climate Finance Needed			Climate Finance Approved			Climate Finance Actually Received	Climate Finance Progress
Conditional	Unconditional	Total (US\$ M)	Conditional	Unconditional	Total (US\$ M)	Total (US \$ M)	Total (USD \$ M)
774.86	156.64	931.5	233.922	10.56	244.482	12.106	-919.394

Uganda's health sector under NDC 2.0 identified a total climate finance need of US\$931.5 million, with US\$774.86 million conditional on external support and US\$156.64 million expected from domestic sources.

To date, only US\$244.48 million has been approved, and a mere US\$12.11 million received—representing just over 1% of the total needed. This results in a significant financing gap of US\$919.39 million, highlighting critical underperformance in climate finance delivery for health-related adaptation actions.

The current trajectory signals the urgent need to accelerate disbursements, improve alignment between approved and received funds, and prioritize health within climate finance strategies to protect vulnerable populations from increasing climate-related health risks.

Table 910:- Funding Status for Health NDC 2.0 Adaptation Actions

Health NDC 2.0 Adaptation Action	Funding Status of Projects relevant to NDC 2.0 Actions							
	NDP Project Title (Code)	NDP III Programme Category for Project	Project Funder	Nature of Instrument	Project Implementer	Project Status	Project Duration	
							Start Year	End Year
Integrate climate considerations into national health plans and strategies	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Improve early warning, surveillance and response system for climate sensitive health hazards	Global Fund for HIV/TB and Malaria Project (220)	Human Capital Development	Global Fund ; GoU	Global Fund (Grant); GoU (Counterpart financing)	Ministry of Health	Ongoing	2024	2026
Strengthen climate resilience of health infrastructure and systems	U.S Agency for International Development (USAID) support to Regional Referral Hospitals (1767)	Human Capital Development	USAID; GoU	Grant; GoU (Counterpart Financing)	RRH Jinja & Mbarara	Ongoing	2021	2026
	Rehabilitation and Construction of General Hospitals- Refurbishing and Equipping Busolwe General Hospital	Human Capital Development	Kingdom of Spain; Government of Uganda	Spanish Debt-Swap Grant; GoU(Counterpart Financing)	Ministry of Health	Several extensions, expected to end in 2021	2012	2021
Implement integrated health related climate interventions taking into account policies on water and sanitation, education, social protection & comprehensive health care	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

In implementing NDC 2.0 adaptation actions in the health sector, Uganda has relied primarily on grant-based financing combined with Government of Uganda counterpart contributions—a strategic choice that aligns well with Uganda’s Public Debt Management Framework, which emphasizes concessional borrowing, grants, and minimizing non-concessional debt.

Key projects under the Human Capital Development programme of NDP III have been funded by development partners such as the Global Fund, USAID, and the Government of Spain, utilizing a mix of instruments:

- Grants (e.g., from Global Fund and USAID)
- Debt-swap grant mechanisms (e.g., Spanish Debt-Swap arrangement)
- Counterpart financing from the Government of Uganda

These instruments reflect a low-debt-risk approach, supporting climate-resilient health infrastructure and early warning systems without adding significant pressure to the national debt burden.

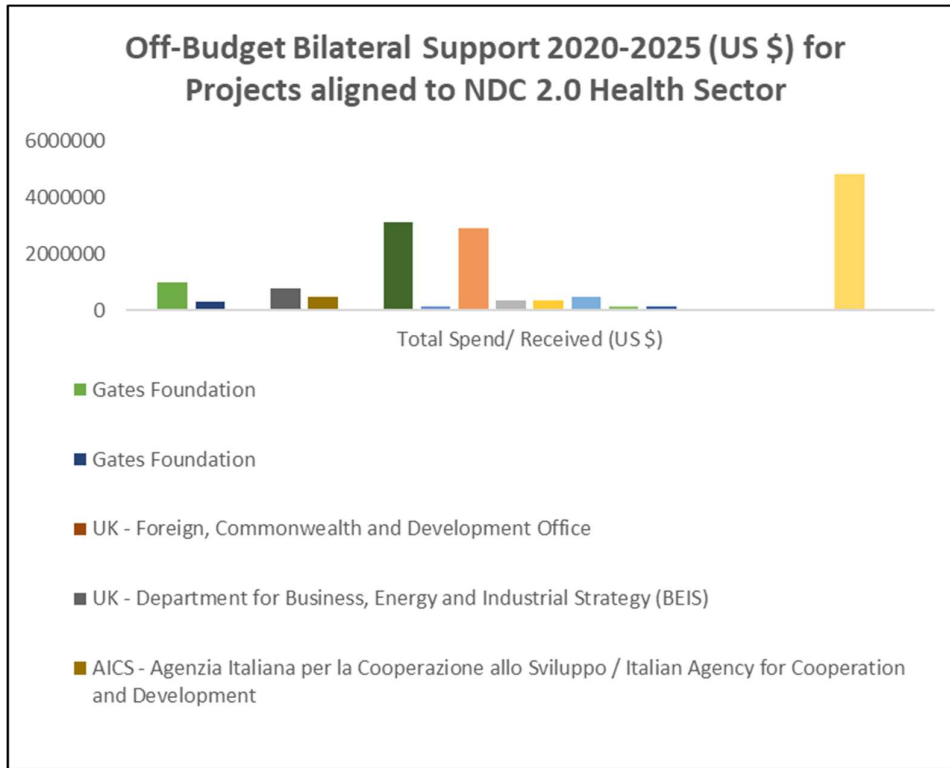
As Uganda prepares its NDC 3.0, this financing model offers a sound precedent. Going forward, expanding access to concessional climate finance and results-based grants, while ensuring alignment with public debt sustainability principles, will be essential for scaling health-sector resilience in the face of growing climate threats.

Off- Budget Support for Health NDC 2.0 Adaptation Action

Only 3.1% of the US \$774.86 Million budget for exclusive Conditional Climate Finance targeted for the implementation of priority NDC 2.0 actions in the Health Sector was committed in the period of 2020-2025. However, 63.9% of committed exclusive Conditional Finance was actually received. This indicates that the Health Sector remained critically underfunded regarding exclusive Conditional Climate Finance.

Off-Budget Bilateral Support for Health NDC 2.0 Adaptation	
Climate Finance Committed (US\$ M)	Climate Finance Received (US\$ M)
23.8	15.2

The Gates Foundation, UK-Foreign, Commonwealth and Development Office, UK-Department for Business, Energy and Industrial Strategy (BEIS), as well as the Italian Agency for Cooperation and Development, were the major sources of exclusive Conditional Climate Finance support for NDC 2.0 related activities in the Health Sector.



Forestry NDC 2.0 Adaptation Actions

National Budget Support for Forestry NDC 2.0 Adaptation Action

Table 911:- Support Needed and Received for Forestry NDC 2.0 Adaptation Actions

NDC-aligned Climate Finance Needed & Received for Forestry NDC 2.0 Adaptation Actions							
Climate Finance Needed			Climate Finance Approved			Climate Finance Actually Received	Climate Finance Progress
Conditional	Unconditional	Total (US\$ M)	Conditional	Unconditional	Total (US\$ M)	Total (US \$ M)	Total (USD \$ M)
5478.75	214.75	5693.5	148.2	30	178.2	20.95	-5672.55

The forestry sector under Uganda's NDC 2.0 requires US\$5.69 billion in climate finance (mostly conditional). However, only US\$178.2 million has been approved, with just US\$20.95 million received to date. This leaves a substantial financing gap of over US\$5.67 billion, highlighting significant challenges in mobilizing and disbursing climate funds critical for forestry adaptation actions.

Table 9-10: Funding Status for Forestry NDC 2.0 Adaptation Actions

Forestry NDC 2.0 Adaptation Action	Funding Status of Projects relevant to NDC 2.0 Actions							
	NDP Project Title (Code)	NDP III Programme Category for Project	Project Funder	Nature of Instrument	Project Implementer	Project Status	Project Duration Start Year	End Year
Promote afforestation and reforestation (including in Urban Areas)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Sustainable forest management to enhance forest ecosystems	Investing in Forest and Protected Areas for Climate Smart Development (1613)	NRE-CCLWRM	International Development Agency (IDA); GoU	Loan & Grant; Counterpart financing	MWE; Ministry of Tourism, Wildlife & Antiquities; Uganda Wildlife Authority; National Forestry Authority	Ongoing	2021	2026
Promote agro-forestry in landscapes	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Promote use of non-timber forest products (NTFPs) to diversify livelihoods and improve resilience of communities	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Climate finance for forestry adaptation under Uganda's NDC 2.0 is primarily channeled through the Investing in Forest and Protected Areas for Climate Smart Development project, aligned with the NRE-CCLWRM programme of NDP III. This initiative is funded by the International Development Association (IDA) and the Government of Uganda, using a mix of loans, grants, and counterpart financing.

Implemented by the Ministry of Water and Environment, Ministry of Tourism, Wildlife & Antiquities, Uganda Wildlife Authority, and the National Forestry Authority, the project runs from 2021 to 2026 and focuses on sustainable forest management and ecosystem enhancement.

Although tourism-related activities are integrated within this project, Tourism NDC actions are not explicitly funded through national budget support, reflecting cross-sectoral financing dynamics.

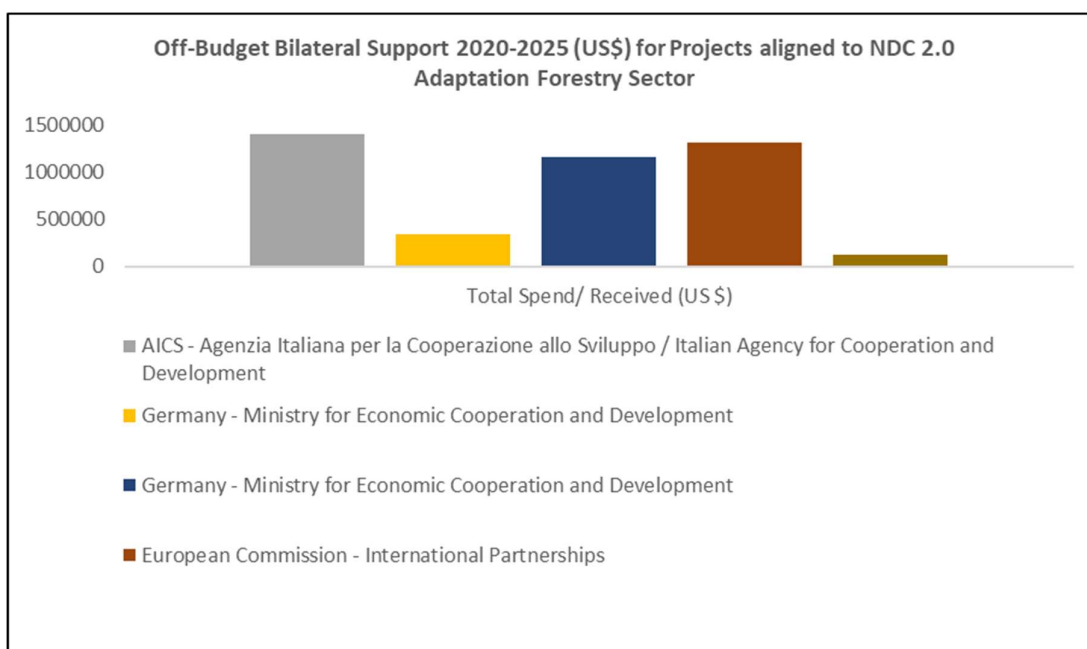
For the next NDC cycle, leveraging blended financing instruments while enhancing collaboration across forestry and tourism sectors presents a key opportunity to broaden resource mobilization and strengthen integrated climate resilience efforts. This approach can help close financing gaps and support sustainable livelihoods linked to forest ecosystems.

Off- Budget Support for Forestry NDC 2.0 Adaptation Action

There's a 99.8% gap between the US \$ 5478.75 Million budgeted as the exclusive Conditional Climate Finance needed and the exclusive Conditional Climate Finance that were committed. Additionally, only 46.3% of the committed exclusive Conditional Climate Finance was received for the implementation of priority NDC 2.0 Forestry Actions.

Off-Budget Bilateral Support for Forestry NDC 2.0 Adaptation	
Climate Finance Committed (US\$ M)	Climate Finance Received (US\$ M)
9.5	4.4

The Italian Agency for Cooperation and Development, European Commission-International Partnerships; and the Germany-Ministry for Cooperation and Development contributed the highest volumes of the exclusive Conditional Climate Finance that was received in the period of 2020-2025 for the implementation of projects whose activities aligned with the priority NDC 2.0 actions in the Adaptation Forestry Sector.



Water & Sanitation NDC 2.0 Adaptation Actions

National Budget Support for Water & Sanitation NDC 2.0 Adaptation Action

Table 9-11; Support Needed and Received for Water & Sanitation NDC 2.0 Adaptation Actions

NDC-aligned Climate Finance Needed & Received for Water & Sanitation NDC 2.0 Adaptation Actions							
Climate Finance Needed			Climate Finance Approved			Climate Finance Actually Received	Climate Finance Progress
Conditional	Unconditional	Total (US\$ M)	Conditional	Unconditional	Total (US\$ M)	Total (US \$ M)	Total (USD \$ M)
2280.32	448.98	2729.3	1185.771	70.556	1256.327	862.96	-1866.34

The Water & Sanitation sector requires US\$2.73 billion in climate finance to support NDC 2.0 adaptation actions, with the majority being conditional funding. So far, US\$1.26 billion has been approved, but only US\$862.96 million has been received. This leaves a financing gap of approximately US\$1.87 billion, underscoring significant challenges in translating approved funds into real disbursements to advance climate resilience in water and sanitation services.

Table 9-12: Funding Status for Water & Sanitation NDC 2.0 Adaptation Actions

Water & Sanitation NDC 2.0 Adaptation Action	Funding Status of Projects relevant to NDC 2.0 Actions							
	NDP Project Title (Code)	NDP III Programme Category for Project	Project Funder	Nature of Instrument	Project Implementer	Project Status	Project Duration Start Year	End Year
Ensure resilient access to water supply for domestic and productive purposes	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Promote sustainable water harvesting, storage and use	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Increase to sanitation and wastewater treatment infrastructure and services	Kampala Water-Lake Victoria Water and Sanitation Project (1193)	Human Capital Development	Mutual Reliance Initiative (KfW Entwicklungsbank; EU Africa Infrastructure Trust Fund-EU-ITF; European Investment Bank; Agence Francaise de Developpement-AFD); GoU	Loan	MWE; National Water and Sewerage Corporation	Ongoing	2011	2025
	Strategic Towns Water Supply and Sanitation Project (1529)	Human Capital Development	African Development Bank; GoU	Loan	MWE; National Water and Sewerage Corporation	Ongoing	2019	2025
	Support to Rural Water Supply and Sanitation Project (1614)	Human Capital Development	EXIM Bank of India; Agence Francaise de Developpement (AFD); GoU (Counterpart Funding)	Loan	MWE	Ongoing	2022	2027
	Integrated Water Management and Development Project (1530)	Human Capital Development	International Development Agency (IDA); KfE; GoU	Loan; Grant; Counterpart funding	MWE; NWSC	Completed	2019	2024
	South Western Cluster Project (1531)	Human Capital Development	AFD; GoU (Counterpart funding)	Loan	National Water and Sewerage Corporation	Ongoing	2019	2024
	Water and Sanitation Development Facility North Phase II (1534)	Human Capital Development	Federal Republic of Germany (KfW Bank); GoU	Grant; GoU (Counterpart Funding)	MWE; NWSC; Umbrella Water Authority	Ongoing	2019	2025
Scale-up Integrated Water Resources Management	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Climate finance for Uganda's Water & Sanitation sector under NDC 2.0 is primarily supported through a combination of loans and grants from multiple international development partners, including major multilateral and bilateral financiers. Funding sources such as European development banks, African development institutions, and international agencies provide these financial instruments, often complemented by Government of Uganda counterpart financing.

These initiatives fall under the Human Capital Development programme of NDP III and are implemented by key agencies like the Ministry of Water and Environment and the National Water and Sewerage Corporation. The projects focus on expanding resilient water supply, improving sanitation infrastructure, and enhancing wastewater treatment services. Most are ongoing, with some nearing completion, collectively strengthening climate adaptation in water and sanitation.

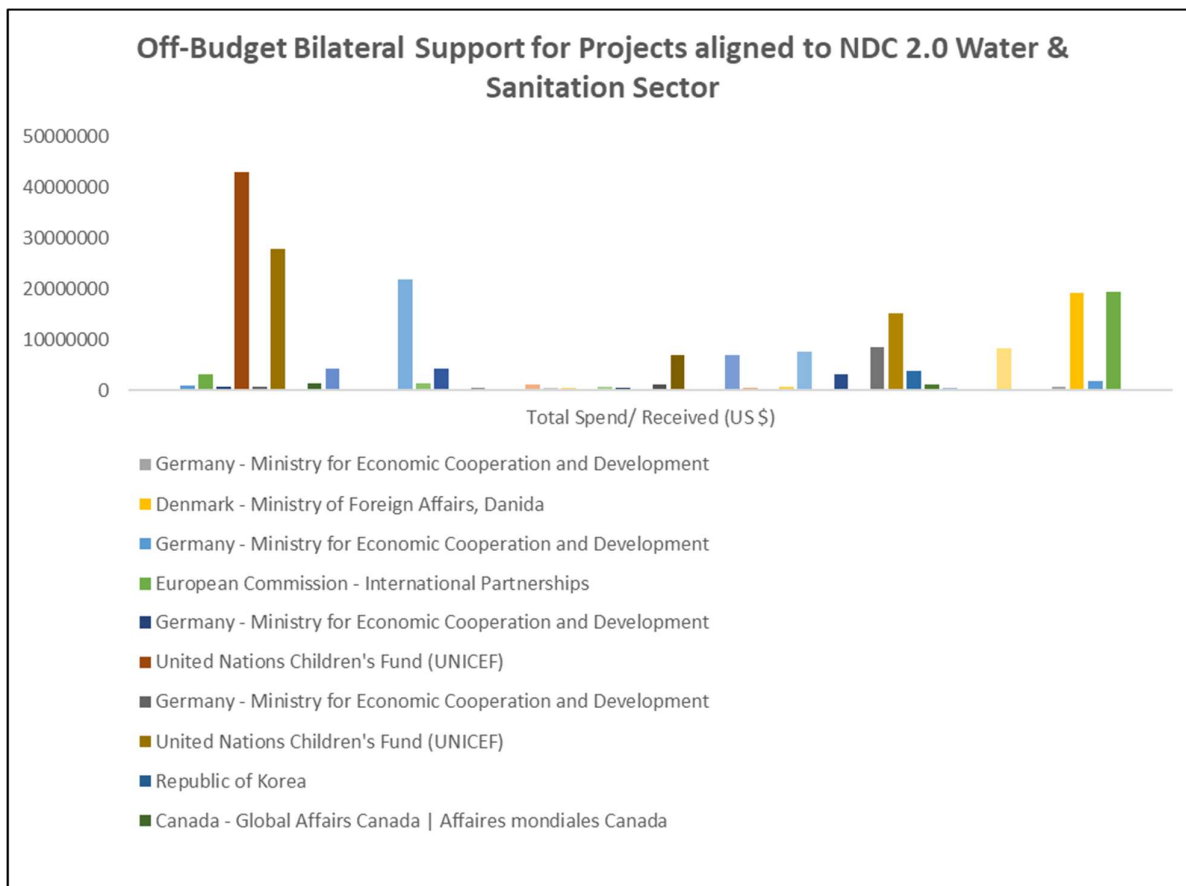
For the next round of NDC, Uganda can harness this established financing framework by continuing to blend concessional loans and grants while ensuring sustainable domestic funding commitments. Enhancing partnerships and diversifying funding instruments will be vital to closing existing finance gaps and scaling up climate-resilient water and sanitation solutions nationwide.

Off- Budget Support for Water & Sanitation NDC 2.0 Adaptation Action

Although only 15.3% of the expected budget for exclusive Conditional Climate Finance was committed, approximately 63.3% of these commitments were received from 2020 to 2025 for the implementation of activities aligned with NDC 2.0 priority actions in the Water & Sanitation Sector. Despite this status, there is still a 90.3% funding gap that is yet to be addressed.

Off-Budget Bilateral Support for Water & Sanitation NDC 2.0 Adaptation	
Climate Finance Committed (US\$ M)	Climate Finance Received (US\$ M)
347.7	220.2

Exclusive Conditional Climate Finance for implementing NDC 2.0 priority actions in Water & Sanitation Sector was mostly channeled through the United Nations Children's Fund (UNICEF), followed by the Germany-Ministry of Economic Cooperation and Development; and in equal measure, the Denmark-Ministry of Foreign Affairs (DANIDA) and the European Commission-International Partnerships



Manufacturing, Industrial Process & Mining NDC 2.0 Adaptation Actions

National Budget Support for Manufacturing, Industrial Process & Mining NDC 2.0 Adaptation Action

Table 9-13: Support Needed and Received for Manufacturing, Industrial Process & Mining NDC 2.0 Adaptation Actions

NDC-aligned Climate Finance Needed & Received for Manufacturing, Industrial Processes & Mining NDC 2.0 Adaptation Actions							
Climate Finance Needed			Climate Finance Approved			Climate Finance Actually Received	Climate Finance Progress
Conditional	Unconditional	Total (US\$ M)	Conditional	Unconditional	Total (US\$ M)	Total (US\$ M)	Total (USD \$ M)
87.08	10.06	97.14	323.57	1.494	325.064	249.92	152.78

The Manufacturing, Industrial Processes, and Mining sector requires US\$97.14 million in climate finance for NDC 2.0 adaptation actions. However, the amount approved is significantly higher, at US\$325.06 million—which is about 335% of the estimated need. Of this approved amount, US\$249.92 million has been received, representing 77% of the approved finance and 257% of the initially estimated need.

While the sector has successfully mobilized and received more than double the initially identified financing requirement, there remains a gap of approximately 23% between approved funds and actual disbursements. This shows strong progress in securing climate finance but also highlights the need to accelerate fund absorption and deployment for effective adaptation outcomes.

Table 9-14: Funding Status for Manufacturing, Industrial Process & Mining NDC 2.0 Adaptation Actions

Manufacturing, Industry, Production & Mining NDC 2.0 Adaptation Action	Funding Status of Projects relevant to NDC 2.0 Actions							
	NDP Project Title (Code)	NDP III Programme Category for Project	Project Funder	Nature of Instrument	Project Implementer	Project Status	Project Start Year	Project End Year
Scale-up adoption of resource efficient technologies	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Build capacity in research and innovation	National Science Technology Engineering Innovation-Skills Enhancement Project (1513)	Innovation Technology and Transfer	Exim Bank of China(85%); GoU (Counterpart funding 15%)	Loan	Uganda National Council of Science and Technology; National Enterprise Corporation	Ongoing (Multiyear project)	2019	2024
	Development of Industrial Parks Projects (0994)	Manufacturing	United Kingdom Export Finance (UKEF); Standard Chartered Bank UK; GoU	Loan	Uganda Investment Authority	Ongoing	2020	2025
Promote circular economy	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

In the Manufacturing, Industry, Production & Mining sector, the bulk of climate finance under NDC 2.0 has been directed towards building capacity in research and innovation, mainly through multiyear projects supported by concessional loans. Key financiers include the Exim Bank of China, providing 85% of funding through loans, complemented by a 15% counterpart contribution from the Government of Uganda. Another significant source is the United Kingdom Export Finance (UKEF) alongside commercial partners like Standard Chartered Bank UK, supporting industrial development projects with loan financing. These initiatives fall under the Innovation Technology and Transfer and Manufacturing programmes of NDP III and are currently ongoing.

However, other critical adaptation actions such as scaling up resource-efficient technologies and promoting circular economy practices remain unfunded, representing a significant 100% funding gap in these areas. This highlights that while substantial financing is concentrated on innovation and industrial park development, other NDC priorities in the sector have yet to attract investment.

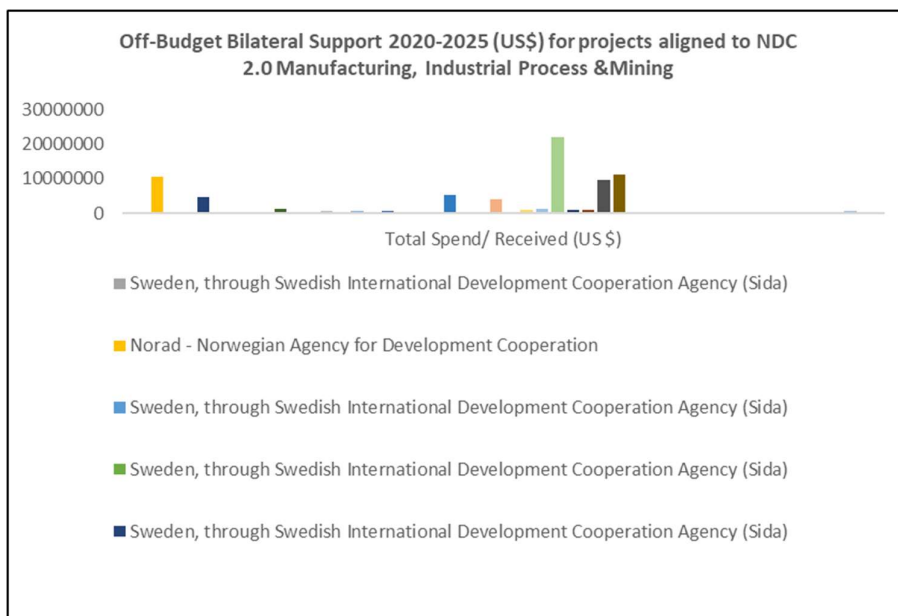
For the next round of the NDC, broadening the financing portfolio to include these unfunded actions will be essential. This could involve leveraging a mix of grants, blended finance, and concessional loans, and actively engaging new sources such as climate funds and private sector investments. Strengthening coordination to ensure balanced funding across all priority actions will maximize impact and accelerate Uganda’s transition to a climate-resilient and resource-efficient industrial sector.

Off-Budget Support for Manufacturing, Industrial Process & Mining NDC 2.0 Adaptation Action

Even when the funding gap under this modality of financing for this sector stands at 11.5%, commitments stood at 99.7% of the exclusive Conditional Climate finance that was needed. Additionally, 88.8% of the committed exclusive Conditional Climate Finance for this Sector was received.

Off-Budget Bilateral Support for Manufacturing, Industrial Process & Mining NDC 2.0 Adaptation	
Climate Finance Committed (US\$ M)	Climate Finance Received (US\$ M)
86.8	77.1

The Swedish International Development Cooperation Agency (SIDA) and the Norwegian Agency for Development and Cooperation (NoRAD) are the main sources of the exclusive Conditional Climate Finance that has been received from 2020 to 2025 for the implementation of activities that are aligned to the priority NDC 2.0 actions for the Manufacturing, Industrial Process & Mining



Transport NDC 2.0 Adaptation Actions

National Budget Support for Transport NDC 2.0 Adaptation Action

Under Uganda’s NDC 2.0, the Transport sector was expected to be primarily funded through conditional sources, with a total estimated need of US\$1.03 billion—of which 77% (US\$793.75 million) was anticipated from external support. However, climate finance approvals for the sector have exceeded expectations by over 300%, reaching US\$4.27 billion. Of this, US\$2.47 billion (58%) has already been received, significantly surpassing the original climate financing need and reflecting strong momentum in mobilizing funds.

This performance highlights Uganda’s readiness and credibility in accessing concessional external debt for productive infrastructure, particularly in transport. This approach aligns with key national frameworks including the National Public Debt Management Policy, which prioritizes concessional borrowing for high-impact development sectors, and the National Climate Finance Strategy, which encourages leveraging external finance for climate-resilient infrastructure. It also resonates with the National Green Taxonomy, which identifies green transport investments as a key area for climate-aligned finance.

Moving into NDC 3.0, Uganda is well-positioned to continue this trajectory by strategically expanding its transport investments through climate-aligned financial instruments—particularly concessional loans, climate bonds, and blended finance. The experience in this sector sets a strong precedent for scaling up finance in other infrastructure-heavy sectors and for deepening the integration of climate finance planning within national development strategies.

Table 9-15: Support Needed and Received for Transport NDC 2.0 Adaptation Actions

NDC-aligned Climate Finance Needed & Received for Transport NDC 2.0 Adaptation Actions							
Climate Finance Needed			Climate Finance Approved			Climate Finance Actually Received	Climate Finance Progress
Conditional	Unconditional	Total (US\$ M)	Conditional	Unconditional	Total (US\$ M)	Total (US \$ M)	Total (USD \$ M)
793.75	237.47	1031.22	3459.832	805.43	4265.26	2473.496	1442.276

Table 9-16: Funding Status for Transport NDC 2.0 Adaptation Actions

Funding Status of Projects relevant to NDC 2.0 Actions							
NDP Project Title (Code)	NDP III Programme Category for Project	Project Funder	Nature of Instrument	Project Implementer	Project Status	Project Duration	
						Start Year	End Year
Atiak-Moyo-Afoji: Atiak-Laropi (66km)-Lot 1 (0265)	Integrated Transport Infrastructure & Services Development	GoU (15%); European Development Fund/ European Union Delegation (85%)	Loan	Uganda National Roads Authority (UNRA)	Ongoing	2020	2024
Namagumba-Budadidiri-Nalugugu Road Project (1794)	Integrated Transport Infrastructure & Services Development	African Development Bank; GoU	Loan	Ministry of Works and Transport; UNRA	Ongoing	2023	2027
Upgrading Rukungiri-Kihihi-Ishasha/Kanungu Road (1311)	Integrated Transport Infrastructure & Services Development	African Development Fund (85.11%); GoU (14.89%)	Loan	Uganda National Roads Authority (UNRA)	Ongoing (with some completed sections)	2015	2024
Upgrading of Tirinyi-Pallisa-Kumi/Kamukoli Road Project (1281)	Integrated Transport Infrastructure & Services Development	Islamic Development Bank (73.9%); GoU (26.1%)	Loan	Uganda National Roads Authority (UNRA)	Completed	2018	2024
The North Eastern Road Corridor Asset Management Project (NERAMP) (1313)	Integrated Transport Infrastructure & Services Development	International Development Agency (IDA 95.6%); GoU 4.4%	Loan	Uganda National Roads Authority (UNRA)	Ongoing	2015	2027
Moyo-Yumbe-Koboko Road Project (1657)	Integrated Transport Infrastructure & Services Development	World Bank; GoU (Availability of road corridor, land acquisition & PAPs resettlement plan)	Grant	Uganda National Roads Authority (UNRA)	Ongoing (Near Completion)	2021	2025
Luwero-Butalangu Road Project (1490)	Integrated Transport Infrastructure & Services Development	GoU (USD 15.5 M); Arab Bank for Economic Development in Africa-BADEA-USD 13.0 M; OPEC Fund for Economic Development (OFID) USD 11.5 M		Uganda National Roads Authority (UNRA)	Ongoing (Near completion)	2017 (BADEA) & 2019 (OFID)	2026 (BADEA); 2024 (OFID)
Upgrading of Muyembe-Nakapiripit Road Project (1322)	Integrated Transport Infrastructure & Services Development	Islamic Development Bank (97%); GoU (3%)	Loan	Uganda National Roads Authority (UNRA)	Ongoing (New Closure)	2019	2025
Kampala City Roads Rehabilitation Project (1253)	Integrated Transport Infrastructure & Services Development	GoU (3.8%); GEF (0.7%); African Development Bank Group (ADB (77.8%) & ADF (17.7%))	Loan; Grant (GEF)	Kampala Capital City Authority (KCCA)	Ongoing	2021	2027
Upgrading of Rwenkuny- Apac-Lira-Acholilur Road Project (1402)	Integrated Transport Infrastructure & Services Development	Islamic Development Bank (100%); GoU (Land acquisition)	Loan	Uganda National Roads Authority (UNRA)	Ongoing (Near completion)	2020	2028
Entebbe Airport Rehabilitation Phase 1 Project (1373)	Integrated Transport Infrastructure & Services Development	EXIM Bank of China; Uganda Civil Aviation Authority	Loan; Internal Revenues	Ministry of Works and Transport; Uganda Civil Aviation Authority (UCAA)	Ongoing (Near completion)	2016	2024
Kampala Institutional and Infrastructure Development Project -2 (1259)	Integrated Transport Infrastructure & Services Development	International Development Agency (IDA); GoU	Loan	Kampala Capital City Authority (KCCA)	Completed	2015	2023
Kampala-Jinja Expressway Project (1278)	Integrated Transport Infrastructure & Services Development	GoU (through a PPP); African Development Bank	Loan (57.14%); PPP (42.86%)	UNRA	Ongoing	2021	2027
Development of Kabaale International Airport Project (1489)	Integrated Transport Infrastructure & Services Development	United Kingdom Export Finance (UKEF); GoU	Loan (85%); Counterpart Financing (15%)	Ministry of Works and Transport; Uganda Civil Aviation Authority (UCAA)	Ongoing (Near completion)	2018	2025
Hoima-Wanseko Road (Critical Oil Roads) Project (1176)	Integrated Transport Infrastructure & Services Development	Exim Bank of China; GoU	Loan	UNRA	Ongoing (some parts completed)	2020	2025
Multinational Kapchorwa-Suam-Kitale and Eldoret Bypass Road Project (1040)	Integrated Transport Infrastructure & Services Development	GoU; African Development Bank (AfDB); African Development Fund (ADF)	Loan	UNRA	Completed	2018	2024
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Uganda's Integrated Transport Infrastructure & Services Development programme under NDP III has seen significant climate-relevant investments aligned with NDC 2.0 adaptation objectives. These investments are primarily structured around loan financing—often concessional—complemented by grants and government counterpart funding. The financing architecture reflects Uganda's strategic recourse to external concessional debt for productive infrastructure, as provided for in the National Public Debt Management Policy, National Climate Finance Strategy, and the National Green Taxonomy.

Financial Instruments and Sources

Across the sector:

- Over 90% of projects are financed through loans, with leading financiers including the African Development Bank (AfDB), Islamic Development Bank (IsDB), World Bank (IDA), European Development Fund, UKEF, and EXIM Bank of China.

- Grants, such as those from the Global Environment Facility (GEF), are used sparingly, often to complement loan packages.
- The Government of Uganda consistently contributes counterpart funding, covering between 3% and 26% of project costs, mostly for land acquisition and resettlement planning.

Project Status Overview

- Many of the projects are ongoing, with several categorized as near completion.
- A smaller number have been fully completed, particularly those initiated before 2020.
- Projects span various modalities including public-private partnerships (PPP), particularly for large-scale expressway developments.

Implications for Climate Finance Audit & NDC 3.0

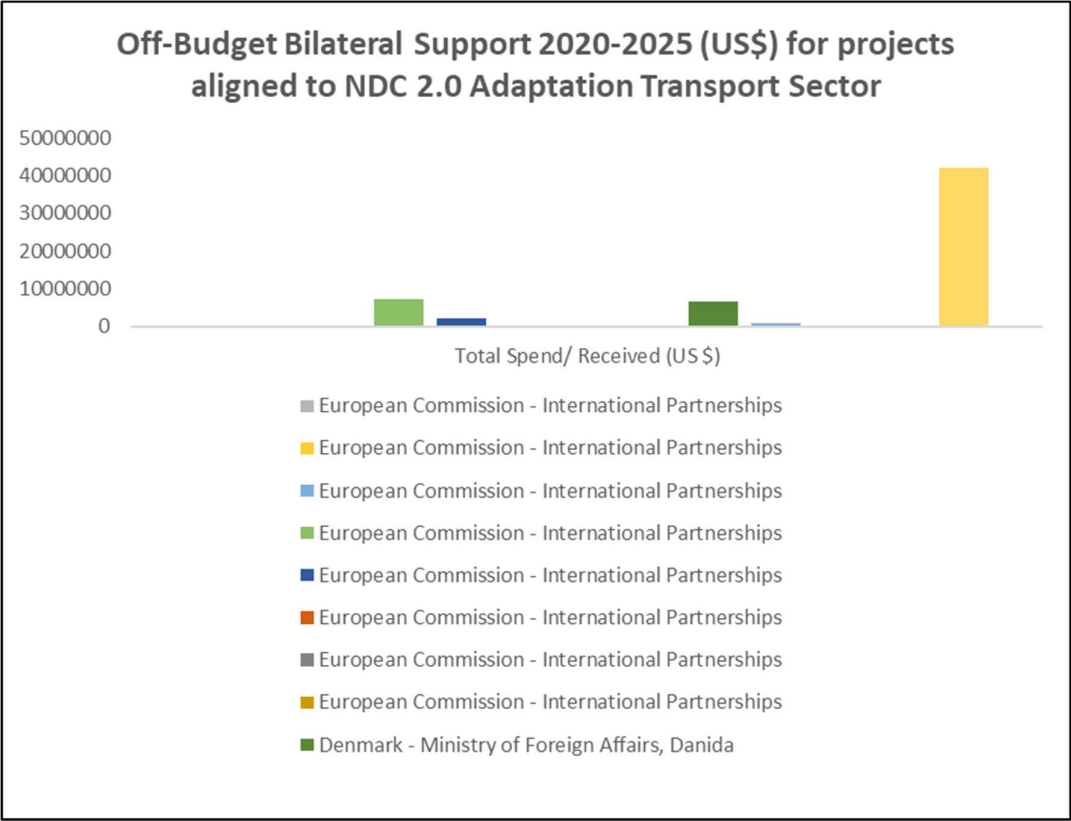
- **Instrument Diversity:** The sector heavily relies on concessional debt, suggesting fiscal space remains aligned with Uganda's debt sustainability thresholds for productive infrastructure. However, there's room to increase the use of grants, climate bonds, and blended finance to diversify financial risk and reduce future debt burdens.
- **Alignment with National Strategies:** The clear use of concessional loans and counterpart contributions confirms alignment with Uganda's public debt and climate finance policies, ensuring compliance with national fiscal prudence and green finance priorities.
- **Project Readiness & Execution:** The strong pipeline of ongoing and nearly completed projects signals institutional capacity for delivery, an essential consideration as Uganda prepares its next NDC (3.0). The challenge remains in ensuring that climate resilience components are fully integrated into infrastructure designs and tracked under climate finance reporting.
- **Audit Considerations:** For those auditing financial instruments, attention should be paid to:
 - Loan terms (e.g., interest rates, grace periods, maturity)
 - Grant-to-loan ratios
 - Timeliness of disbursements
 - Climate tagging and alignment of expenditures with NDC priorities

In conclusion, Uganda's transport sector provides a robust example of climate-aligned infrastructure financing through strategic use of concessional debt. As the country moves into NDC 3.0, embedding climate-smart criteria across all transport projects and leveraging more climate-specific funding sources will be essential for deepening adaptation impact and maintaining fiscal sustainability.

Off-Budget Support for Transport NDC 2.0 Adaptation Action

Committed Conditional Climate Finance exceeded that needed for this sector by 41%. However, only 7.5% of the exclusive Conditional Climate Finance needed was received.

The European Commission-International Partnerships and the Denmark-Ministry of Foreign Affairs, DANIDA- have contributed the hugest volumes of the exclusive Conditional Climate Finance that was received for implementing activities in line with the priority NDC 2.0 Adaptation actions for the Transport Sector.



Energy NDC 2.0 Adaptation Actions

National Budget Support for Energy NDC 2.0 Adaptation Action

Table 9-17: Support Needed and Received for Energy NDC 2.0 Adaptation Actions

NDC-aligned Climate Finance Needed and Received for Energy NDC 2.0 Adaptation Actions							
Climate Finance Needed			Climate Finance Approved			Climate Finance Actually Received	Climate Finance Progress
Conditional	Unconditional	Total (US\$ M)	Conditional	Unconditional	Total (US\$ M)	Total (US \$ M)	Total (USD \$ M)
457.26	34.08	491.34	932.72	46.975	979.695	307.5948	-183.7452

Under Uganda's NDC 2.0, the Energy sector identified a total climate finance need of US\$491.34 million, with 93% of that expected from conditional (external) sources. However, as of the latest report, a total of US\$979.70 million has been approved, which is nearly 200% of the originally estimated need.

Despite this over-commitment in approvals, only US\$307.59 million (about 31% of the approved amount and 63% of the needed amount) has been received. This leaves a shortfall of US\$183.75 million relative to the original financing needs, pointing to ongoing disbursement delays or pipeline implementation issues.

This mixed picture reflects both strong success in mobilizing commitments, and the need for strengthened finance absorption capacity, project readiness, and delivery systems to fully capitalize on available resources for climate-resilient energy adaptation.

Table 9-18: Funding Status for Energy NDC 2.0 Adaptation Actions

Energy NDC 2.0 Adaptation Action	Funding Status of Projects relevant to NDC 2.0 Actions							
	NDP Project Title (Code)	NDP III Programme Category for Project	Project Funder	Nature of Instrument	Project Implementer	Project Status	Project Duration Start Year	Project Duration End Year
Improve access and utilization of electricity from sustainable sources	Uganda Rural Electricity Access Project/UREAP (1518)	Sustainable Energy Development	GoU; African Development Bank (AfDB); EU-Africa Infrastructure Trust Fund (EU-AITF)	Counterpart funding; Loan; Grant	Ministry of Energy and Mineral Development (MEMD)	Ongoing (Near Completion)	2016	Q2 2025
	Electricity Access Scale Up Project (1775)	Sustainable Energy Development	World Bank; Private Sector; GoU	Loan & Grant; Private Sector; Counterpart funding	Ministry of Energy and Mineral Development (MEMD)	Ongoing	2022	2027
	Bridging the Demand Supply Gap through the Accelerated Rural Electrification Programme/AREP (1517)	Sustainable Energy Development	EXIM Bank of China; GoU	Loan; Counterpart funding	Ministry of Energy and Mineral Development (MEMD)	Near completion	2019	2025
Promote use of renewable energy sources and energy efficient technologies	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Increase access to clean energy cooking technologies	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Rehabilitate and climate proof electricity transmission infrastructure	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Uganda's energy sector under NDC 2.0 has made progress in mobilizing climate finance for adaptation, particularly around rural electrification and access to sustainable energy. Three major projects are currently ongoing or nearing completion, implemented under the Sustainable Energy Development programme of NDP III. These projects are financed through a mix of loans, grants, government counterpart funding, and

private sector contributions. Key financiers include the African Development Bank (AfDB), World Bank, EXIM Bank of China, and the EU-Africa Infrastructure Trust Fund.

Despite this, several priority NDC adaptation actions remain unfunded, including:

- Promotion of renewable energy sources and energy-efficient technologies
- Expansion of clean energy cooking technologies
- Rehabilitation and climate-proofing of electricity transmission infrastructure

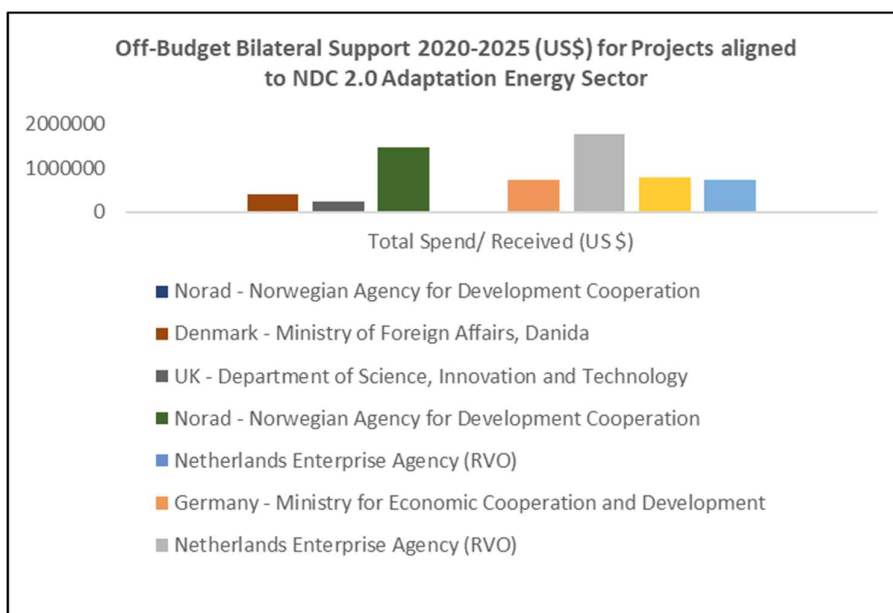
The current focus of financing—about 100% of funded activities—is on access to electricity, leaving critical resilience and efficiency actions unaddressed. As Uganda prepares for NDC 3.0, there's a clear opportunity to:

- Broaden financing instruments to include green bonds, results-based finance, and climate venture capital
- Engage new sources, including climate-focused private equity and philanthropic funds
- Prioritize funding for climate-proofing infrastructure and decarbonizing energy consumption, especially in cooking
- Closing these gaps will be key to aligning Uganda's energy sector with both climate adaptation goals and the country's National Climate Finance Strategy.

Off-Budget Support for Energy NDC 2.0 Adaptation Action

Commitments for exclusive Conditional Climate Finance fell short by 97.4% of the exclusive Conditional Climate finance needed for the Adaptation Actions in the Energy Sector. However, 52.1% of the commitments were received for the period 2020 to 2025.

Off-Budget Bilateral Support for Energy NDC 2.0 Adaptation	
Climate Finance Committed (US\$ M)	Climate Finance Received (US\$ M)
11.9	6.2



Cities & Built Environment NDC 2.0 Adaptation Actions

National Budget Support for Cities & Built Environment NDC 2.0 Adaptation Action

Table 9-19: Support Needed and Received for Cities & Built Environment NDC 2.0 Adaptation Actions

NDC-aligned Climate Finance Needed & Received for Cities & Built Environment NDC 2.0 Adaptation Actions							
Climate Finance Needed			Climate Finance Approved			Climate Finance Actually Received	Climate Finance Progress
Conditional	Unconditional	Total (US\$ M)	Conditional	Unconditional	Total (US\$ M)	Total (US \$ M)	Total (USD \$ M)
1784.4	1140.8	2925.18	968.66	571.31	1539.97	327.1	-2598.08

Under Uganda's NDC 2.0, the Cities & Built Environment sector identified a total climate finance need of US\$2.93 billion, with 61% expected from conditional (external) sources and 39% from unconditional (domestic) sources.

As of the latest status:

- Approved climate finance totals US\$1.54 billion, covering only 53% of the estimated need.
- Actual disbursements stand at just US\$327.1 million, which is only 11% of the required funding and 21% of what has been approved.
- This leaves a funding gap of US\$2.6 billion, highlighting a significant shortfall in both commitments and actual flows.

The slow pace of disbursement and low absorption relative to approved amounts signals the need for improved project readiness, execution capacity, and alignment with climate finance mechanisms. As Uganda prepares for NDC 3.0, strengthening financial instruments for urban resilience, infrastructure upgrading, and climate-smart urban planning will be critical. Unlocking both domestic and concessional climate finance will also be essential to meet adaptation goals in fast-growing urban areas.

Table 9-20: Funding Status for Cities & Built Environment NDC 2.0 Adaptation Action

Cities & Built Environment NDC 2.0 Adaptation Action	Funding Status of Projects relevant to NDC 2.0 Actions							
	NDP Project Title (Code)	NDP III Programme Category for Project	Project Funder	Nature of Instrument	Project Implementer	Project Status	Project Duration	
							Start Year	End Year
Promote sustainable urbanization and housing	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Expand and maintain cities with greenbelts	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Promote efficient mobility in cities	Uganda Support to Municipal Infrastructure Development- Additional Financing (1514)	Sustainable Housing & Urbanization	World Bank	Loan & Grant	Ministry of Lands, Housing and Urban Development	Ongoing (Near Completion)	2019	2024
	Greater Kampala Metropolitan Area Urban Development Program (1798)	Sustainable Housing & Urbanization	World Bank; Agence Francaise de Development (AFD); GoU	Loan & Grant; Counterpart Financing	Kampala Capital City Authority; neighbouring district local governments of Mpigi, Wakiso, Mukono; and respective urban authorities of Nansana, Kira, Mukono, Makindye-Ssabagabo and Entebbe	Ongoing	2023	2027
Improve waste management	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Climate finance for Uganda’s Cities & Built Environment adaptation actions has mainly been mobilized through a mix of loans, grants, and government counterpart funding from sources such as the World Bank, African Development Bank, and EU partners. Most financing has supported urban infrastructure development, especially in stormwater drainage, roads, and housing.

While some projects are ongoing, disbursement remains low relative to approvals. Key NDC actions—such as climate-resilient urban planning and green public spaces—remain largely unfunded.

As Uganda moves toward NDC 3.0, there’s an opportunity to:

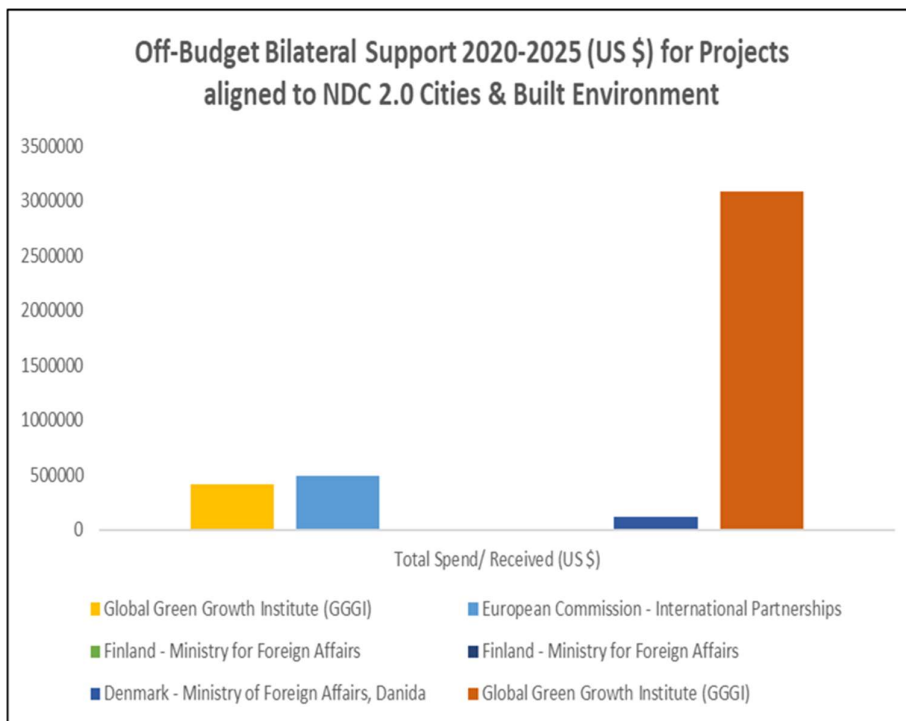
- Expand financing to cover broader urban resilience priorities,
- Integrate municipal green bonds and public-private partnerships, and
- Align future investments with the National Green Taxonomy and Climate Finance Strategy for more targeted impact.

Off Budget Support for Cities & Built Environment NDC 2.0 Adaptation Action

Only 0.3% of the budgeted exclusive Conditional Climate Finance was committed from 2020 to 2025. However, 85.4% of these commitments were received, in the same period.

Off-Budget Bilateral Support for Cities & Built Environment NDC 2.0 Adaptation	
Climate Finance Committed (US\$ M)	Climate Finance Received (US\$ M)
4.8	4.1

The Global Green Growth Institute (GGGI); European Commission-International Partnerships; and Denmark-Ministry of Foreign Affairs (DANIDA) were the main sources, by volume of the total exclusive Conditional Climate Finance received for the implementation of Adaptation Activities for the Cities & Built Environment Sector from 2020-2025.



Off Budget Support for NDC 2.0 Adaptation Sectors unfunded under National Budget Support

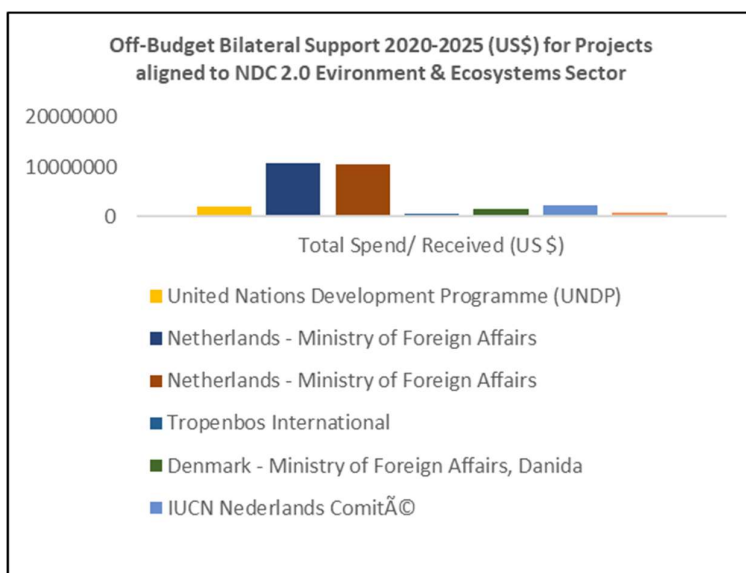
Three priority NDC 2.0 Adaptation Sectors appeared not to have been funded with financial resources channeled through the National Budget, namely, Disaster Risk Reduction; Tourism; and Environment & Ecosystems. These were, however, financed with monies channeled through Off-Budget Support sources. The details are provided in the sections that follow.

Off Budget Support for Environment & Ecosystems NDC 2.0 Adaptation Actions

90.8% of the budgeted exclusive Conditional Climate Finance for the implementation of NDC 2.0 Environment & Ecosystems actions, was committed for the period 2020-2025. Approximately 75% of these commitments have been received, during this period

Off-Budget Bilateral Support for Environment & Ecosystems NDC 2.0 Adaptation	
Climate Finance Committed (US\$ M)	Climate Finance Received (US\$ M)
38.2	28.7

Most of the received exclusive Conditional Climate Finance for Environment & Ecosystems was channeled through the Netherlands-Ministry of Foreign Affairs; Tropenbos International; IUCN Netherlands; Denmark-Ministry of Foreign Affairs (DANIDA); and the United Nations Development Programme (UNDP)



Off Budget Support for Tourism NDC 2.0 Adaptation Actions

37.3% of Tourism's budget for exclusive Conditional Climate Finance was committed from 2020 to 2025. However, during this duration, only 29.8% of these commitments have been received.

Off-Budget Bilateral Support for Tourism NDC 2.0 Adaptation	
Climate Finance Committed (US\$ M)	Climate Finance Received (US\$ M)
28.2	8.4

The main channels for this modality of climate financing targeting Tourism Adaptation Actions were; the European Commission-International Partnerships; and the Germany-Ministry for Economic Cooperation and Development.

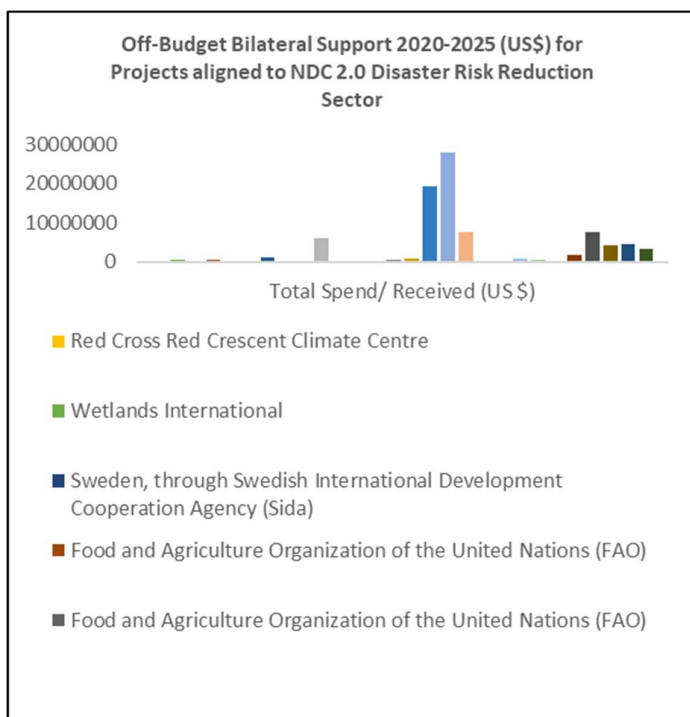


Off Budget Support for Disaster Risk Reduction NDC 2.0 Adaptation Actions

Commitments for exclusive Conditional Climate Finance for activities relevant for the Disaster Risk Reduction Sector, were 238.2% of the budget for this modality of climate finance. Up to 85.5% of the commitments have been received in the duration from 2020 to 2025.

Off-Budget Bilateral Support for Disaster Risk Reduction NDC 2.0 Adaptation	
Climate Finance Committed (US\$ M)	Climate Finance Received (US\$ M)
104.8	89.6

Food and Agriculture Organization of the United Nations (FAO); Sweden, through Swedish International Development Cooperation Agency (SIDA); Red Cross Red Crescent Climate Centre; and Wetlands International, have been critical channels of this modality of financing.



9.1.3 Support for NDC 2.0 Sectoral Mitigation Actions

Three (3) of the five (5) priority NDC 2.0 Mitigation sectors appear to have actions and activities funded via national budget support. The Mitigation sectors whose actions and activities are not funded, include; Waste and Industrial Processes & Production Units (IPPU).

AFOLU NDC 2.0 Mitigation Actions

National Budget Support for AFOLU NDC 2.0 Mitigation Actions

Table 9-21: Support Needed and Received for AFOLU NDC 2.0 Mitigation Actions

NDC-aligned Climate Finance Needed and Received for AFOLU NDC 2.0 Mitigation Actions							
Climate Finance Needed			Climate Finance Approved			Climate Finance Actually Received	Climate Finance Progress
Conditional	Unconditional	Total (US\$ M)	Conditional	Unconditional	Total (US\$ M)	Total (US\$ M)	Total (USD \$ M)
6465.69	1127.01	7592.7	5.86	0.858	6.718	2.33	-7590.37

Uganda's AFOLU sector under NDC 2.0 identified a climate finance need of US\$7.59 billion for mitigation actions. However, only US\$6.72 million (0.09%) has been approved, with just US\$2.33 million received—leaving a funding gap of US\$7.59 billion.

This stark shortfall highlights the urgent need to prioritize AFOLU in resource mobilization for NDC 3.0 and improve access to targeted mitigation financing for land use and carbon sinks.

Table 9-22: Funding Status for AFOLU NDC 2.0 Mitigation Actions

AFOLU NDC 2.0 Mitigation Action	Funding Status of Projects relevant to NDC 2.0 Actions							
	NDP Project Title (Code)	NDP III Programme Category for Project	Project Funder	Nature of Instrument	Project Implementer	Project Status	Project Duration	
							Start Year	End Year
Climate Smart Agriculture – Sustainable land Management/ use and Agroforestry Practices	Coffee and Cocoa Value Chains Development Project/ EU-EAC Market Access Upgrade Programme MARKUP(1202)	Agro-Industrialization	European Union (European Development Fund-EDF)	Grant	Uganda Coffee Development Authority	Ongoing	2022	2027
Climate Smart Agriculture – vegetables production in greenhouses	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Small Scale Commercial small-holder and community bioenergy woodlots	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Small Scale Commercial small-holder and community poles & timber plantations (with coffee agroforestry)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Small scale fuel production -Improved charcoal kilns linked to bioenergy woodlots	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Large scale Commercial transmission pole and timber plantations	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Large scale Commercial pole and sawlog plantations	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Designated areas for natural forest regeneration	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Restoration of degraded protected natural forest	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Energy Efficient fuelwood and charcoal stoves	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Climate Smart Livestock rearing in the cattle corridor	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Wetland and Peatland management	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

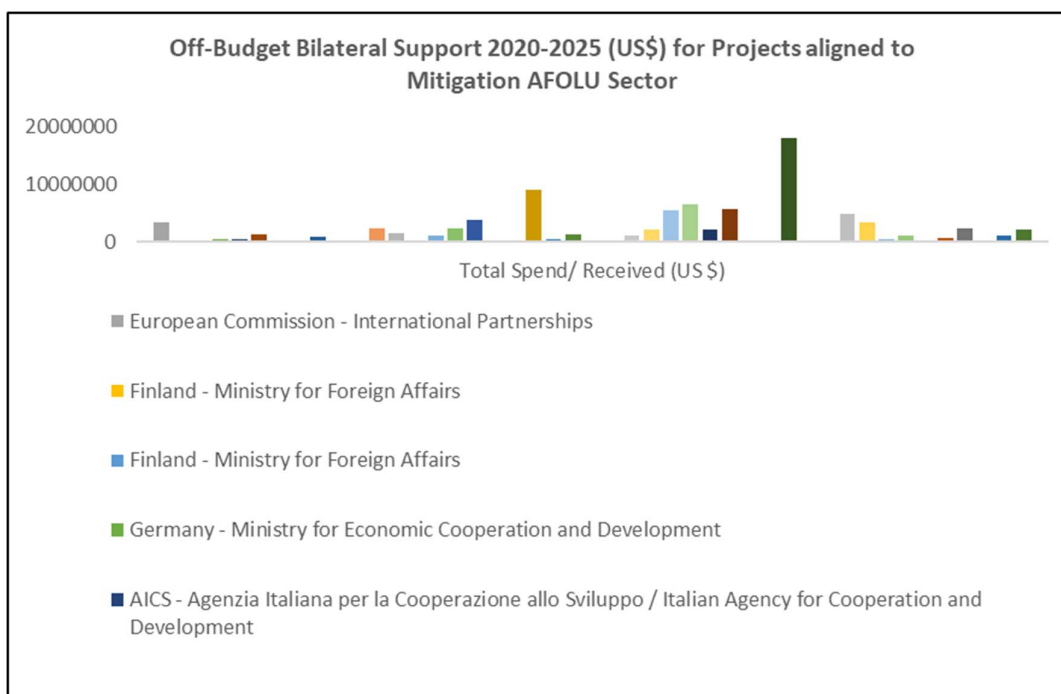
The AFOLU sector's climate-smart agriculture initiatives, such as sustainable land management and agroforestry, are currently supported through ongoing projects funded by grants from the European Union. However, many critical NDC 2.0 mitigation actions—including greenhouse vegetable production, bioenergy woodlots, timber plantations, natural forest restoration, and efficient fuelwood technologies—lack dedicated funding and project support. Strengthening financing and project implementation in these areas will be key to advancing Uganda's climate goals in the next NDC cycle.

Off-Budget Support for AFOLU NDC 2.0 Mitigation Actions

From 2020 to 2025, only 5.5% of the budgeted exclusive Conditional Climate Finance has been committed for the implementation of activities relevant to the NDC 2.0 AFOLU actions. Moreover, only 24.3% of these commitments have been received during this period.

Off-Budget Bilateral Support for AFOLU NDC 2.0 Mitigation	
Climate Finance Committed (US\$ M)	Climate Finance Received (US\$ M)
358.6	87.3

The main channels for this modality of financing have included; Finland-Ministry of Foreign Affairs, Italian Agency for Cooperation and Development, Germany-Ministry for Economic Cooperation and Development; and the European Commission-International Partnerships.



Energy NDC 2.0 Mitigation Actions

National Budget Support for Energy NDC 2.0 Mitigation Actions

Table 9-23: Support Needed and Received for Energy NDC 2.0 Mitigation Actions

NDC-aligned Climate Finance Needed and Received for Energy NDC 2.0 Mitigation Actions							
Climate Finance Needed			Climate Finance Approved			Climate Finance Actually Received	Climate Finance Progress
Conditional	Unconditional	Total (US\$ M)	Conditional	Unconditional	Total (US\$ M)	Total (US\$ M)	Total (USD \$ M)
1017.45	310.05	1327.5	1679.077	383.82	2062.897	1625.21	297.71

For Energy NDC 2.0 Mitigation Actions, the total climate finance needed was approximately \$1.33 billion, with \$1.68 billion approved and \$1.63 billion received. While funds received exceed initial needs, there remains a gap of about \$298 million to fully support all planned activities, highlighting the need for continued mobilization and efficient use of resources moving forward.

Table 9-24: Funding Status for Energy NDC 2.0 Mitigation Actions

Energy NDC 2.0 Mitigation Action	Funding Status of Projects relevant to NDC 2.0 Actions							
	NDP Project Title (Code)	NDP III Programme Category for	Project Funder	Nature of Instrument	Project Implementer	Project Status	Project Duration Start Year	End Year
Renewable energy generation	Karuma Hydropower Project (1183)	Sustainable Energy Development	EXIM Bank-China (85%); GoU Counterpart Funding (15%)	Loan; Counterpart Funding	Ministry of Energy and Mineral Development; Uganda Electricity Generation Company Limited	Completed	2013	2024
Reduction in transmission and distribution losses	Gulu- Agago Transmission Project (1391)	Sustainable Energy Development	GoU; KfW	Loan; Counterpart Funding	Uganda Electricity Transmission Company Limited (UETCL)	Appears to be Ongoing	2019	2023
	Mirama-Kabale Transmission Project (1409)	Sustainable Energy Development	GoU; Islamic Development Bank	Counterpart Funding; Loan	Uganda Electricity Transmission Company Limited (UETCL)	Ongoing	2015	2024
	Kampala Metropolitan Transmission System Improvement Project (1492)	Sustainable Energy Development	GoU; JICA	Counterpart Funding; Loan	Uganda Electricity Transmission Company Limited (UETCL)	Completed	2017	2023
	Masaka-Mbarara 400 kV Transmission Line Project (1497)	Sustainable Energy Development	GoU; KfW; Agence Francaise de Development (AFD)	Counterpart Funding; Loan	Uganda Electricity Transmission Company Limited (UETCL)	Ongoing	2018	2024
	Grid Expansion and Re-enforcement Project (1426)	Sustainable Energy Development	GoU; International Development	Counterpart Funding; Loan	Uganda Electricity Transmission Company Limited (UETCL)	Ongoing	2018	2024
	Kampala-Entebbe Transmission expansion (1259)	Sustainable Energy Development	GoU; KfW	Grant & Loan; Counterpart Funding	Uganda Electricity Transmission Company Limited (UETCL)	Implementation period was 2014-2017 but the project seems to have continued	2013	2023
Improved efficiency of charcoal production	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Industrial Energy efficiency	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Industrial Fuel switching	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Increased electricity access for households	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Lighting energy efficiency in households	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Cooking mitigation measures, incl. energy efficiency and fuel switch	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Uganda's AFOLU sector under NDC 2.0 identified a climate finance need of US\$7.59 billion for mitigation actions. However, only US\$6.72 million (0.09%) has been approved, with just US\$2.33 million actually received—leaving a funding gap of US\$7.59 billion.

This stark shortfall highlights the urgent need to prioritize AFOLU in resource mobilization for NDC 3.0 and improve access to targeted mitigation financing for land use and carbon sinks.

The Energy NDC 2.0 Mitigation Actions under the NDP III focus heavily on sustainable energy development, primarily through renewable energy generation and transmission infrastructure projects. Key funding sources include concessional loans and counterpart funding from multilateral and bilateral partners such as EXIM Bank-China, KfW, Islamic Development Bank, JICA, AFD, and the World Bank's International Development Agency.

Projects like the Karuma Hydropower have been largely funded through a loan agreement (85% EXIM Bank-China) and are near completion. Transmission projects aimed at reducing losses and expanding the grid are ongoing, funded through a mix of loans, grants, and counterpart funding, implemented by the Uganda Electricity Transmission Company Limited (UETCL).

Notably, areas such as improved charcoal production, industrial energy efficiency, fuel switching, and household-level interventions currently lack dedicated financing under this plan, highlighting opportunities

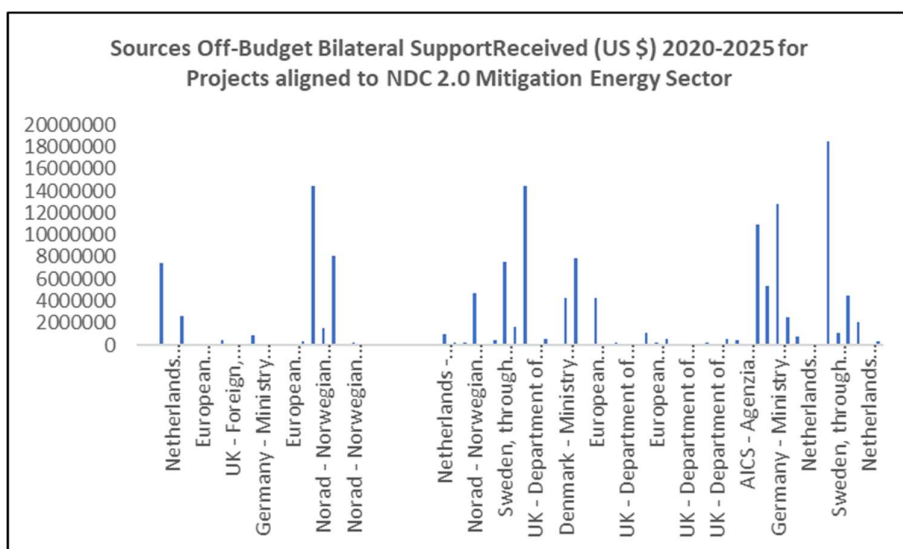
for the next round of NDC to broaden financial instruments and sources to support these critical mitigation actions.

Off-Budget Support for Energy NDC 2.0 Mitigation Actions

Commitments for exclusive Conditional Climate Finance, for the implementation of activities relevant to the NDC 2.0 Mitigation actions in the Energy Sector stood at 104.5% of the budget of this financing modality. However, only 13.9% of these commitments were received in the period from 2020 to 2025.

Off-Budget Bilateral Support for Energy NDC 2.0 Mitigation	
Climate Finance Committed (US\$ M)	Climate Finance Received (US\$ M)
1063.3	147.3

Several bilateral Development Partners have been instrumental in the reception of this modality of financing for the Energy-Mitigation Sector. These have included; Germany-Ministry of Development Cooperation; Netherlands-Ministry of Foreign Affairs; Sweden-through SIDA; Norwegian Agency for Development Cooperation (NORAD); UK Foreign Commonwealth and Development Office; Italian Agency for Development and Cooperation; and the European Commission-International Partnerships.



Transport NDC 2.0 Mitigation Actions

National Budget Support for Transport NDC 2.0 Mitigation Actions

Table 9-25: Support Needed and Received for Transport NDC 2.0 Mitigation Actions

NDC-aligned Climate Finance Needed & Received for Transport NDC 2.0 Mitigation Actions							
Climate Finance Needed			Climate Finance Approved			Climate Finance Actually Received	Climate Finance Progress
Conditional	Unconditional	Total (US\$ M)	Conditional	Unconditional	Total (US\$ M)	Total (US \$ M)	Total (USD \$ M)
613.54	38.81	652.35	526.83	8.99	535.82	346.93	-305.42

For the Transport sector's NDC 2.0 Mitigation Actions, a total of \$652.35 million in climate finance is needed, with \$613.54 million conditional and \$38.81 million unconditional. So far, \$535.82 million has been approved, primarily from conditional sources, but only \$346.93 million has been received. This indicates a significant funding gap of approximately \$305 million remains to fully meet the sector's climate goals. Strengthening disbursement mechanisms and mobilizing additional resources will be critical to accelerate progress in the next NDC cycle.

Table 9-26: Funding Status for Transport NDC 2.0 Mitigation Actions

Transport NDC 2.0 Mitigation Action	Funding Status of Projects relevant to NDC 2.0 Actions							
	NDP Project Title (Code)	NDP III Programme Category for Project	Project Funder	Nature of Instrument	Project Implementer	Project Status	Start Year	End Year
Road transport fuel efficiency	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Alternative fuel switch (use of less carbon-intensive fuels (including compressed natural gas, ethanol, and liquefied petroleum gas) in vehicles, instead of relying heavily on gasoline and diesel fuels)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Modal shift of passengers to Mass Rapid & Non-Motorized Transport (modes of transport that consider greenhouse gas emissions reduction such as use of non-motorised transport and increase access to high-volume public transport options to replace single-car commuter trips in cities and urban areas)	Kampala Flyover Construction and Road Upgrading Project (1319)	Integrated Transport Infrastructure & Services Development	Japan International Cooperation Agency (JICA)	Loan	UNRA	Ongoing	2016	2027
	Busega-Mpigi Express project (1404)	Integrated Transport Infrastructure & Services Development	African Development Bank (AfDB); Africa Development Fund (ADF); GoU	Loan		Ongoing	2017	2025
	Development of the New Bukasa Port Project (1248)	Integrated Transport Infrastructure & Services Development	European Export and Trade Bank (42.48%); Commerz Bank (42.48); GoU (15.04%)	Loan	Ministry of Works and Transport	Ongoing (Towards completion)	2016	2025
MGR-Meter Gauge Railway rehabilitation for freight transit	Uganda Railway Corporation Capacity Building Project (1563)	Integrated Transport Infrastructure & Services Development	African Development Bank (AfDB) 92.13%; Spanish Export Credit Agencies (CESCE) 7.87%	Loan	Uganda Railway Corporation	Ongoing	2021	2026
Promote the development, approval, and effective implementation of a long-term national transport policy and plan that will take greenhouse gas mitigation concerns into account	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

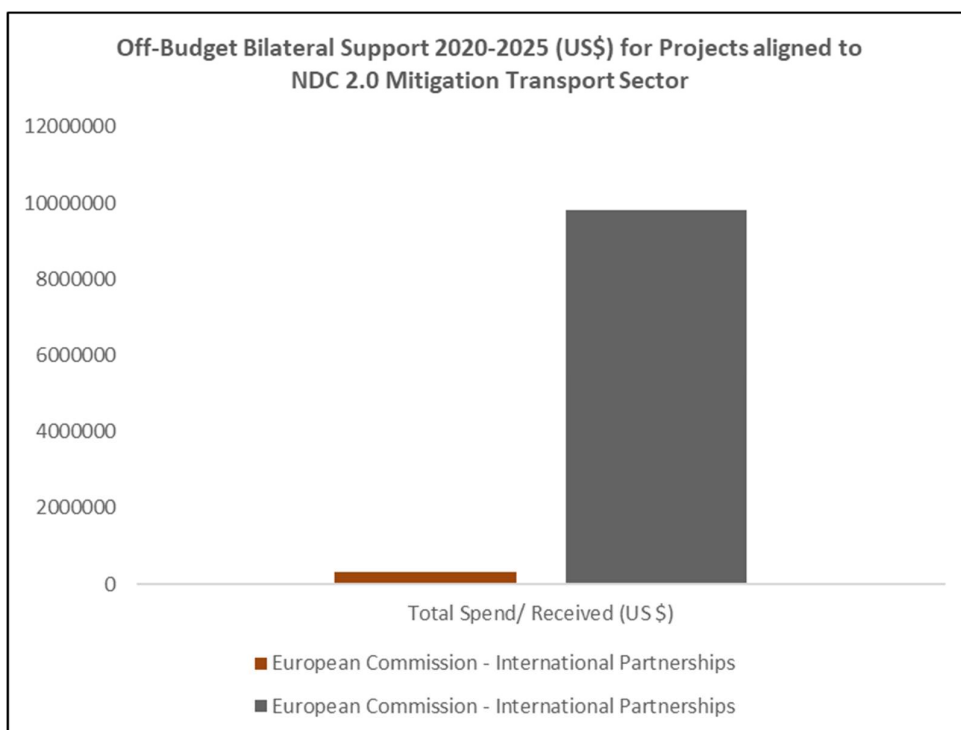
The Transport sector's NDC 2.0 Mitigation Actions focus on sustainable infrastructure and modal shifts to reduce greenhouse gas emissions. Key ongoing projects include urban mass transit improvements like the Kampala Flyover and Busega-Mpigi Expressway, funded primarily through loans from JICA and the African Development Bank. Additionally, the New Bukasa Port development, supported by European financial institutions, is progressing towards completion, enhancing trade infrastructure. Efforts to rehabilitate the meter gauge railway for freight transit, backed by the African Development Bank and Spanish export credit agencies, are also underway. However, there are currently no funded initiatives addressing road transport fuel efficiency, alternative low-carbon fuels, or the formal development of a long-term national transport policy integrating climate mitigation. These gaps highlight opportunities for targeted financing and policy development in the next NDC cycle.

Off-Budget Support for Transport NDC 2.0 Mitigation Actions

Only 3.4% of the budgeted exclusive Conditional Climate Finance was committed from 2020-2025. However, 47.9% of these commitments have materialized into exclusive Conditional Climate Finance needed during this period.

Off-Budget Bilateral Support for Transport NDC 2.0 Mitigation	
Climate Finance Committed (US\$ M)	Climate Finance Received (US\$ M)
21.1	10.1

The European Commission-International Partnerships has been a pivotal source for this modality of Financing for mitigation actions in the Transport Sector



Off-Budget Support for NDC 2.0 Mitigation Sectors unfunded through National Budget Support

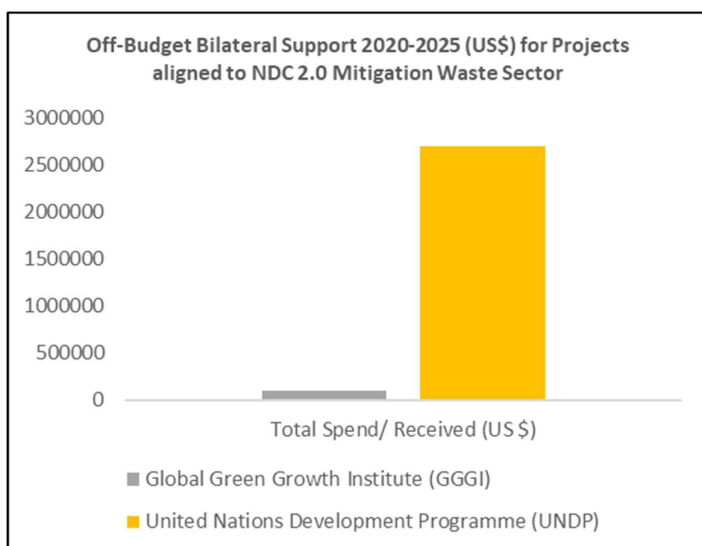
Both the Waste and IPPU Mitigation Sectors were not funded via national budget support. The IPPU Sector wasn't also funded via the Off-Budget Support. The following section provides details about off-budget support provided for the Waste NDC 2.0 Mitigation Sector.

Off-Budget Support for Waste NDC 2.0 Mitigation Actions

Only 0.46% of the budgeted exclusive Conditional Climate Finance for the NDC 2.0 Waste Sector was committed from 2020 to 2025. However, 100% of the funds committed were received during this period.

Off-Budget Bilateral Support for Waste NDC 2.0 Mitigation	
Climate Finance Committed (US\$ M)	Climate Finance Received (US\$ M)
2.8	2.8

The main sources of this modality of financing for the implementation of activities aligned to Waste Sector priority mitigation actions were; United Nations Development Program (UNDP) and the Global Green Growth Institute (GGGI).



9.2 Tracking Technological and capacity building support using CTF Table 7-9- aligned framework

9.2.1 Technology Transfer and Development

With regards to Technology Transfer and Development, even whilst there weren't any targets set in Uganda's NDC 2.0, the review of the projects that were identified as aligned to the priority NDC 2.0 Sectoral actions and activities revealed that in Uganda's NDC context, technology transfer and development generally encompassed the following;

Deployment of physical infrastructure and technologies such as solar-powered irrigation systems, water and sanitation facilities, renewable energy generation and transmission networks, climate-resilient agricultural inputs, and climate-adaptive transport infrastructure.

Integration of imported specialized equipment with local technical expertise, facilitating the construction, operation, and maintenance of resilient health, energy, water, transport, and forestry infrastructure.

Utilization of digital tools and data management systems to enhance technology uptake, monitoring, and transparency, as in the case of property valuation for taxation by Kampala Capital City Authority.

Innovation and local development of new technologies tailored to Uganda's specific environmental, social, and economic contexts, such as sustainable fish feeds, diversified crop and livestock varieties, and ecosystem restoration techniques.

Establishment of an enabling institutional and regulatory environment that supports the diffusion, uptake, and sustainable management of climate technologies across sectors.

The details of the components of Technology Transfer and Development for the reviewed projects are included in the Annex of this report. The sectoral highlights are provided in the sections below.

Adaptation

The Agricultural sector has recorded instances of transferring climate-resilient technologies that support adaptation and productivity among smallholder farmers. Critical investments in infrastructure—over 1,200 kilometres of community and farm access roads, bridges, valley dams, and solar-powered irrigation systems—were reported to have contributed to improved water availability and market access. This infrastructure expansion enables enhanced adoption of climate-smart practices and inputs.

A network of over 21,000 demonstration gardens serves as practical learning sites for climate-adaptive crops, livestock breeds, and clean energy solutions like improved cookstoves and biogas units. The introduction of 96 new crop varieties and livestock breeds, alongside dissemination of over 36,000 tonnes of quality seed, diversifies production systems and boosts resilience.

Research institutions such as NARO and NACCRI are pivotal in developing locally adapted technologies and updating training manuals and curricula, ensuring extension workers and farmers are equipped with current climate-smart knowledge.

Integrated water and natural resource management strategies, including micro-catchment planning and landscape restoration, support ecosystem services and mitigate climate impacts. Rehabilitation of degraded lands and reforestation initiatives complement these efforts by sustaining agricultural productivity.

The Fisheries sector has similarly advanced through upgrading the Aquaculture Research Development Centre (ARDC), enhancing research capacity and infrastructure including breeding tanks and hatcheries. Innovations in sustainable fish feed—such as substituting traditional ingredients with black soldier fly larvae and adopting multi-enzyme treatment protocols—have improved feed efficiency and environmental sustainability.

In the Water and Sanitation sector, technology transfer has enabled the development of regulatory tools, solar-powered water supply systems, and gender- and disability-friendly sanitation facilities. Modernized data management platforms and climate-resilient infrastructure designs support improved service delivery and resilience.

The Forestry sector has implemented technology-driven restoration efforts, successfully rehabilitating 1,705 hectares of degraded forest land within key reserves and controlling invasive species in National Parks. Innovative conservation techniques such as buffalo trenching have been introduced to protect biodiversity and enhance ecosystem resilience.

The Health sector exemplifies a hybrid approach to technology transfer, combining local construction expertise (e.g., Excel Construction and UPDF Engineering Brigade) with the importation of specialized equipment (sanitary fittings, electrical panels, transformers) to build resilient health infrastructure. This synergy strengthens climate-proofing facilities, ensuring continuity of essential services amidst environmental stresses.

Technology transfer in the Transport sector includes advanced engineering works—road improvements, drainage upgrades, traffic signalization, and construction of bridges and underpasses. Rail infrastructure development, notably the Kampala-Namanve railway section, introduces low-carbon transport alternatives that reduce emissions and enhance climate resilience.

In Cities and the built environment, technologies applied in paved and community access roads, bus parks, markets, and parking facilities enhance connectivity and urban climate adaptation, including in refugee-hosting districts.

Mitigation

The Energy sector has leveraged technology transfer through the commissioning of major hydropower plants (e.g., Karuma HPP) and expansion of transmission lines and substations. These developments integrate renewable energy technologies critical for decarbonizing Uganda's electricity grid.

The Transport sector's mitigation actions focused on infrastructure modernization and strategic planning. Completion of bridges, expressway sections, and road widening, alongside traffic management systems and

the Port Master Plan, promote efficient, low-emission transport modes. The advancement of railway infrastructure further supports Uganda's shift toward sustainable transport.

9.2.2 Capacity-Building and Gender

This stocktake categorized capacity building as Institutional capacity-building and Innovation capacity-building. A review of the priority actions listed in Uganda's updated NDC for gender and capacity-building showed that all these actions related to Institutional Capacity-building.

Institutional Capacity Building and Gender			
Dimension of Institutional Capacity Building from International Best Practice	Priority NDC 2.0 Capacity Building Action	Priority NDC 2.0 Gender Action	Status of NDC 2.0 Capacity Building & Gender Action
1. Resources to train relevant government agency staff and possibly non-government agency staff too with the view to increasing technical and managerial skills of these individuals	1. Reviewing and implementing the Uganda National Climate Change Learning Strategy	1. Develop skills and capacity on gender and climate change	Representatives of MDAs, Development Partners, NGOs and Academia acknowledged that there were very many initiatives in the country that focused on building capacity of actors on various aspects of climate change and gender. The challenge was that reporting on these activities was very poor, uncoordinated, and lacked transparency.
	2. Establish climate change knowledge and learning centers		MOFPED in lead together with NPA, MWE undertaking Climate Change Budget Tagging (CCBT). This is a key element which enables Ministries, Departments, Agencies, and Local Governments to identify, classify, and track spending associated with climate action. Starting in FY 2021/22, the Government through the National Budget Call Circular provided explicit instructions to all accounting officers to ensure that their budgets were aligned with the climate change interventions proposed in the NDP III. The Circulars also indicated that the alignment was to be achieved through Climate Change Budget Tagging (CCBT). by (1) deploying high-resolution tools for climate-economy modelling; (2) strengthening national and local capacity in Climate Vulnerability Assessment; (3) aligning public investment programming with Uganda's NDC and adaptation priorities; and (4) improving institutional capability to prepare, structure, and mobilize financing for bankable, climate-resilient development projects. Votes, Programs and DLGs are being trained on CCBT and MOFPED has two codes (Adaptation and Mitigation) in the PBS (Public Budget System) to guide budgeting. 30 DLG out of 130 so far trained- shall train all
	3. Expanding access to tools and support for capacity building and awareness raising		
2. Capacity to engage all relevant stakeholders, through consultations designed to elicit their input	4. Building Long-term strategic multi stakeholder partnerships	No related priority NDC 2.0 Gender Action. This gap can be explored in NDC 3.0	
	5. Strengthening coordination and institutional arrangement to enhance collaboration and promote knowledge sharing		
3. Aptitude to monitor progress and report on it making best use of existing data collection mechanisms	6. Strengthening monitoring, reporting and verification of the implementation and achievement of the updated NDC	2. Designate gender and climate change specialists across MDAs and Local Governments to strengthen gender and	Ministry of Gender, Labour and Social Development (MGLSD) -National coordination of all gender mainstreaming across all government programs and policies.

	7. Technology Development and Transfer	climate change mainstreaming in the formulation, implementation and monitoring of NDC Actions	The NPA and CCD had set up quarterly meetings to update each other on the progress of implementation of NDC activities viz a viz the finance mobilization efforts. These meetings have not happened frequently, so possibly twice a year.
4. Competence to conduct a regulatory framework revision to streamline and complement existing laws and regulations, and strengthen related governmental processes and entities	No related priority NDC 2.0 capacity building action. This gap can be explored in NDC 3.0	No related priority NDC 2.0 Gender action. This gap can be explored in NDC 3.1	Circular from Attorney General cautioning unguided generation of MOUs between MDAs. This MO appears to be characteristic of planning and budgeting at Sector level and yet Uganda has moved on to Program based planning and budgeting; various MDAs that hadn't worked together before find that they have no option, work is delayed because of unclear definition of roles that has come as a result of planning and budgeting as programmes. The Judiciary must adjust to expedite this new mode of working otherwise the existing MO no longer fits the new suit of government working.
5. Capacity to integrate NDC priorities into sectoral and cross-sectoral programs and projects to ensure that the latter don't undermine efforts to achieve the former and vice versa	No related priority NDC 2.0 capacity building action. This gap can be explored in NDC 3.0	3. Enhance women participation in decision making processes across all levels to enable their contribution in the assessment of needs and prioritization of climate actions 4. Expand the mainstreaming of gender and climate change in all budget framework papers and local government plans 5. Develop gender disaggregated data for the updated NDC Implementation plan	Ministry of Gender, Labour and Social Development (MGLSD) -National coordination of all gender mainstreaming across all government programs and policies.

However, there was evidence of Innovation capacity-building in projects supported through national budget support and were aligned to NDC 2.0 priority Adaptation and Mitigation Actions. The details of the capacity-building components of these projects are shown in the Annex of this report. Highlights of sectoral evidence are presented in the sections below.

Innovative Capacity-building within NDC 2.0 Adaptation Sectors

In the Agriculture sector, large-scale capacity building initiatives reached over 1,800 farmer groups and 15,000 individuals, focusing on sustainable production, enterprise development, and climate-smart agronomic practices. Community-based natural resource management plans, mentoring for vulnerable households, and training in irrigation governance further demonstrate systemic investments in local resilience.

In the Fisheries sector, efforts included the establishment of aquaculture training hubs, development of technical regulations, and training of over 230 farmers in integrated systems like rice-fish farming. Institutional support was provided through the development of aquaculture strategies and regulatory frameworks, enhancing the sector's capacity to adapt to climate change.

The Forestry sector-initiated curriculum development at Nyabyeya Forestry College, laying out the groundwork for future workforce development in sustainable forestry management.

In Water and Sanitation, over 850 women and youth were trained in WASH entrepreneurship, and District Local Governments received continuous technical support through 127 Regional Support Units to ensure compliance with sector standards and hygiene promotion.

In the Transport sector, capacity building focused on UNRA staff, equipping them with skills in procurement, financial management, and environmental safeguards, with additional staff trained in project management and data collection technologies.

The Health sector enhanced coordination through technical working groups and strengthened collaboration with institutions such as the Ministry of Education, Ministry of Gender, and WHO, ensuring integrated responses to climate-sensitive health risks.

The Cities and Built Environment sector benefited from skills development in urban planning, integrated revenue systems, and retooling of municipal offices, contributing to institutional readiness for climate-resilient service delivery.

Innovative Capacity-building with NDC 2.0 Mitigation Sectors

In the AFOLU sector, capacity building emphasized stakeholder awareness and regulatory development, particularly around the National Coffee Act 2021 and its implementation through stakeholder engagement in key agricultural regions.

The Energy sector focused on institutional strengthening through a USD 2.2 million capacity assessment and USD 11.4 million investment in system modernization for the Uganda Electricity Transmission Company Ltd (UETCL), supporting a transition to low-emission grid infrastructure.

In the Transport sector, mitigation-focused capacity building included digital innovations such as 3D modelling and animation for infrastructure planning and partial training (27.8%) of Uganda Railways Corporation staff to support a modal shift to lower-emission rail transport.

The Manufacturing, Industrial Processes and Mining sector recorded specialized capacity-building activities such as training UPDF officers in heavy equipment operation, technical staff training in China, and operationalization of an equipment leasing program to improve resilience and emissions management in infrastructure and extractives.

10 Assessment of Gaps and Barriers

Persistent gaps in data, MRV systems, institutional coordination, technical skills, and finance continue to limit comprehensive tracking and full-scale implementation of NDC 2.0. Key barriers such as weak regulatory enforcement, inadequate funding, procurement delays, and socio-economic constraints have slowed project execution and restricted adoption of climate-resilient practices, especially in energy, AFOLU, and transport sectors.

This section synthesizes the key gaps and barriers that have impeded or could impede the effective implementation of Uganda's NDC 2.0 and subsequent NDCs, if not addressed. These were identified through the stock take process and the six-pillar assessment, which highlighted weaknesses in data, institutional capacity, financing, technical skills, and enabling conditions. Addressing these gaps is critical to meeting current NDC targets, monitoring and tracking implementation, and enhancing ambition in NDC 3.0.

10.1 Data Gaps and MRV Limitations

The six-pillar assessment underscored data and MRV as a cross-cutting weakness. While Uganda has initiated improvements through the GHG Inventory Improvement Plan, the MRV system is not yet fully ETF-aligned. Current gaps include:

- The centralized digital GHG inventory portal for MDAs/NDAs to directly submit data, is not yet operationalized, which limits systematic consolidation of information.
- Limited country-specific emission factors; reliance remains largely on IPCC default values and planned work in this area has been slow.
- Weak QA/QC systems and incomplete documentation of methodologies and uncertainty ranges.

These structural gaps compound sector-specific data challenges. For instance:

- In the transport sector, GKMA and Meter Gauge Railway (MGR) projects are underway, but no emission reductions have been computed due to lack of activity data and emission factors.
- In the AFOLU sector, expected reductions from interventions such as agroforestry and irrigation are quantified in principle, but no adoption data is available to verify actual uptake.
- In the waste sector, some PAMs (biogas recovery, aerobic composting) have had progress reported, but others, such as school bio-latrines, lack systematic monitoring data.

As a result, only a limited number of sub-measures currently provide the verified data necessary for tracking Uganda's NDC implementation. Although Uganda's NDC includes quantified expected reductions (~30 MtCO₂e by 2030), the actual evidence base for measuring progress is still partial and fragmented.

The NDC Tracking Framework, developed by the consultants for this stock take, has helped consolidate the best available data from MDAs and other sources and highlights progress where it exists. However, the exercise also underscored persistent gaps stemming from irregular or missing sectoral reporting. While the GHG Inventory Improvement Plan is a positive step, without mandatory, systematic reporting protocols and a functional centralized portal to enable direct submission in ETF-compatible formats, Uganda will continue to face challenges in consistently meeting ETF requirements and effectively tracking NDC implementation progress.

10.2 Institutional and coordination challenges

The stocktake assessment framework and the six-pillar assessment review highlighted persistent weaknesses in inter-ministerial coordination and uneven engagement of ministries, departments, and agencies (MDAs). While the Climate Change Act (2021) established important structures such as the Climate Change Department (CCD) and the National Climate Change Advisory Committee (NCCAC), their full operationalization remains incomplete. As a result, roles and reporting lines across MDAs are not always clear, and coordination for NDC implementation continues to rely heavily on ad hoc efforts.

At the sub-national level, local governments are mandated by the Climate Change Act to integrate climate change into district development plans. However, alignment between local plans and national processes remains limited, partly due to capacity constraints and weak coordination channels. This limits the ability to scale up adaptation and mitigation measures at community level and hampers systematic bottom-up reporting of progress.

On Parliamentary engagement and accountability, Uganda's Parliamentary Forum on Climate Change provides a valuable platform for raising awareness and mobilizing political will on climate change issues. However, the Forum is not yet systematically linked to sectoral accountability mechanisms for NDC delivery. Sector-level mandates, monitoring frameworks, and clear lines of responsibility for implementation are still evolving hence strengthening these linkages would allow parliamentary oversight to play a more consistent role in holding MDAs accountable for their climate commitments.

Finally, looking towards the NDC 3.0, it would be advisable to have a multi-sectoral and medium-term coordinating body in order to coordinate efforts between the multiple institutions and partners involved. This single entity could oversee action, monitor progress, and address cross-sectoral bottlenecks. Establishing such a mechanism, which could be built on the provisions of the Climate Change Act, would provide the institutional anchor needed for effective and sustained NDC 3.0 delivery.

In the AFOLU and waste sectors, MDAs lack the dedicated staff and systems needed for data collection, verification, and reporting. Consequently, PAMs such as agroforestry, irrigation, and school bio-latrines can't be properly tracked. On the adaptation side, ministries and districts often lack personnel with the specialized skills needed to conduct vulnerability assessments, design bankable adaptation projects, and effectively integrate climate risk into sectoral planning. This skill gap hinders the country's ability to prepare for and respond to the impacts of climate change.

10.3 Financial Constraints

Uganda's NDC implementation cost is estimated at USD 28.1 billion to 2030, with 85% conditional on external finance. Actual inflows remain far below these needs, limiting implementation of NDC commitments such as renewable energy expansion, clean cooking scale-up, and large adaptation programmes. At the local level, adaptation financing remains scarce. Even where funding is secured, absorption capacity challenges (procurement bottlenecks, limited project preparation) delay implementation. The six-pillar assessment also noted that the NDC lacks a pipeline of bankable projects, and private sector engagement is minimal. Without enhanced access to international funds and development of domestic finance tools, NDC targets will remain at risk.

10.4 Technical and Human Capacity Gaps

Across sectors, persistent technical capacity shortages within MDAs continue to undermine effective NDC implementation.

In the transport sector, for example, efforts to estimate emission reductions from the rehabilitation of the Meter Gauge Railway are stalled because Uganda has not yet developed context-specific emission factors or in-country modeling expertise; sector officers rely on default values and external consultants, which limits accuracy and timeliness.

Uganda's MRV system also suffers from a critical shortage of trained GHG inventory compilers, modelers, and QA/QC specialists. Very few MDAs have designated climate officers, and where they exist, they are often overstretched with multiple portfolios. This leaves Uganda heavily reliant on external consultants for essential technical tasks such as developing sector emission baselines, applying IPCC methodologies, or drafting transparency reports, a dependence that raises costs and undermines long-term institutional ownership of the MRV process.

10.5 Policy and Regulatory barriers

Several sectoral policies and regulatory frameworks remain misaligned with Uganda's NDC objectives, creating friction for implementation. In the energy sector, Uganda's continued heavy reliance on biomass (over 80% of household energy use) directly undermines the NDC's clean cooking targets. Despite policy references to modern energy access, subsidies and incentives for efficient cookstoves or electric cooking technologies remain weak. As a result, only 3.8% of households had adopted clean and sustainable cooking solutions by 2024, far below the 10% target for 2025.

In the AFOLU sector, weak enforcement of forestry regulations and persistent land tenure uncertainties make it difficult to scale up reforestation and agroforestry programmes, even though they are central to Uganda's mitigation pathway.

On transport, Uganda has taken steps such as banning the import of very old vehicles and applying higher environmental levies on cars older than 9 years, which indirectly discourages high-emitting vehicles. However, beyond these age- and import-based measures, there are still no binding fuel-efficiency standards for the existing fleet, and fiscal incentives for electric vehicle (EV) uptake remain minimal. As a result, while the draft National E-Mobility Policy provides a starting point, the regulatory environment is not yet sufficient to drive large-scale EV deployment or systemic emissions reductions in the sector.

The Climate Change Act (2021) provides a strong overarching legal foundation, but complementary regulations and sector-specific instruments have not yet been fully operationalized. For example, clear and consistent feed-in tariffs and incentives for renewables are lacking, investment approval processes remain slow, and climate-proofing of infrastructure and public investments is not yet mandated across sectors.

This patchwork of partially aligned or weakly enforced policies signals that while Uganda now has the legal framework in place, the regulatory detail, enforcement mechanisms, and targeted incentives needed to deliver transformative sectoral change remain underdeveloped.

10.6 Social Economic and Behavioural Barriers

At the household level, high upfront costs and affordability constraints remain one of the biggest barriers to adopting clean energy solutions. For example, efficient cookstoves and electric cooking devices remain out of reach for many families, even where long-term savings exist. Financing schemes for households are limited, and distribution networks in rural areas are thin, creating both an economic and accessibility challenge. Cultural preferences and traditional cooking practices further slow behavioral change, with households often reluctant to shift to new technologies unless clear advantages—such as reduced cooking time or fuel savings—are demonstrated.

In the AFOLU sector, adoption of climate-smart agriculture (CSA) and sustainable land management practices is similarly constrained. Many farmers prioritize short-term yields over long-term resilience, particularly when immediate livelihood needs are pressing. Without clear evidence of profitability, or incentives such as subsidies, insurance, or guaranteed markets, farmers are hesitant to shift practices. Land tenure insecurity further discourages investment in tree planting or soil restoration, as households fear losing benefits from long-term improvements.

Transport behaviors also present challenges. Public transport remains underdeveloped in many areas, and commuters often rely on individual motorcycles or second-hand imported vehicles. Shifting habits towards cleaner mobility options is difficult without affordable and reliable alternatives. The limited availability of charging infrastructure for electric vehicles exacerbates these barriers, making early adoption largely confined to small pilot initiatives.

Underlying all these issues is widespread poverty and competing development priorities, which constrain adaptive capacity. Households often lack the resources to invest in climate-resilient technologies, even when they recognize the risks of climate change. Furthermore, public awareness of climate risks and the co-benefits of mitigation and adaptation measures remain low, reducing momentum for voluntary behavior change. Awareness campaigns are fragmented, and climate education has yet to be fully integrated into community outreach, schools, or extension services.

Addressing these barriers will require targeted financial mechanisms (e.g. household subsidies, microfinance for clean energy, insurance for CSA), improved extension and demonstration programs to build trust in new practices, integration of climate awareness into national education and communication strategies, and stronger regulatory and market signals that make climate-friendly options the default rather than the exception.

10.7 Coverage Gaps in Economy-wide Targeting

Uganda's updated NDC 2.0 is formally considered economy-wide, since it covers the major emissions sources and removals in Uganda including 4 IPCC sectors—energy, AFOLU, waste, and IPPU—and includes the three principal gases (CO₂, CH₄, N₂O) and HFCs. However, certain high-impact subsectors are not yet fully represented in either the GHG inventory or corresponding NDC actions. Most notably, the emerging oil and gas industry, expected to become a significant source of emissions, is not explicitly included in mitigation planning. In addition, domestic and international aviation remains outside the scope of reported measures, despite its growing contribution to both stationary and transport energy-related emissions. These omissions create a gap between the formal classification of the NDC as economy-wide and its practical coverage of key emission sources. Addressing these gaps will be critical to safeguarding the achievement of Uganda's mitigation targets and ensuring a low-carbon development pathway for the oil and gas sector.

10.8 Gaps Identified from the Six-Pillar Assessment of NDC 2.0

Ambition Level of Targets (Pillar 1 - Ambition)

Despite the raising ambition and the broadening of sectoral coverage from INDC to NDC 2.0, Uganda's target remains modest considering global temperature goals under BAU, national emissions were projected to grow significantly, so a 24.7% cut still allows absolute emissions to increase. There is no explicit peak emissions year, or long-term emissions goal stated. Future NDCs could articulate a peaking trajectory or a net-zero year in line with its LT-strategy, to strengthen credibility. Additionally, increasing the share of unconditional commitment (currently only ~6% of BAU) would demonstrate greater domestic resolve and could spur more support. Ambition could also be enhanced by including more non-GHG targets (e.g. renewable energy share, forest cover).

Pillar2 – Quality

Uganda's NDC 3.0 should improve the clarity, transparency, and robustness of its content. All key assumptions — such as BAU projections, emission factors, and modelling methodologies, should be presented in the main text, not confined to annexes, to allow stakeholders and partners to assess the basis of Uganda's commitments.

Strengthening the MRV framework is also essential. Indicators should be consistent with UNFCCC Common Tabular Formats (CTFs) and linked to Uganda's national data systems, with sector ministries clearly assigned responsibility for data provision. Making MRV arrangements more explicit will improve transparency, comparability, and accountability.

In short, NDC 3.0 should prioritize clearer assumptions, transparent methodologies, and stronger MRV systems, ensuring that targets are credible, and progress can be consistently measured.

Pillar 3 - Process

To strengthen the process dimension of Uganda's next NDC, the government should build on the inclusive approach adopted in NDC 2.0 and further institutionalize mechanisms for transparency, participation, and fairness. A just transition lens should be systematically embedded across all sectors, ensuring that measures in energy, AFOLU, and transport account for potential social and economic impacts on workers, women, youth, and vulnerable groups. Gender-responsive planning should go beyond mainstreaming language in sectoral texts to include specific gender indicators and targeted support measures (e.g., access to finance, training, and technology for women-led enterprises in clean energy and climate-smart agriculture).

Stakeholder engagement must also shift from one-off consultations to continuous dialogue platforms that enable regular feedback from civil society, private sector, academia, and local governments. Finally, documentation of how stakeholder inputs influence the design and revision of NDC commitments should be made standard practice, improving transparency and credibility in the process.

Pillar 4 Implementability

Uganda's NDC 3.0 should strengthen implementability by converting targets into actionable delivery plans. While NDC 2.0 listed measures and numeric goals, it lacked milestones, timelines, and clearly assigned responsibilities, making annual tracking and accountability difficult.

Each sectoral commitment in NDC 3.0 should therefore include:

- Implementation timelines and interim checkpoints between 2025 and 2030.
- Responsible institutions with defined mandates and accountability.
- Resource needs and indicative funding sources.
- Specific actions or measures tagged as conditional or unconditional

Overly ambitious targets should be backed by practical strategies. For example, the clean cooking goal of 50% access by 2025 lacked mechanisms for stove distribution, financing, or private sector engagement. NDC 3.0 should ensure that each major target is tied to specific policy instruments or programmes.

Institutional coordination also needs strengthening. The NDC should clarify how the Climate Change Department, sector ministries, and local governments will share roles, supported by a mechanism for joint accountability.

Finally, a reinforced MRV framework should be embedded in the institutional design to ensure results are tracked and reported regularly, aligned with the Enhanced Transparency Framework (ETF).

Therefore, NDC 3.0 must be underpinned by a clear roadmap with roles, milestones, and accountability structures to translate ambition into results.

Pillar 5 - Investability

Uganda's NDC 3.0 should go beyond costing to outline a clear financing strategy. While NDC 2.0 estimated USD 28.1 billion in total needs, it did not show how resources would be mobilized. The next NDC should:

- Identify domestic resource options (e.g., green bonds, fiscal reforms, local financial institutions).
- Outline pathways for accessing international finance (e.g., Green Climate Fund, Article 6 markets, bilateral partnerships).
- Develop a pipeline of bankable projects in priority sectors such as renewables, forestry, and resilient agriculture to attract private capital.

The NDC should also include a contingency plan to prioritize actions if only partial funding is secured. Strengthening links between MRV outputs and investment promotion will further enhance Uganda's ability to mobilize resources.

Hence, NDC 3.0 must demonstrate not only the scale of costs, but also how financing will be mobilized and directed toward implementable, investment-ready projects.

Pillar 6- Credibility

Uganda's NDC 3.0 must demonstrate that commitments are realistic, evidence-based, and aligned with long-term strategies. This means:

- Ensuring close alignment with the Long-Term Low Emissions Development Strategy (LTS) that is under development and consistent with the Global Stocktake (GST).
- Linking adaptation priorities directly to the forthcoming National Adaptation Plan (NAP), with measurable indicators and milestones.
- Publishing regular domestic progress reports to build trust with citizens and international partners.
- Addressing emerging risks such as oil and gas lock-in, which could undermine the mitigation pathway, through transparent disclosure and clear mitigation measures to ensure a low-carbon development pathway for the oil and gas sector.

Credibility will therefore be strengthened by combining realistic targets, transparent reporting, and risk management, ensuring Uganda's NDC 3.0 is both ambitious and realistically deliverable.

The implementation of adaptation and mitigation projects across various sectors has encountered a range of persistent barriers that have collectively impacted project timelines, costs, and outcomes. The details of the sector-specific project barriers are illustrated in Annex 3 of this report. Below are highlights of the key thematic challenges observed across sectors, providing a basis for informed decision-making and targeted interventions.

1. Delays in Procurement and Contracting

Prolonged procurement processes, including delayed contractor recruitment, administrative reviews, and responses to complaints, have been a significant impediment. These delays have affected the timely commencement of civil works and consultancy services, leading to cascading delays in project execution and completion across sectors such as energy, transport, forestry, and water sanitation.

2. Funding and Financial Management Challenges

Inadequate and delayed disbursement of funds, coupled with complexities in fund tracking due to insufficient coding in financial management systems, have constrained project activities. Over-expenditure, delayed payment of contractors, and currency exchange fluctuations have further exacerbated financial management issues, impacting contractor cash flow and project sustainability.

3. Land Acquisition and Resettlement Complexities

Delays in acquiring land and compensating Persons Affected by Projects (PAPs) have been a recurrent challenge. Protracted negotiations, resistance from communities, unclear valuation rates, and late approval of safeguard instruments have slowed progress, particularly in infrastructure-heavy projects. These issues have led to redesigns, scope creep, and in some cases, significant disputes.

4. Coordination and Institutional Capacity Constraints

Inadequate coordination among implementing agencies and stakeholders, understaffing, and weak project management capacities have hindered adherence to timelines and effective project execution. The lack of clear operational frameworks and guidelines, especially concerning Resettlement Action Plans, has led to further inefficiencies.

5. Environmental and External Factors

Unpredictable environmental challenges such as floods, landslides, difficult soil conditions, and impacts of the COVID-19 pandemic have disrupted project activities, causing additional delays and cost overruns.

6. Scope Changes and Technical Challenges

Increased scope of work, often due to design revisions, unforeseen technical difficulties, or inadequate initial planning, have resulted in budget overruns and extended timelines. Quality issues and the need for remedial works have further delayed project completion in several sectors.

Implications for Future Implementation

Addressing these barriers requires a multifaceted approach including streamlining procurement procedures, enhancing financial management and accountability frameworks, strengthening institutional coordination, and proactive community engagement on land and resettlement issues. Moreover, building resilience to environmental disruptions and improving initial project planning and design accuracy will be critical to improving delivery outcomes.

This stocktake underscores the need for strategic interventions and close monitoring to mitigate risks and accelerate progress towards climate adaptation and mitigation goals across sectors.

11 Lessons learnt and Recommendations for NDC 3.0

Lessons learnt emphasize that progress toward NDC targets requires strengthening institutional, technical, and data systems, including an ETF-aligned MRV framework, robust GHG inventory management, and centralized adaptation tracking with clear indicators. Recommendations highlight raising mitigation ambition while leveraging co-benefits, integrating adaptation and resilience into national development planning, designing bankable projects with private sector engagement, and deploying whole-of-society approaches that embed gender and social inclusion throughout planning, budgeting, and implementation. Effective coordination, clear governance, timely stakeholder engagement, and risk-managed NDC update processes—aligned with long-term strategies and supported by targeted capacity building—are essential to translating ambition into measurable, investable, and sustainable climate action. In addition, while Uganda’s NDC 2.0 is considered economy-wide because it covers all IPCC sectors and major gases, future iterations should explicitly incorporate subsectors such as the emerging oil and gas industry and international aviation into the GHG inventory and NDC framework, in order to close coverage gaps and ensure full consistency with the ETF.

Based on our literature review of both national documents and international best practices, as well as drawing on the discussions with key stakeholders and on our extensive experience supporting countries developing, updating or staking stock of their NDCs, this chapter synthesizes lessons learnt and key recommendations towards Uganda’s NDC stocktake and NDC 3.0 development.

Strengthening ETF and BTR readiness

Institutionalize MRV: Update the MRV system to ensure it is ETF-aligned with standardized templates, QA/QC steps, a fixed reporting calendar, and named data custodians in each MDA; designate CCD as the national system owner. This should include binding reporting obligations under the Climate Change Act, linked to NDP and budget reporting cycles.

Adopt and operationalize CTF and CRT tables: Begin systematic use of UNFCCC Common Tabular Format (CTF) tables for mitigation, adaptation and tracking support received. Training MDAs and focal points on CTF completion will not only streamline the stocktake process but also ensure that Uganda’s first and subsequent BTRs are ETF-compliant. The NDC tracking framework should therefore mirror CTF structures to create continuity between domestic MRV and international reporting. Additionally, updating the GHG inventory to align with the Common Reporting Tables (CRT) under the ETF reporting requirements with subsequent restructuring of IPCC categories for example AFOLU to align with Land Use Land Use Change and Forestry and Agriculture in the enhanced NDC will strengthen consistency and comparability in alignment with the ETF.

GHG Inventory improvements: Implement the GHG Inventory Improvement Plan, including the development of country-specific emission factors for key categories (e.g. AFOLU, transport, waste), and ensure full documentation of methods, assumptions, and uncertainties. This will strengthen both ETF consistency and credibility of future NDC baselines.

Build national review capacity: Enroll national experts in UNFCCC Technical Expert Review training and run domestic “mock reviews” of draft BTRs. These exercises will raise quality, consistency, and preparedness for

international review. In parallel, strengthen financing strategies by aligning MRV outputs with investment roadmaps (e.g. showing where mitigation or adaptation actions generate bankable opportunities).

Operationalize MRV Tool: The existing national MRV Tool should be fully activated through sustained financing, staff training, and integration into national planning systems. Each MDA should appoint a focal point to submit quarterly data in standardized formats, with CCD assigning a dedicated officer to manage aggregation and follow-up. Automated notifications for overdue submissions would strengthen accountability and ensure continuous MRV data flow.

Updating mitigation targets, baselines and ensuring an economy-wide scope

Uganda's next NDC update provides an opportunity to raise mitigation ambition while consolidating credible baselines and closing coverage gaps to achieve true economy-wide alignment.

Enhancing ambition and co-benefits: While Uganda's emissions remain relatively low, the NDC 3.0 can raise ambition without imposing economic strain by emphasizing mitigation co-benefits—cleaner energy, improved efficiency, biodiversity protection, job creation, and alignment with green growth and SDGs. Framing mitigation within Uganda's broader development agenda will attract financing and strengthen political ownership.

Strengthening baselines and sectoral coverage: Future baselines should be updated using improved activity data and country-specific emission factors. Importantly, the scope of mitigation planning must expand to explicitly include emerging high-impact sectors such as oil and gas, as well as domestic and international aviation, both currently underrepresented in the GHG inventory and mitigation actions. Their inclusion is critical to ensuring the NDC's formal economy-wide classification is matched by effective coverage under the Enhanced Transparency Framework.

Mobilizing finance for ambition: Higher ambition can position Uganda as an attractive investment destination for international climate finance. Conditional mitigation commitments should be linked with clear policies or measures, with clear implementation roadmaps and investment strategies that identify funding sources, private sector partners, and enabling mechanisms.

Avoiding maladaptation and ensuring social safeguards: Mitigation measures should be assessed for potential social or environmental trade-offs, with cost-benefit analyses to prevent adverse impacts on vulnerable groups.

Improving adaptation and mitigation tracking

There is need to integrate adaptation tracking into national M&E frameworks such as the NDP reviews for the NDP programs. All actions should be aligned with NDP targets for easy tracking and monitoring their progression. There is need to have a dedicated national adaptation tracking unit within the Climate Change Department in the Ministry of Water and Environment for collecting relevant data on a continuous process during the NDC implementation period. The adaptation tracking unit should build a centralized digital platform to record adaptation actions, indicators, and progress, and this platform should be regularly updated. This will help in tracking the progress made and ease data collection for NDC3.0 stock take process.

For proper tracking of adaptation progress, the NDC implementation framework should be aligned with the actual indicators provided in the NDC for example the NDC2.0 implementation framework had 150 actions categorized as short-term, medium term and long term with no target indicators for tracking their progress. On the other hand, the actual NDC had 59 priority adaptation actions with indicators for tracking their progress. Therefore, the actions in the implementation plan of NDC3.0 should be aligned properly with the actions in NDC3.0.

In addition, we recommend formulating the National greenhouse gas emissions and adaptation reporting regulations to improve data collection for Uganda's GHG inventory and adaptation reporting. This will facilitate effective fulfilment of Uganda's reporting obligations under the convention and enhance monitoring, reporting and verification arrangements nationally. The regulations will provide legally binding institutional frameworks for activity data providers to collect and report data in alignment with IPCC and UNFCCC guidelines supporting implementation of the Uganda's climate change act and laying a foundation for improvement of reporting under the Enhanced Transparency Framework by enhancing transparency, accuracy, consistency, completeness, and comparability. Moreover, regular updating of the inventory will enhance data availability and facilitate adoption of an inventory-based mitigation tracking system for the NDCs, ensuring the NIR processes are aligned with NDC update processes and each feed into the other.

Building resilience and addressing loss and damage

Experience from other countries shows that framing climate action as a binary choice between adaptation and mitigation often creates competition for attention and resources. Progress accelerates when stakeholders adopt an integrated resilience lens that links both agendas to human development outcomes. Positioning resilience as a holistic vision—“leaving no one behind”—helps stakeholders grasp that climate action is part of national development, just like poverty reduction or improvements in health and education. This language shift builds trust, facilitates commitment, and encourages coherent planning and investment.

Under Article 8 of the Paris Agreement, Parties are called to take action on loss and damage (L&D). Uganda, as an active member of the LDC Group, supports L&D action under the Paris Agreement and the Warsaw International Mechanism. Uganda is highly exposed to extreme temperatures, droughts, storms, floods, and landslides, which translate into water scarcity, crop failure, livestock loss, and food insecurity—making climate change a significant threat to the economy and livelihoods.

While both economic and non-economic losses are evident, documentation remains insufficient to inform policy and guide comprehensive responses. Recent examples illustrate the scale and recurrence of climate-related disasters: in November 2024, heavy rainfall triggered a landslide in Bulambuli District, causing 28 deaths and burying more than 45 houses; in May 2023, intense rains in Kisoro led to landslides that destroyed roads and houses, with six fatalities; in September 2022, severe rains in Kasese caused landslides that claimed 16 lives, damaged infrastructure, and displaced many residents; in July 2022, flooding along Rivers Nabuyonga and Namatala in Eastern Uganda resulted in over 24 deaths; and in January 2022, a landslide on Mount Muhavura in Kisoro caused nine deaths and affected more than 800 people. These events, occurring across multiple years and regions, underscore escalating risks and the need for systematic management.

Despite the frequency and severity of such events, policy and institutional gaps hinder effective L&D management. Although the Climate Change Act (2021) references loss and damage among national response measures, an operational implementation framework has yet to be developed. The next NDC should therefore (i) set explicit targets for addressing loss and damage, and (ii) establish robust mechanisms for L&D data collection and reporting, improving evidence for decision-making and enabling alignment with national planning, disaster-risk management, and the Enhanced Transparency Framework.

Enhancing financing strategies, designing bankable projects and private sector engagement

A key barrier to scaling climate action is the limited capacity to structure bankable projects and mobilize private finance. Advancing Uganda's climate goals across NDC coordination, mitigation, and adaptation requires robust financing strategies, targeted capacity building, and policies that lower transaction costs and risks for investors. With the support of government incentives and enabling regulations, financing long-term,

low-emission, climate-resilient pathways is achievable. In this context, unpacking recently launched national climate-finance strategies and reports can clarify modalities, instruments, and institutional responsibilities that NDC 3.0 can leverage.

To strengthen the financing pillar of NDC 3.0, the following recommendations apply:

- **Comprehensive breakdown of climate-finance needs:** Provide an itemized budget that specifies the financial requirements for priority mitigation and adaptation measures **by sector**—including technology transfer, capacity building, infrastructure, and implementation costs. This clarity increases credibility and improves the ability of domestic and international partners to align resources.
- **Bankability and project preparation:** Develop standardized investment-grade templates (e.g., concept notes, pre-feasibility/feasibility studies, financial models, risk matrices) and establish project preparation facilities to help MDAs and local governments bring pipelines to investment readiness. Pair technical assistance with transaction advisory to translate policy intent into investable projects.
- **Private-sector involvement:** Use risk-sharing instruments (guarantees, first-loss tranches, political-risk cover), fiscal incentives (tax credits, accelerated depreciation), and regulatory clarity (PPAs, clear licensing) to attract private participation—particularly in renewable energy, energy efficiency, nature-based solutions, climate-smart agriculture, and resilient infrastructure.
- **Blended and Innovative Finance:** Combine concessional and commercial capital through blended finance, and explore green bonds, sustainability-linked loans, results-based finance, and credit lines for local financial institutions to on-lend to SMEs and communities. Align instruments with sectoral pipelines to maximize leverage.
- **Explicit Mechanisms to Reach Vulnerable Groups:** Define pathways to ensure climate finance reaches the most vulnerable—for example, micro-finance schemes, community-managed funds, or ring-fenced quotas within national programs. Transparently articulating these delivery mechanisms will promote equity, strengthen local ownership, and maximize the social returns of climate investments.
- **Governance, MRV, and Investment Tracking:** Link financing strategies to clear KPIs and integrate them with the national MRV system so that financiers can track outcomes and performance. This improves transparency, supports results-based disbursements, and builds investor confidence over time.

Taken together, these measures will increase the bankability of Uganda’s climate pipeline, de-risk private investment, and ensure that resources flow efficiently—particularly to vulnerable communities—thereby enhancing the feasibility and credibility of NDC 3.0

Adaptation

Below we recommend indicators to be included in NDC3.0 and to improve tracking in some sectors.

Agriculture

- For the action of scaling up climate smart agriculture including agro-ecology, we propose an additional indicator of ‘number of farmers trained on CSA techniques. This will supplement information on the adoption of CSA practices for resilience building.
- For the action of strengthening water harvesting and irrigation farming, a targeted action for drought-prone areas is recommended. The proposed action is to promote small-scale irrigation, water harvesting and catchment protection in drought-prone areas with two indicators: (i) number of water harvesting systems installed; and (ii) area under irrigation.

DRR

- For the action of building effective early warning systems, the proposal is an additional indicator of the number of functional weather stations installed or upgraded in addition to the automation weather station network.
- For the action of strengthening policy linkage and actions on climate change, migration and disaster risk reduction, an additional indicator is suggested to track the percentage of districts with updated DRR and climate response plans.

Health

- For the action of strengthening climate resilience of health infrastructure and systems, an additional indicator of the number of health facilities with climate-resilient infrastructure is proposed. This will supplement data obtained from regional referral hospitals.
- For the action of improving early warning, surveillance and response system for climate sensitive health hazards, the proposal is two additional indicators: (i) number of health workers trained on climate-health risks, and (ii) percentage of districts with functional climate-health surveillance systems.

Water and Sanitation

For the action of Increase to sanitation and wastewater treatment infrastructure and services in addition to percentage of Sewerage Service coverage we propose two additional indicators:

- Number of total Sewer Connections and
- Km of total Sewer Extensions. These two indicators are already reported annually by NWSC.

Fisheries Sector

For the action of Promotion of ecosystem approach to aquaculture management that did not have an indicator in NDC2.0, we propose two indicators

- Number of farmers practicing aquaculture
- Number of fishponds established per year

Other Adaptation recommendations

- Strengthen data systems by prioritizing baseline establishment, regular reporting, and digitization
- Harmonize indicators by ensuring consistency and coordination across sectors, for example transport and Cities.
- Direct attention and resources to sectors such as cities, fisheries, forestry, and education where progress is below average or unmeasurable.

Mitigation -Sectoral Recommendations

Transport

- Scale up PPP models for transport infrastructure that incorporate climate mitigation technologies such as electric vehicles and low-carbon construction materials.
- Design bankable projects emphasizing sustainable urban mobility and freight efficiency with clear economic returns to attract private investment.

Energy

- Accelerate private sector participation through regulatory reforms and incentives for renewable energy generation, grid modernization, and energy efficiency projects.
- Develop bankable projects with clear cash flow models based on power purchase agreements (PPAs) to attract independent power producers and investors.
- Leverage blended finance and risk guarantees, to lower investment risks in clean energy infrastructure.

AFOLU

- Design integrated AFOLU projects that combine sustainable agriculture with forestry restoration and carbon sequestration to access both adaptation and mitigation finance.
- Engage private sector actors, including agribusinesses and cooperatives, to co-invest in climate-smart agriculture and carbon credit schemes.

NDC Coordination Financing and Engagement Strategies

- Integrated Financing Frameworks: Establish national platforms that coordinate diverse funding instruments—grants, loans, private equity, and impact investments—across sectors to optimize resource use.
- Strengthen Data and Monitoring: Improve project data collection, performance monitoring, and impact reporting to enhance transparency and attract performance-based financing.
- Capacity Building: Invest in institutional capacity to design, appraise, and implement bankable projects with clear climate, social, and economic benefits.
- Private Sector Mobilization: Create enabling policy and regulatory environments that reduce barriers for private investment, including clear risk mitigation tools, tax incentives, and streamlined approval processes.
- Innovative Financing Mechanisms: Explore green bonds, climate funds, and blended finance to mobilize additional resources, particularly in underfunded sectors like fisheries, education, and manufacturing.

Building institutional capacity and whole-of-government approaches and engagement mechanisms

There were several unaccounted institutional capacity building efforts by Development Partners and NGOs, yet these were delivered to representatives of MDAs to facilitate the coordination, planning and monitoring of Uganda's NDC implementation in line with the country's commitments to the Paris Agreement. To strengthen institutional capacity-building for effective, coordinated implementation of Uganda's future NDCs, we recommend that the Government of Uganda establishes a centralized and functional national climate institutional capacity-building registry under the leadership of Climate Change Department (CCD), with technical support from the National Planning Authority (NPA), Ministry of Finance, Planning and Economic Development; and the Office of the Prime Minister (OPM). For this to be operational, it's also important to acknowledge that facets of such a centralized or synchronized system fall within the jurisdiction of various mandated MDAs. The Office of the State Solicitor General should be actively involved in advising on the generation of operational Memoranda of Understanding (MOU) to guide this inter government agency collaboration endeavour.

Key actions of the Centralized National Institutional Capacity-building registry could include:

- Develop and operationalize the registry with clear data protocols for reporting on capacity-building activities, including details on scope, budgets, personnel trained, skill areas, implementing institutions, and linkages to NDC targets.
- Institutionalize mandatory reporting by all MDAs and development actors (through MOUs or legal provisions) to feed into the centralized system, ensuring transparency and reducing double counting.
- Integrate the registry with existing planning and budgeting systems, such as the Programme-Based Budgeting System (PBS), to align capacity-building initiatives with national climate priorities and NDC implementation.
- Establish cross-sectoral coordination mechanisms (e.g. inter-ministerial climate task forces) to enhance collaboration between key MDAs, especially in the six identified thematic areas of institutional capacity-building, with regular capacity needs assessments to guide targeted interventions.
- Provide a dedicated budget line for institutional capacity-building within the national climate financing framework, ensuring resources are available for technical training, stakeholder engagement, and regulatory reviews.
- Leverage digital tools and platforms for stakeholder engagement and knowledge sharing, particularly to enhance inclusivity and enable bottom-up feedback into national planning processes.

It is recommended that Uganda invests in building more institutional capacity throughout the NDC process and beyond. This must translate into detailed and available data, shorter action and response timeframes, better tracking and monitoring systems, and overall, into a solid governmental capacity to lead the NDC process, coordinate the different stakeholders, monitor and report, among other key activities.

A stronger public capacity can also be reflected on a decentralized system, with ability and capacity to generate and manage data locally and regionally.

Recommendations for a whole-of-society approach and stakeholder engagement

The Whole-of-Society (WoS) approach must draw on two crucial pillars: on the one hand, a clear governance structure for stakeholder engagement led by the responsible governmental organization, and on the other, a robust stakeholder mapping a key starting piece.

A clear governance structure led by the government is essential because the government sets priorities and timelines for the engagement process. Implementing a WoS approach for NDC development shall come as a national government decision, and it's expected that the process will further respect principles such as inclusiveness and transparency. For a successful implementation of a WoS approach, the NDC process shall consider issues and interactions with social inclusion, climate justice and just transition, and make explicit the level of involvement that is expected of key stakeholders such as women, the elderly, youth, indigenous groups and minorities.

Further, the government should demonstrate ownership of the NDC and climate processes overall – this is expected by all stakeholders involved in the process. The government also has key roles in data provision and in coordination among stakeholders, in particular public sector, private sector, NGOs, civil society, academia and international supporting partners. These roles shall be defined and reflected in a governance structure for WoS engagement in NDC processes. From our experience, we can add with certainty that the NDC can only be successful if the people feel that their needs and priorities are reflected in the NDC and its process.

The engagement of partners, such as development organizations, NGOs, or consultants, to design and update the NDC can be a key strategy to leverage in-depth expertise and accelerate the technical studies or stakeholder engagement. However, governmental leadership is essential here as well to mainstream the process, define the roles and priorities, and facilitate data exchange.

The stakeholder mapping is essential to identify all stakeholder groups and key institutions, to ensure that no one is left behind, but also to ensure proper data collection. To ensure a national identification with the NDC, its ownership and relevance, an initial document review and a thorough stakeholder mapping are the base. Further, it is crucial to engage with the right stakeholders from the inception of the process. The responsible governmental structure is essential for pointing out which stakeholder groups need to be engaged. Developing a stakeholder engagement plan with a tracking mechanism or format to collect systematic feedback, as well as implementing awareness raising and communication campaigns on the NDC process, are highly valuable to engage stakeholders and achieve meaningful contributions.

Critical to the process is also to maintain regular contact and show responsiveness to stakeholders' inquiries during the whole process. This helps to build the necessary trust, make people feel they are taken into account, learn about their needs, and ensure sustained feedback.

Consultants, either national or international, can support the incorporation and implementation of WoS approaches in NDC development. However, the leadership and timely contributions of the responsible governmental organization are necessary, as the NDC is a national process.

Finally, there are many international best practices in the design and implementation of a WoS approach. Reaching out to development partners or think tanks, such as the NDC Partnership, would be a key step to ensure mainstreaming and following of international best practices on WoS and WoG approaches.

There are guidelines and indicators to enhance and measuring the inclusivity or the success of the WoS approach. However, we can recommend that this be done considering the governmental goals and priorities for WoS approaches for the particular national context, before applying broader international metrics. Thus, the development of NDC 3.0 needs to align with NDPIV.

Gender and social justice

Uganda's national strategies and plans, including the National Development Plan IV (NDPIV), the Uganda Green Growth Development Strategy (UGGDS), and the Updated Nationally Determined Contribution (NDC), have explicitly incorporated gender and social justice. The Fourth National Development Plan (NDPIV) for Uganda adopts a mainstreaming approach to gender equality, integrating it as a core principle across all sectors rather than treating it as a separate issue. This is achieved by aligning the plan with existing national and international frameworks, including Uganda Vision 2040 and the Sustainable Development Goals (SDGs), particularly SDG 5 on gender equality. The Fourth National Development Plan (NDPIV) for Uganda adopts a mainstreaming approach to gender equality, integrating it as a core principle across all sectors rather than treating it as a separate issue. This is achieved by aligning the plan with existing national and international frameworks, including Uganda Vision 2040 and the Sustainable Development Goals (SDGs), particularly SDG 5 on gender equality. The UGGDS is built on the core principle of "social inclusiveness" to address income inequalities and regional imbalances. The updated NDC for 2022 was designed with a "whole-of-society gender responsive approach," and a gender analysis was conducted to inform its measures and actions. It also explicitly set a goal to identify gender equity targets with concrete actions and indicators to measure progress.

Despite challenges, implementation of gender mainstreaming in climate change interventions is considered "on track" (BUR 2025). Key achievements described within the MRV framework implemented under the Capacity Building Initiative for Transparency (CBIT) project include: The Water and Environment sector has

implemented specific measures to address constraints women face in accessing water services; the CBIT project has integrated gender into its Monitoring, Reporting, and Verification (MRV) operations; Gender focal points have been established in five key sector hubs; a procedure manual was developed to generate gender-disaggregated information for the Greenhouse Gas Inventory; and capacity-building initiatives and international participation have shown progress, with women making up 35% of those who completed a training course on the GHG inventory and MRV system.

Progress in implementing and tracking gender in the NDC is affected by institutional arrangements and mechanisms. The key gaps and challenges are:

- Financial Constraints: There is a lack of adequate funding for gender-responsive climate programs and limited knowledge on how to access such funds.
- Institutional Gaps: These include a lack of dedicated gender experts in government ministries, departments, and agencies (MDAs).
- Awareness and Linkages: Some technical staff lack a full understanding of the relevance of gender to their work, and there are weak links between national gender policies and climate change action.
- Monitoring and Reporting: A lack of gender-specific indicators and tracking mechanisms makes it difficult to measure results and progress accurately.
- The transition to NDC 3.0 presents an opportunity to embed climate action within a broader framework and improve gender and social inclusion, through the following recommendations:
- Capacity Building: Implement gender-focused training for technical staff to enhance their understanding of gender analysis.
- Policy and Strategy: Develop sector-specific gender policies and strategies to guide interventions.
- Budgeting: Adopt gender and equity budgeting to ensure resources are properly allocated.
- Institutionalization: Institutionalize the position of a Gender Officer within sectors to lead mainstreaming efforts.
- Data and Indicators: Develop gender-disaggregated data and indicators to better track the impacts of programs on men and women.

Use of resources and timeline for NDC update

It is a best practice that Uganda draws on the resources and networks of international cooperation and partners to draft and conduct the NDC update. This is an excellent strategy to access accumulated knowledge and innovations on NDC processes. Further, it alleviates the public sector from additional workload and sets an example of efficiently mobilizing the resources of international cooperation or private funds.

Further, it is excellent that the Government of Uganda foresaw and aimed at aligning the update of the NDC with the stocktake and the LTS. Aligning the processes optimizes the use of time, alleviates stakeholder fatigue and enhances understanding and ownership of the processes for those involved.

These processes, however, come with the challenges of inadequate time planning and proper coordination and concertation of those actors involved, especially those conducting the NDC update. On the one hand, those in charge of collecting data and conducting the NDC update require knowledge on where and whom to get the data from, and those providing the data, say the sectoral institutions and technical working groups, need to be informed on the expected contributions on their end, and the expected timing for their

engagement. When international partners are involved in the process, it needs to be considered that they need much time in advance for the preparation of licitation or procurement processes (even for small procurements, say, booking venues for NDC meetings). The public, private, and diverse stakeholders have different timings and rhythms, which need to be recognized in the NDC process. Due to these different needs of information and timings, the leading government agency needs to provide information timely and clearly to all involved parties.

The coordination among high-level stakeholders, the definition of communication processes, data flows and the approval of deliverables, are all aspects that need to rely on the lead government agency. The involved stakeholders require and expect the government to define these topics to allow smooth coordination and work towards the NDC update.

The time required to update an NDC is related to these issues described above, as well as the available capacity and data. A timeframe of 6 months should suffice if factors such as robust coordination structures, clear communication and data flows, available data and dedicated capacities and resources are given. As we have seen, the limitations in data availability and accessibility, together with the different timings of stakeholders (e.g. for reviewing documents as well as for procurement), suggest that a longer timeframe for updating the NDC would be optimal. Due to the relevant and importance of the NDC, we suggest planning an NDC update with at least 1 year in advance of the expected delivery date.

In addition, we suggest conducting a risk management exercise at the beginning of the NDC process. This would allow identifying early on the possible bottlenecks or risks, as well as external factors that may be difficult to control and may lead to altered timeframes or conflicts. Such an exercise can allow to identify and plan mitigation or alternative measures to prevent and manage factors leading to delays or time constraints.

Conclusion

Uganda has made measurable progress toward some NDC 2.0 targets, with notable achievements in renewable electricity expansion, REDD+ initiatives, and adaptation in sectors such as transport, energy (electricity access and transmission), and health (via H-NAP). However, overall progress is uneven: many sectors—including clean energy access, agriculture, water and sanitation, forestry, cities, and disaster risk reduction—lack sufficient data to assess outcomes, while persistent gaps in finance, MRV systems, institutional clarity, and private sector engagement continue to limit full implementation of NDC 2.0 goals. Moving forward, addressing these gaps, integrating the National Adaptation Plan, strengthening monitoring and reporting, and leveraging international and private finance will be critical to achieving the stated NDC 2.0 targets.

Uganda's Updated NDC (NDC 2.0) reflects clear progress in scope, ambition, and institutionalization of climate action. The economy-wide mitigation target of 24.7% relative to BAU, expansion of adaptation coverage to 13 sectors, and the establishment of a robust legal framework under the Climate Change Act (2021) collectively demonstrate Uganda's commitment to advancing climate action while pursuing socioeconomic development.

The stocktake reveals notable achievements: renewable electricity expansion contributing measurable emission reductions, REDD+ initiatives reversing deforestation trends in selected landscapes, and climate-smart agriculture practices delivering quantifiable mitigation and adaptation benefits. Adaptation planning has broadened in scope, reflecting Uganda's vulnerability context and aligning with national priorities such as Vision 2040 and NDP III. Progress has also been made in mainstreaming climate considerations into national development planning and budgeting processes.

Uganda has made steady but uneven progress in implementing its NDC adaptation actions over the 5-year period. The absence of an updated National Adaptation Plan weakens coherence of adaptation efforts. More robust data systems and inter-sectoral coordination are critical to drive meaningful adaptation outcomes in the next NDC cycle. The implementation of H-Nap should be a basis for adaptation in the health sector in NDC3.0.

Persistent barriers remain. Critical gaps in data and monitoring systems impede accurate tracking of progress across transport, waste, and IPPU sectors. The financial gap is stark: while total NDC costs are estimated at USD 28.1 billion, Uganda has mobilized only 15% domestically, leaving its ambition heavily dependent on international finance. Limited private sector engagement, weak MRV arrangements, and lack of clarity in institutional mandates further constrain implementation.

Overall, the stocktake shows that Uganda's NDC 2.0 is **largely not on track** to achieve its 2030 mitigation targets. While AFOLU and waste sectors demonstrate measurable progress, high-impact areas such as transport, stationary energy efficiency, and IPPU remain unquantified or delayed. The lack of timely and comprehensive data, particularly the absence of GHG inventory updates beyond 2019, further impedes a full assessment of NDC implementation. Urgent acceleration of sectoral implementation, operationalization of the national MRV Tool, and stronger reporting systems are required to close these gaps and credibly demonstrate results under the Enhanced Transparency Framework.

Looking ahead, the preparation of NDC 3.0 presents a crucial opportunity to address these gaps. Uganda should define a clear emissions peaking year, expand unconditional contributions, and align its targets with a long-term net-zero strategy. The integration of the National Adaptation Plan will be critical to strengthen

adaptation objectives and indicators. Establishing a centralized MRV and climate data management system is an urgent priority to ensure transparency and accountability. At the same time, Uganda must develop a prioritized investment pipeline and leverage international mechanisms such as the Green Climate Fund, bilateral climate finance, and Article 6 cooperative approaches.

The transition to NDC 3.0 should also place stronger emphasis on cross-cutting themes: ensuring gender equality, social justice, and a just transition for communities dependent on high-emission livelihoods. Whole-of-society engagement – from local governments to the private sector, academia, and civil society – will be essential to secure broad ownership and durable outcomes.

In conclusion, Uganda is at a pivotal moment in its climate journey. By consolidating lessons from NDC 2.0, addressing persistent data and finance gaps, and embedding resilience and inclusivity into its climate agenda, Uganda can position NDC 3.0 as a more ambitious, credible, and implementable roadmap. This will enable the country not only to honor its Paris Agreement commitments but also to build a climate-resilient, low-carbon future that supports sustainable national development.

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13 Annexes

Annex 1: Methodologies and Accounting Approaches

Annex 2: List of Stakeholders Contacted and/or Met

Annex 3: NDC Comprehensive Assessment Framework

